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भारतीय प्रौद्योगिकी संस्थान मुंबई
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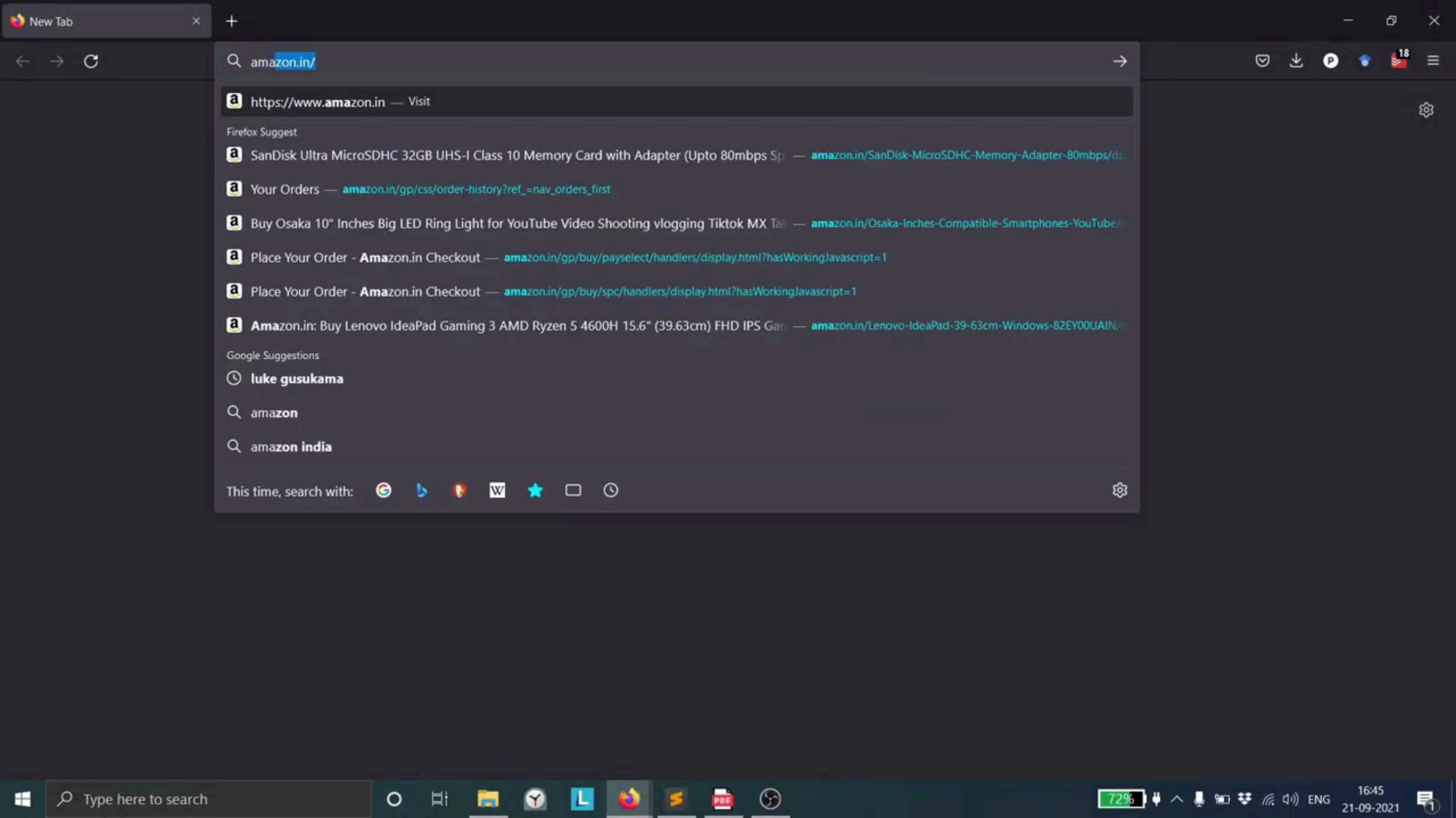
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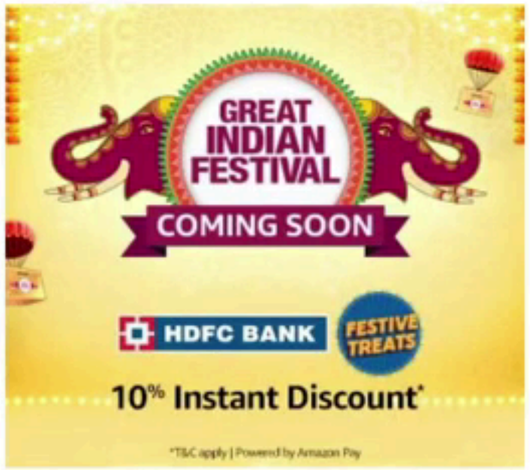

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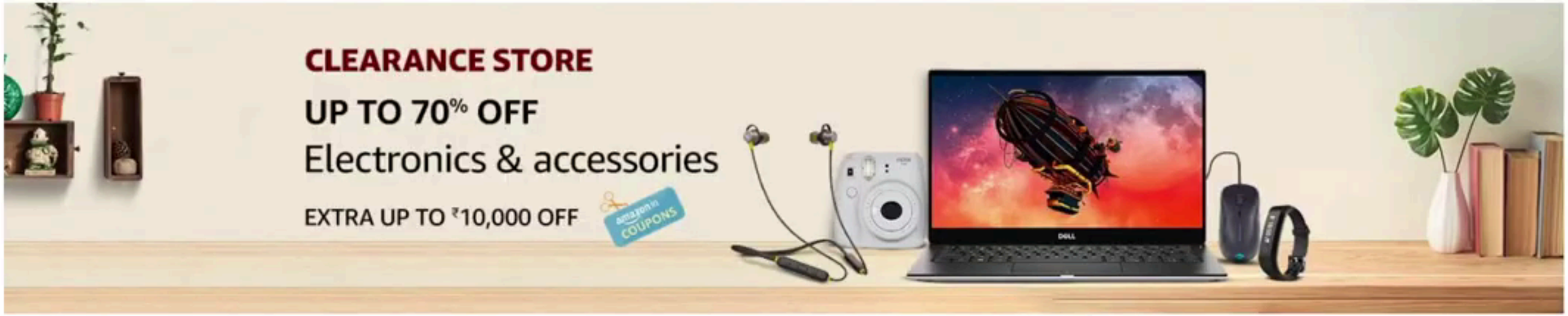

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INFINITY
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INFINITY
4.3 out of 5 stars (43,091)

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Carvaan
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Harman Kardon Omni 20+ 60 Watt Wireless...
Harman Kardon
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FLiX (Beetel) Blaze Wireless Bluetooth v5....
FLiX
₹799.00 ~~₹1,499.00~~
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EDICT by Boat EWE01 in-Ear Wireless Earphone...
EDICT
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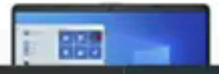
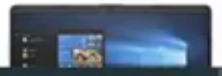
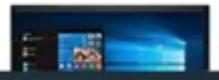
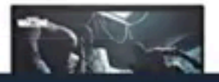
Handpicked laptops for multi-tasking









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(43,091)

(43,091)

₹1,199.00 ~~₹1,599.00~~
(439)

₹21,489.00 ~~₹22,999.00~~
(112)

₹799.00 ~~₹1,499.00~~
(86)

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Handpicked laptops for multi-tasking

HP 15 10th Gen Intel Core i3 Thin & Light 1...
HP
₹40,990.00 ~~₹43,090.00~~
(710)

HP 15 11th Gen Intel Core i3 Processor 15.6"...
HP
₹42,490.00 ~~₹46,805.00~~
(357)

HP Pavilion Gaming 15.6-inch (39.62 cms)...
HP
₹70,990.00 ~~₹79,529.00~~
(493)

HP Pavilion Gaming 10th Gen Intel Core i5 Proce...
HP
₹74,900.00 ~~₹83,038.00~~
(11)

ASUS ROG Zephyrus G14, 14" FHD 120Hz,...
ASUS
(10)

HP Pavilion Gaming 10th Gen Intel Core i5 Proce...
HP
₹69,890.00 ~~₹81,868.00~~
(83)

Desktops & monitors for home office setup

Samsung 27 inch M5

Acer EB321HQU 31.5

DELL Monitor E2720HS

Acer Aopen 21.5 inch

HP Slim Tower Desktop

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Routers & components

AMD Ryzen 7 3800X Desktop Processor 8 C...
AMD
₹32,986.00 ~~₹55,000.00~~
(5,988)

Seagate 240 GB 2.5x7mm SATA_6_0_g...
Seagate
₹2,998.00 ~~₹3,999.00~~
(1,118)

Corsair Vengeance LPX 16GB (1x16GB) DDR4...
Corsair
₹7,050.00 ~~₹21,150.00~~
(76)

Linksys E5600 AC 1200 Dual-Band (2.4 GHz &...
Linksys
₹3,699.00 ~~₹4,999.00~~
(3,952)

Micron 5300 PRO I MTFDDAK3T8TDS I PR...
Micron
₹57,008.00 ~~₹74,600.00~~
(₹814.40 / Grams)

Corsair Vengeance LPX 8GB (1x8GB) DDR4 32...
Corsair
₹3,449.00 ~~₹10,125.00~~
(832)

Cameras & accessories for photo enthusiasts

Fujifilm Instax Mini 9 Instant Camera (Smok...
Fujifilm

Photron 13 Inch Flexible GorillaPod Tripod Octo...
Photron

Fujifilm Instax Square SQ1 Camera - Chalk W...
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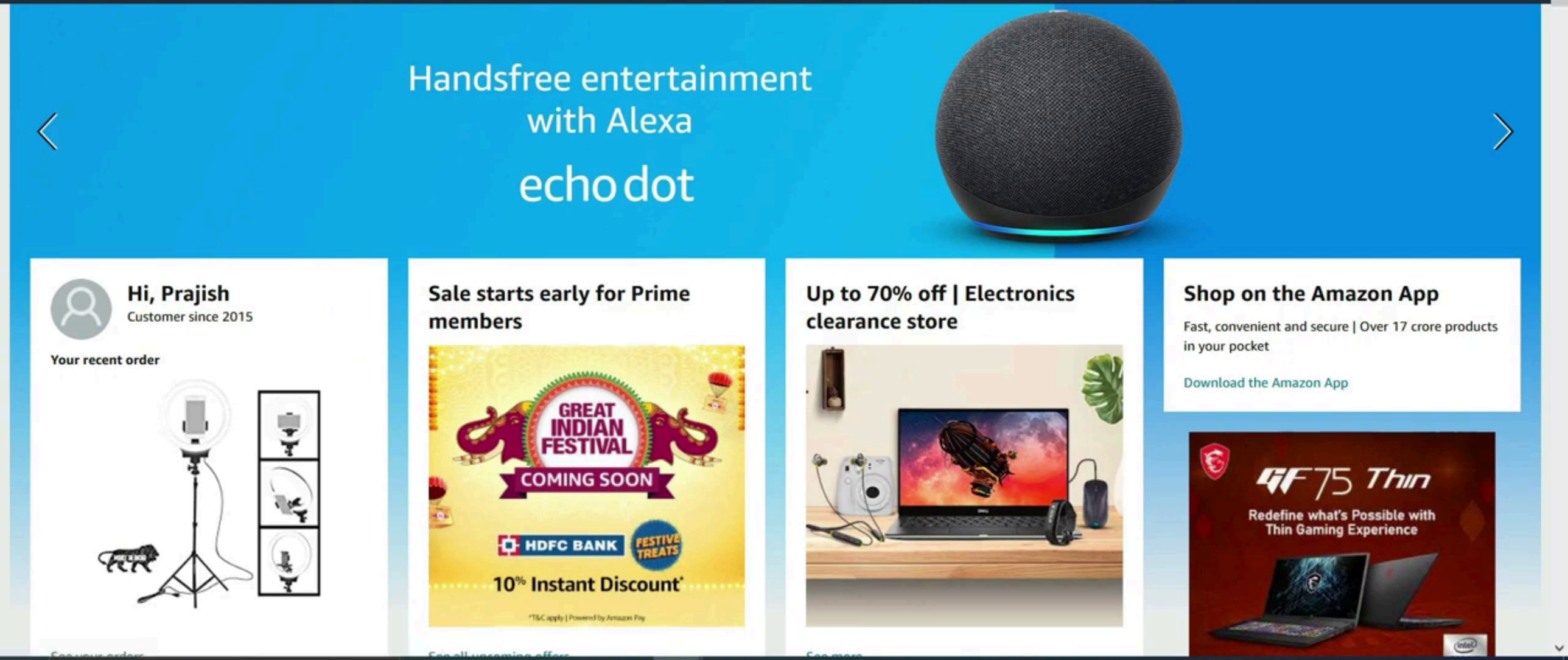

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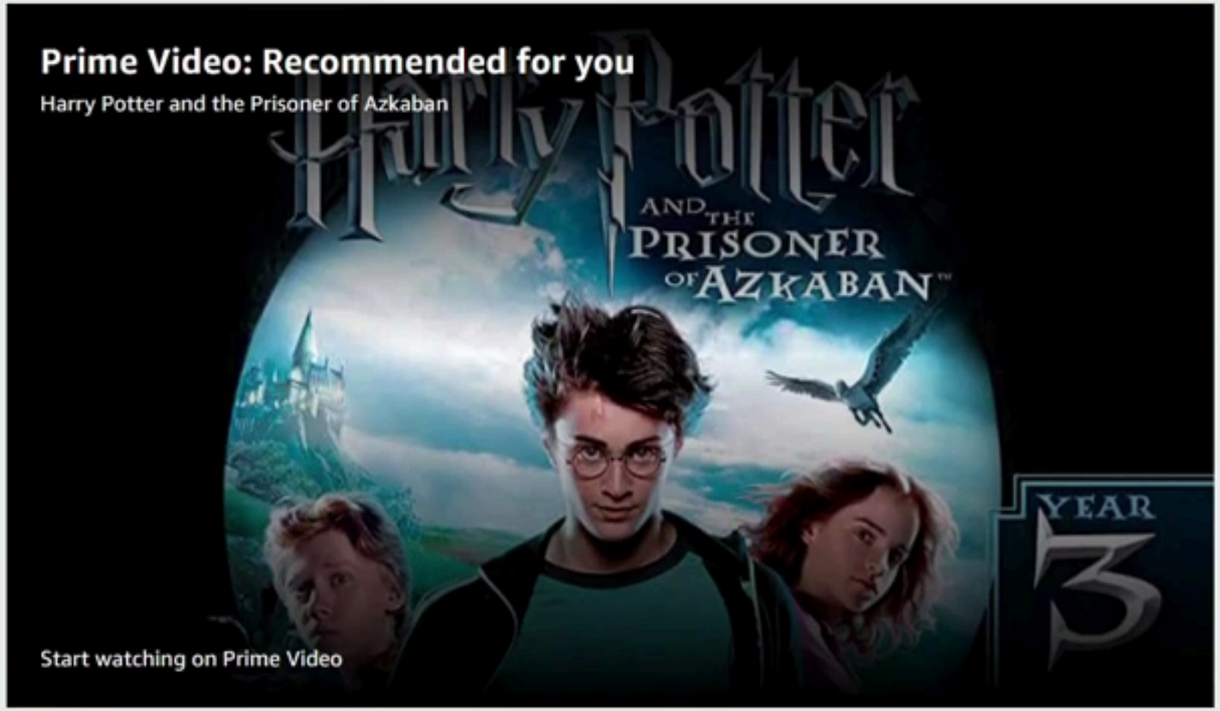
Prime Video: Recommended for you

Harry Potter and the Prisoner of Azkaban



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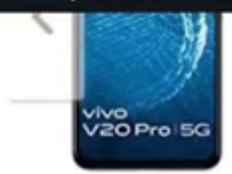

₹29,990.00
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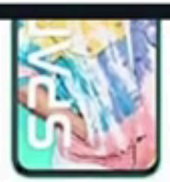

₹9,499.00 - ₹10,599.00
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₹89.00 - ₹3,099.00
Ends in 07:13:59

₹800.00 - ₹16,060.28

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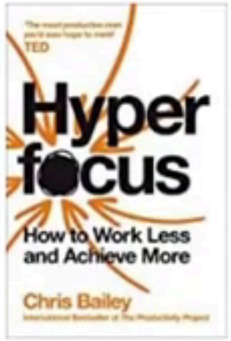

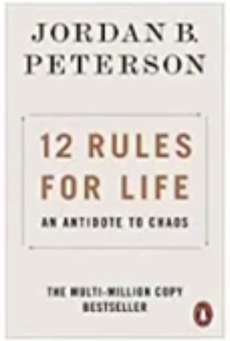

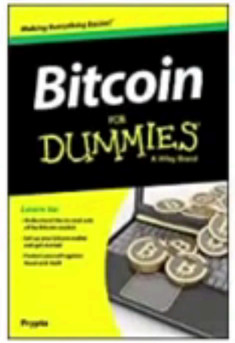

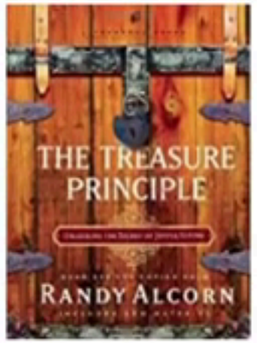

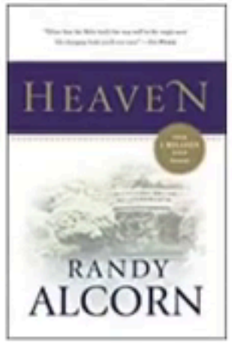

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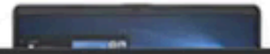

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Stay protected | Masks, PPE kits & more from startup...

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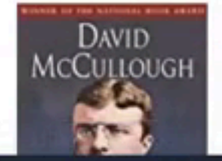

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₹840.00 ~~₹1,050.00~~

LP-E8 (1200 mAh) Rec...
₹892.00 ~~₹1,450.00~~

Tyfy LPE8 (Canon) Batt...
₹750.00 ~~₹1,095.00~~

Powerextra 2 Pack 7.4V...
₹3,898.00 ~~₹5,570.00~~

Masks

PPE kits & face shield

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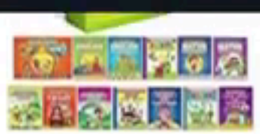
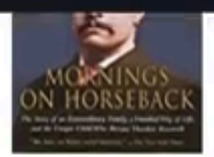
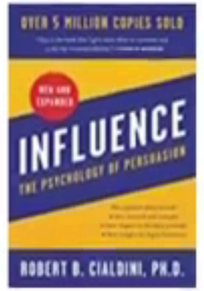
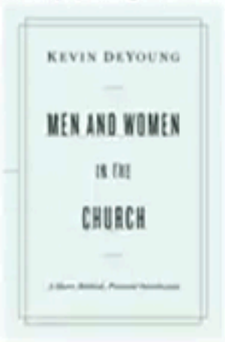
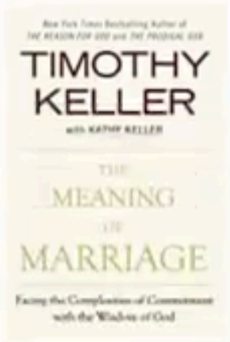


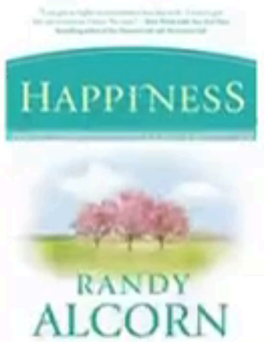


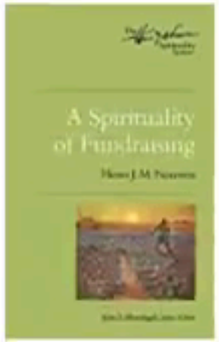


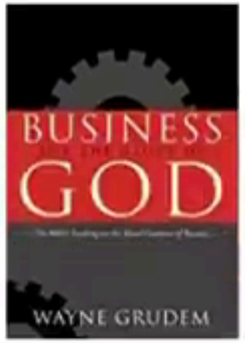
Inspired by your shopping trends

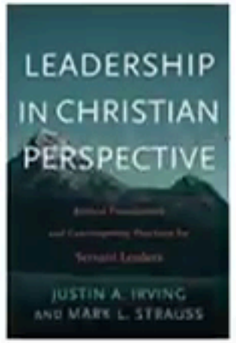


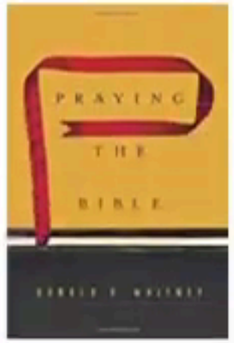


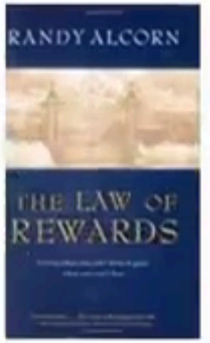




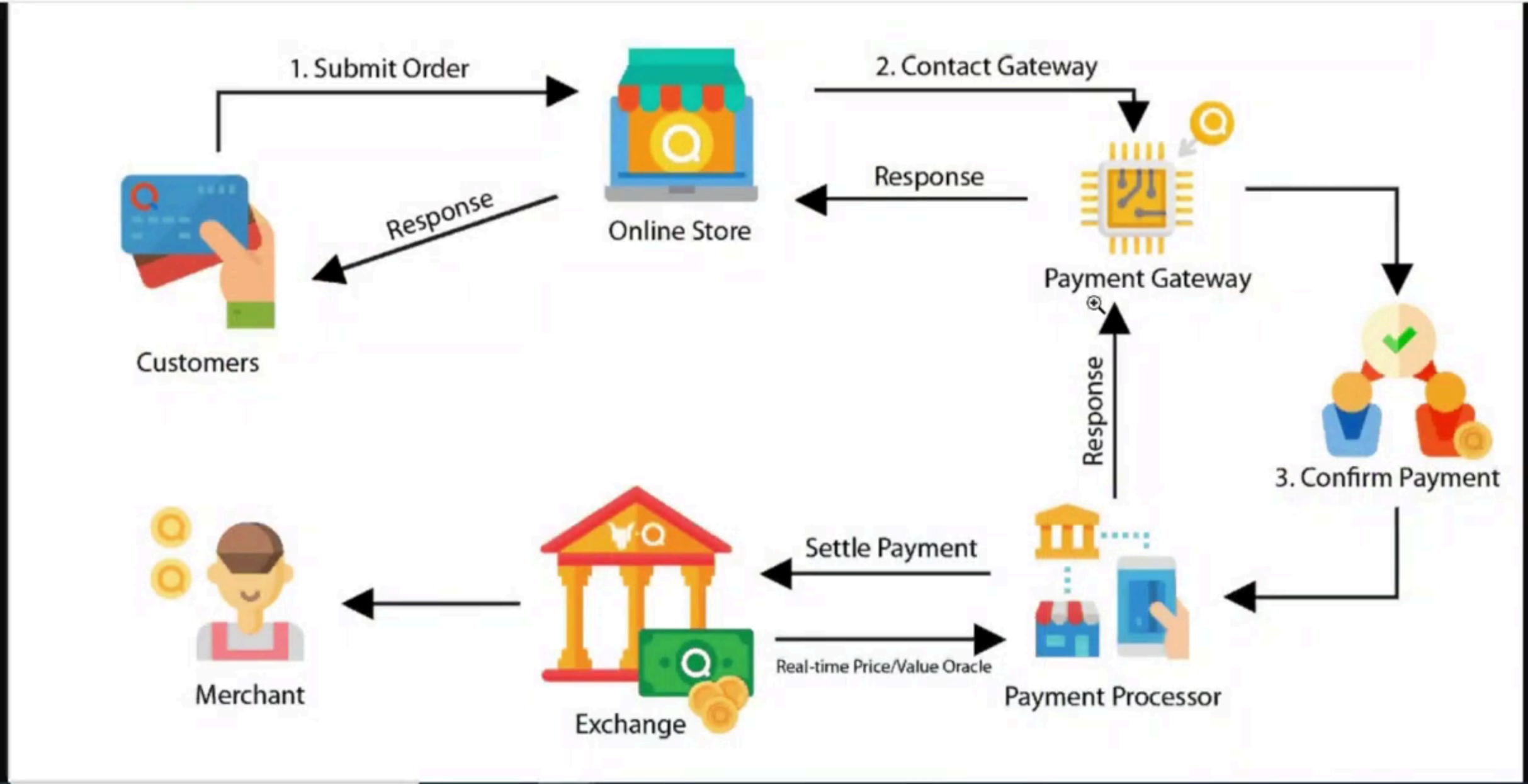








Everyday essentials that you might like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2. Contact Gateway
Response
Response
Online Store
Payment Gateway
Customers
Response
3. Confirm Payment
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Real-time Price/Value Oracle
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Exchange
Payment Processor

Reflection Spot

IIT BOMBAY

We discussed that Amazon has several components. Could you list a few of them ?

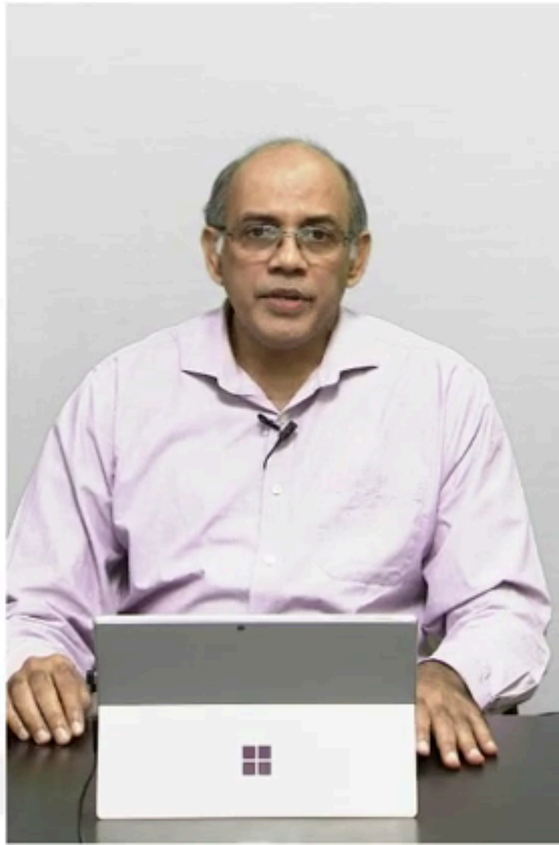
Please pause the video and write down your responses

Summary

IIT BOMBAY

- Discussed various components of a large software system E.g. Amazon
  - Inventory Management System
  - Payment Gateway

- Software can be divided into separately addressable components called **modules** that are integrated to satisfy requirements



amazon pay

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- Add Money
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- Auto Reload
- Account Settings

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Recharges, Bill Payments

Mobile Recharge | Electricity | DTH Recharge | Mobile Postpaid | Credit Card Bill | Gas Cylinder | Insurance Premium

Piped Gas | Water Bill | Landline | Broadband | Municipal Tax | Google Play Recharge | Subscriptions

Cable TV | Education Fees

Travel and insurance

Flights | Bus Tickets | Car Insurance | Bike Insurance

Rewards

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Flights | Bus Tickets | Car Insurance | Bike Insurance

Rewards

All categories | Food orders | Stores Near You | Shopping | Travel | Entertainment | Health & fitness

Gift Cards & Vouchers

Gift Cards | Add Gift Card | Brand Vouchers | Birthday Giftcards | Wedding Giftcards | Corporate Gifting

Pay in 1 click on 10,000+ apps
with **Amazon Pay balance**

Reflection Spot

IIT BOMBAY

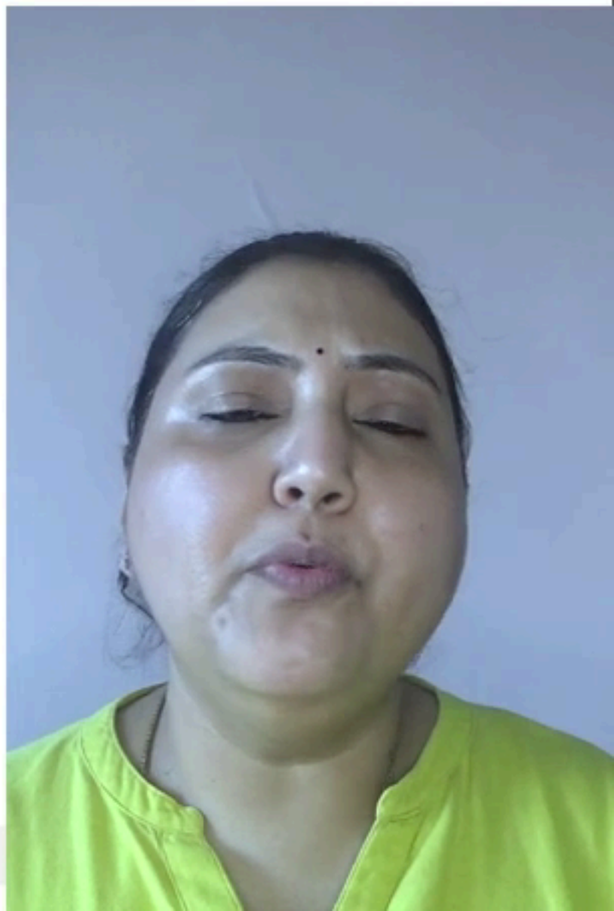
What do you think is the first step in creating a new software component?

Please pause the video and write down your responses

First Step in Creating Software

IIT BOMBAY

- Learn a programming language?



First Step in Creating Software

IIT BOMBAY

- Study existing components of the system -
  to understand how the new component will interact with existing components

First Step in Creating Software

IIT BOMBAY

- Look at similar systems to understand features
  e.g. PayTM, PhonePe

Requirement Specification

IIT BOMBAY

Requirements

- Goals the implemented system should have
- Should cater to the need of clients

Client - External User

IIT BOMBAY

Example - Mobile banking software serves bank customers

Client - Internal to your Company

IIT BOMBAY

Example -
Building an internal employee resource portal

- Internal products team → To build this portal

Client - Another Software

IIT BOMBAY

Example -
A payment gateway interfaces with another ecommerce system

Clients

IIT BOMBAY

- Think about **who** is going to use your software, for **what purpose**, and in **what way**

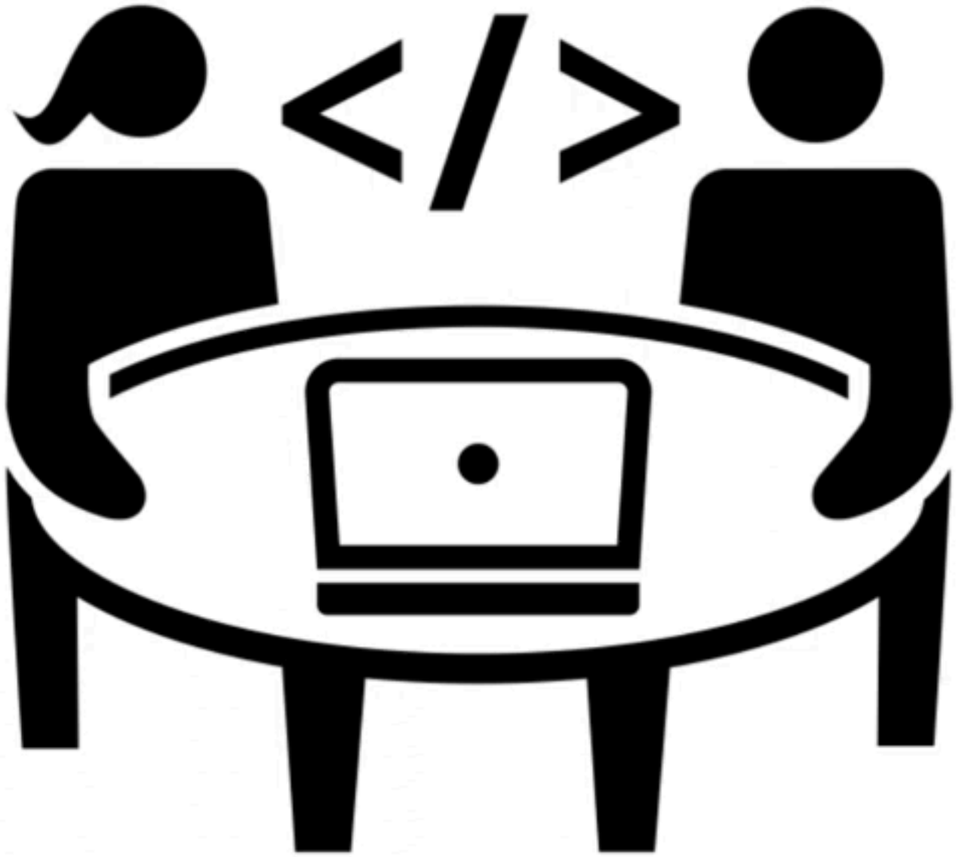
Summary: Requirement Specification

IIT BOMBAY

- **Requirement specification** - First step in the software development process
- **Clients** - end users of the software
- Need to ensure that the requirements capture clients' needs

Software Development Team

IIT BOMBAY



NPTEL

Reflection Spot

IIT BOMBAY

What difficulties are you likely to encounter if you directly start coding?

Please pause the video and write down your responses

Issues during integration

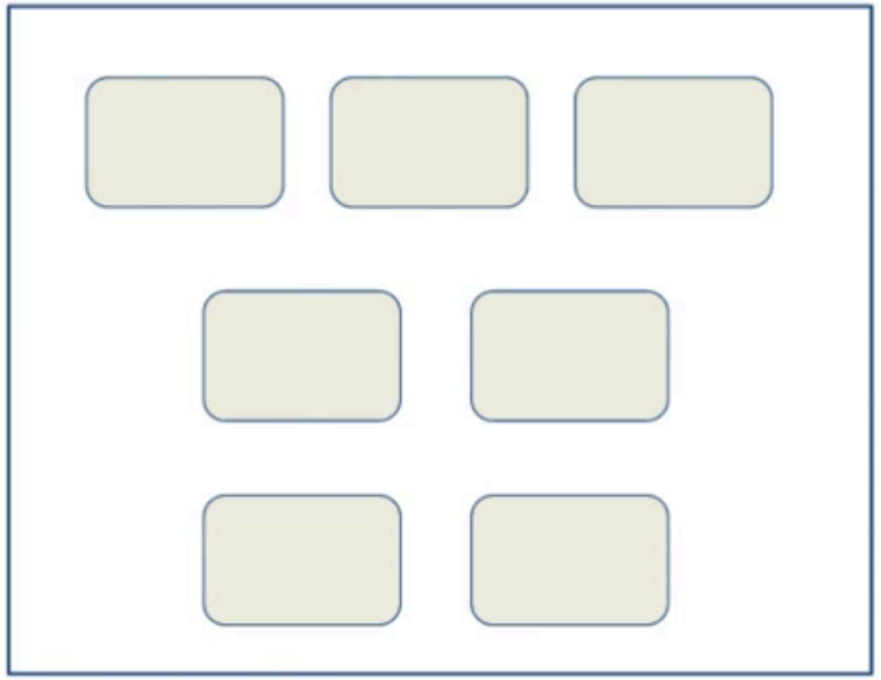
IIT BOMBAY

- Different developers may have different ideas about how the functionality should be implemented

Difficulties while adding new features

IIT BOMBAY

- Adding new features - big picture view of the system is necessary



Software Design

IIT BOMBAY

Requirements

- Goals the implemented system should have
- Should cater to the need of clients

Design

- Big picture view of the software system
- Provides a structure to the software system

Reflection Spot

IIT BOMBAY

How do you think people work in the development phase?

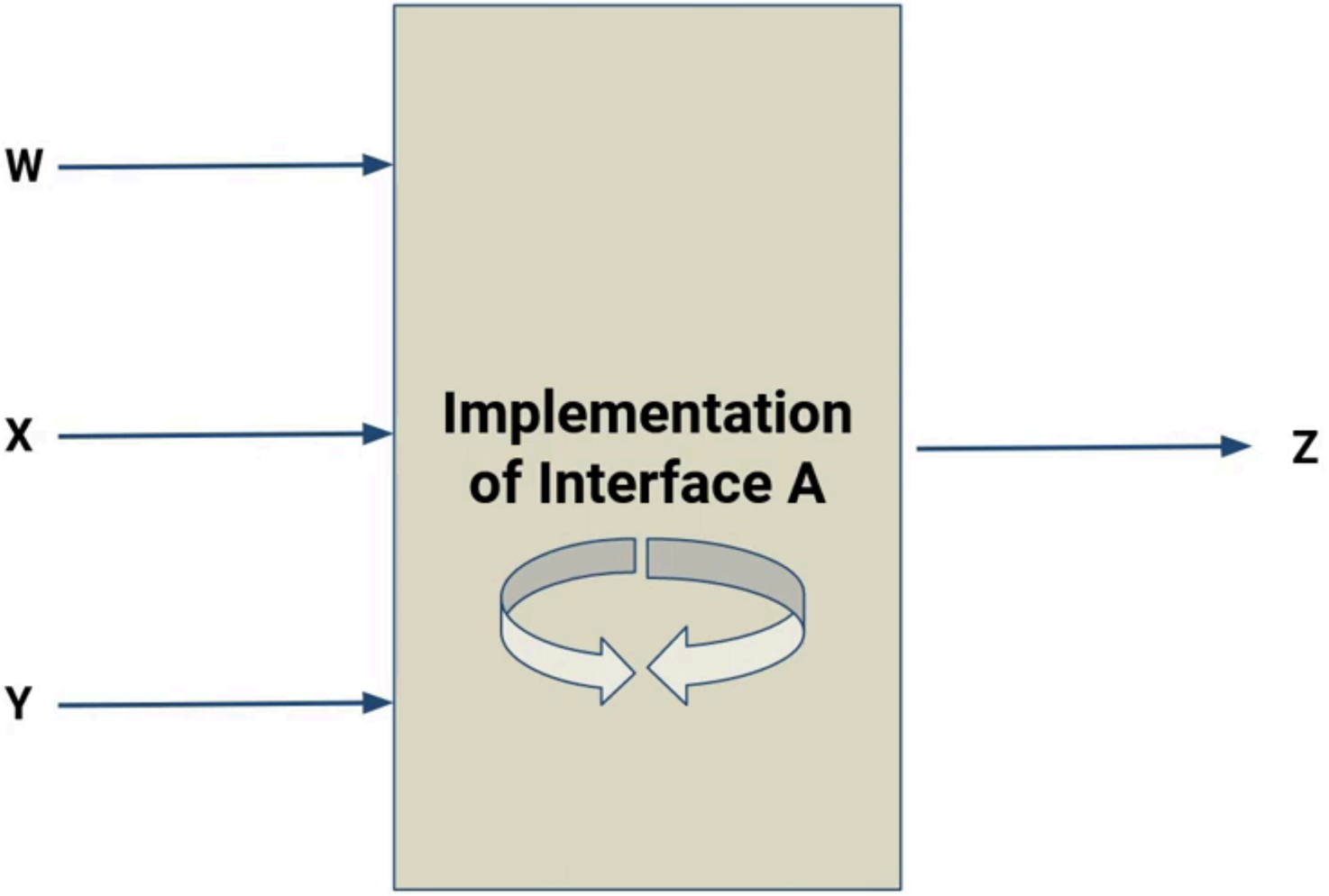
Please pause the video and write down your responses

Development Phase

Development Phase

IIT BOMBAY

Development Phase



Software Development

IIT BOMBAY

Requirements

- Goals the implemented system should have
- Should cater to the need of clients

Design

- Big picture view of the software system
- Provides a structure to the software system

Development

- Write code based on the requirements and the design
- Usually distributed
- Developer documentation and precise interface definitions

Reflection Spot

IIT BOMBAY

Why do you think testing is necessary? What can go wrong if we release the software directly?

Please pause the video and write down your responses

Importance of Testing

IIT BOMBAY

- Testing is done to ensure that the software behaves according to the requirements

- Many bugs might still exist in the system

Importance of Testing

IIT BOMBAY

- In 2002 - $59.5 billion in losses [1]
  In 2016 - $1.1 trillion [2]

- A failure to address bugs can even cause severe catastrophes

[1] Newman, M., 2002. Software errors cost us economy 59.5 billion annually. NIST Assesses Technical Needs of Industry to Improve Software-Testing
[2] https://medium.com/@ryancohane/financial-cost-of-software-bugs-51b4d193f107

Testing Methodologies

IIT BOMBAY

Alpha Testing

Conducted by: internal employees in a lab/staging environment

Goal: catch as many issues as possible before the product has been released to the public

Beta Testing

Conducted by actual users in a real-live setting

Maintenance Phase

IIT BOMBAY

Maintenance - After the feature is rolled out, monitor how users are using the feature

Purposes of maintenance

- Monitor what users are doing, and how they are using the software.
- Change the code for upgrades/updates
- Add features

Software Development Process

IIT BOMBAY

Requirements

- Goals the implemented system should have
- Should cater to the need of clients

Design

- Big picture view of the software system
- Provides a structure to the software system

Development

- Write code based on the requirements and the design
- Usually distributed

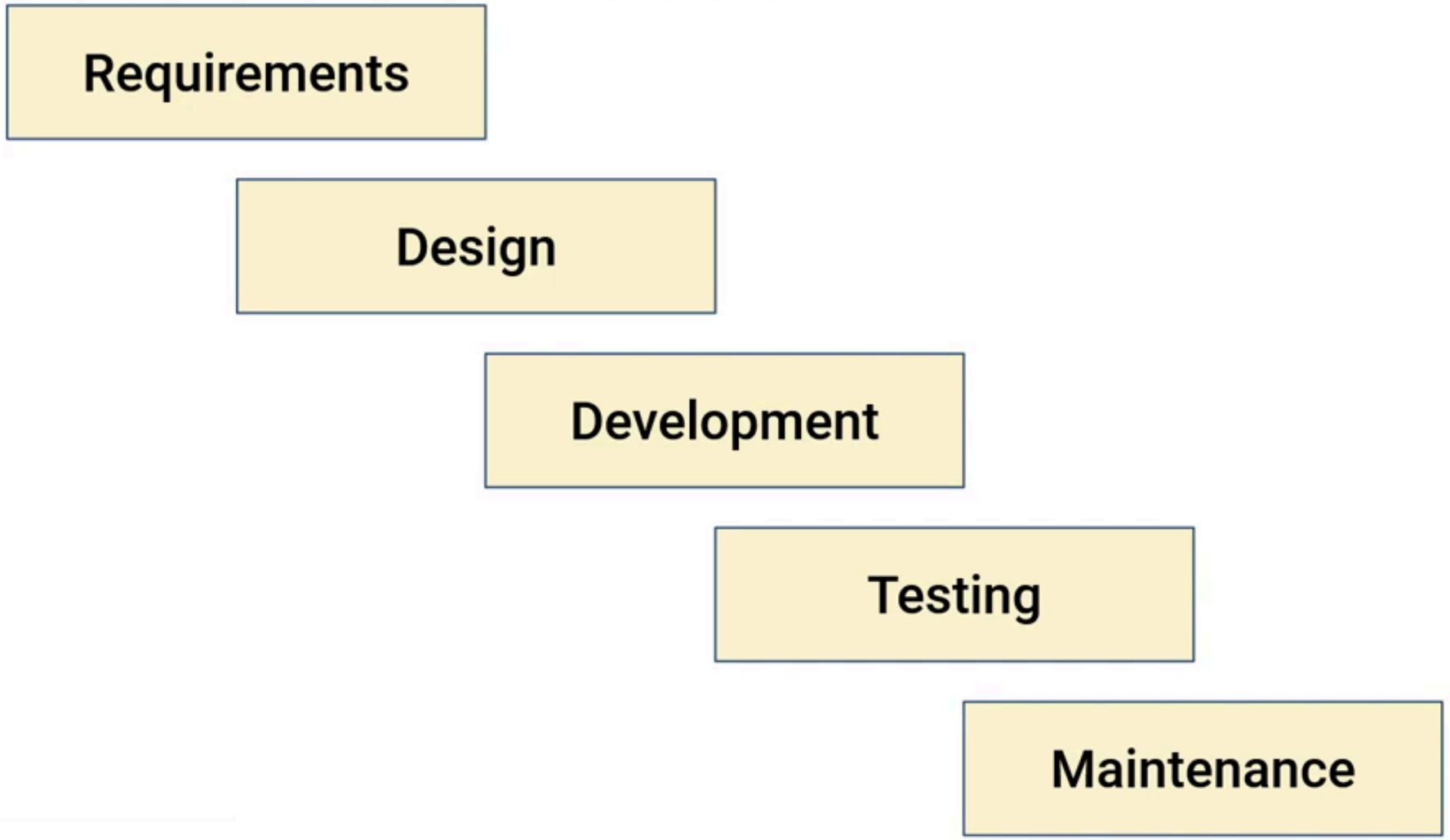
Testing

- Ensures that the software behaves according to the requirements

Maintenance

- Monitor what users are doing
- Change code for updates

Waterfall Model



Reflection Spot

IIT BOMBAY

What could go wrong if we follow these phases sequentially?

Please pause the video and written down your responses

Drawbacks of the Waterfall Model

IIT BOMBAY

2 months — Requirements

3 months — Design

4 months — Development

3 months — Testing

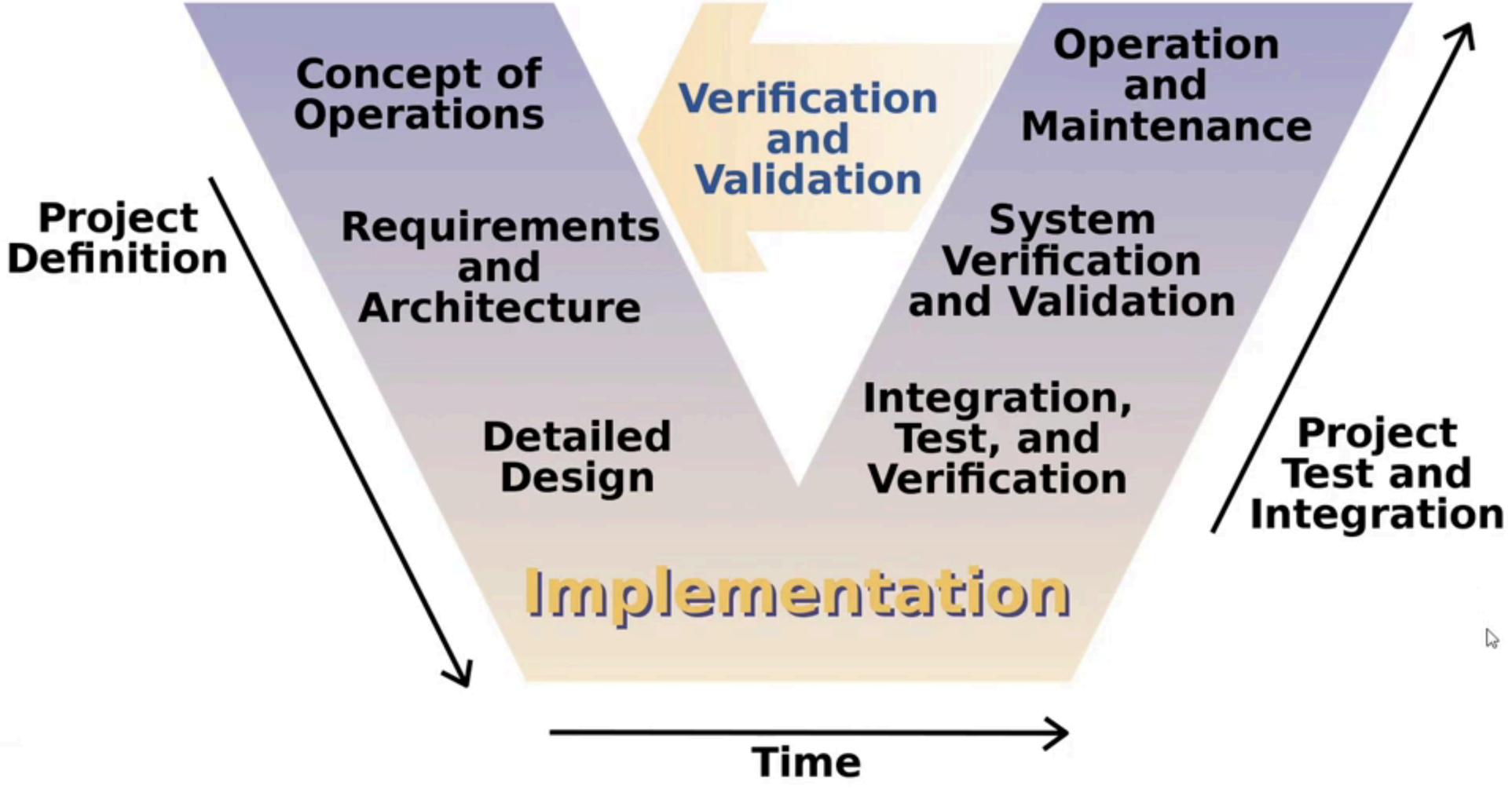
Drawbacks of the Waterfall Model

IIT BOMBAY

- Increase in cost, time if changes are required later on
- Clients may not know what they need!
- Designers may not know which design might be the most feasible/usable by clients

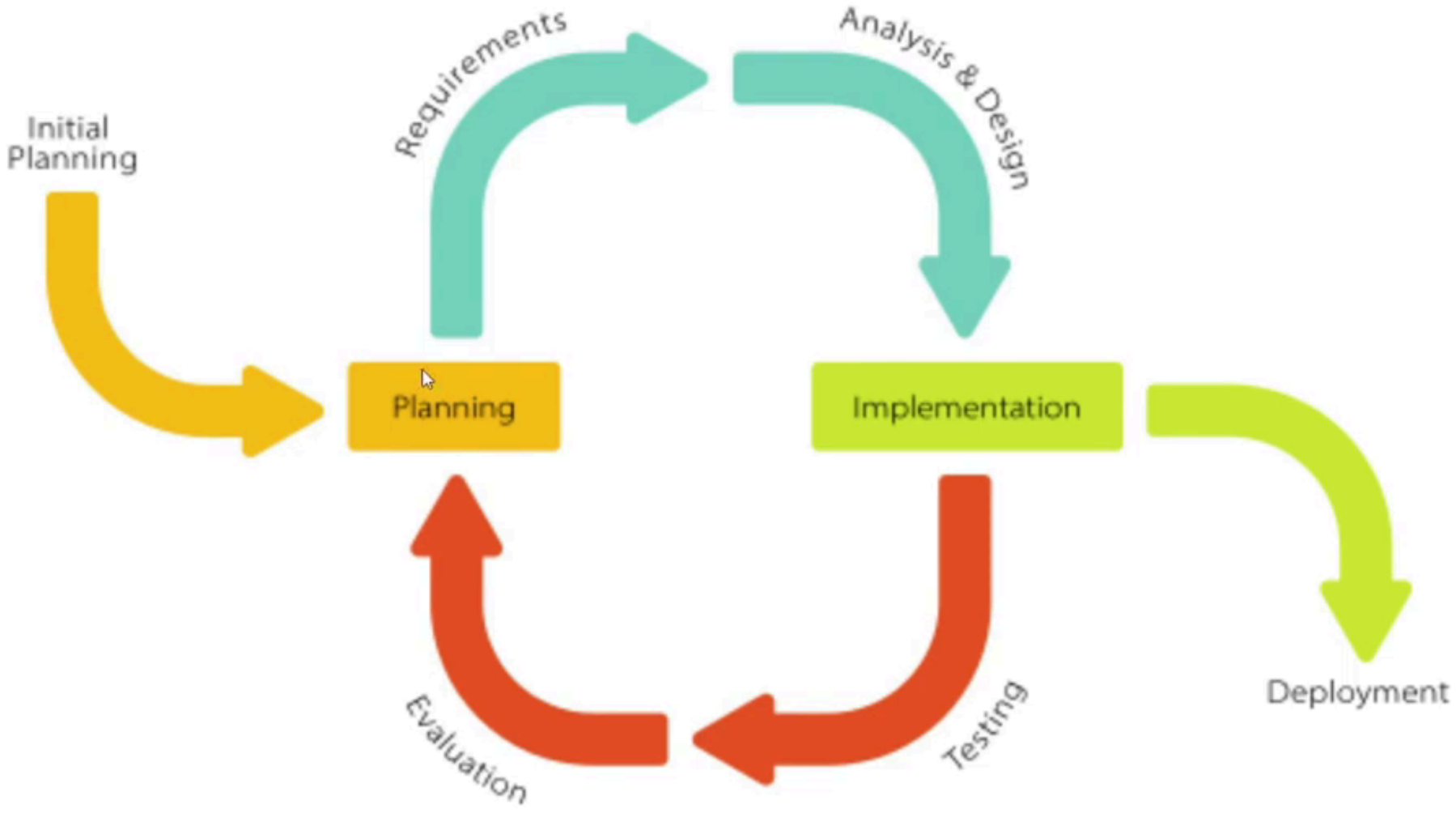
V-Model of Software Development

IIT BOMBAY



Iterative Development

IIT BOMBAY



Agile Methodology

IIT BOMBAY

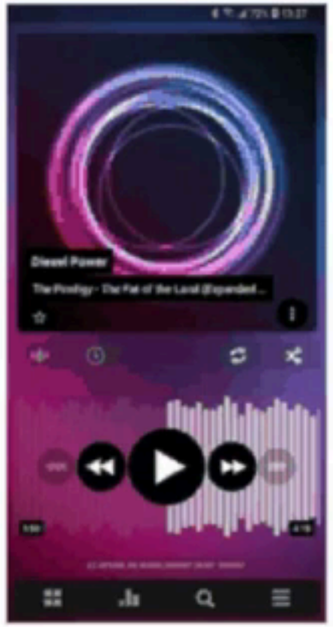
- Iterative software development model - teams work to deliver the product in small increments
- Frameworks - Scrum, Kanban
- Practices - Sprints, Test-driven development
- More than frameworks, practices - **“Agile Philosophy”**

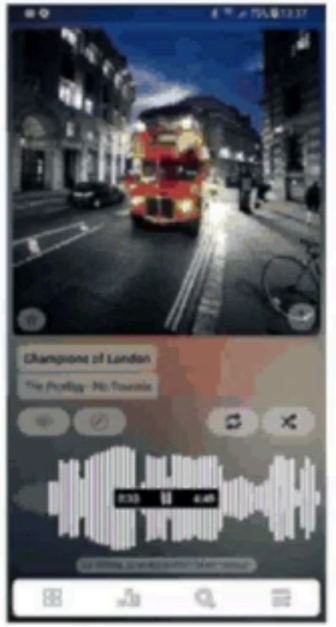
Abstract Nature of Software

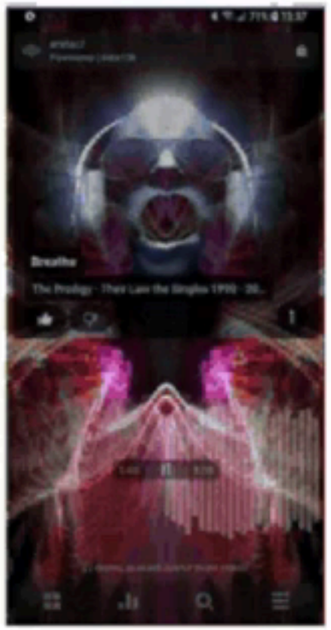
IIT BOMBAY

- Software - logical components - abstract in nature
- Software is adaptable

Dynamic Nature of Software







Reflection Spot

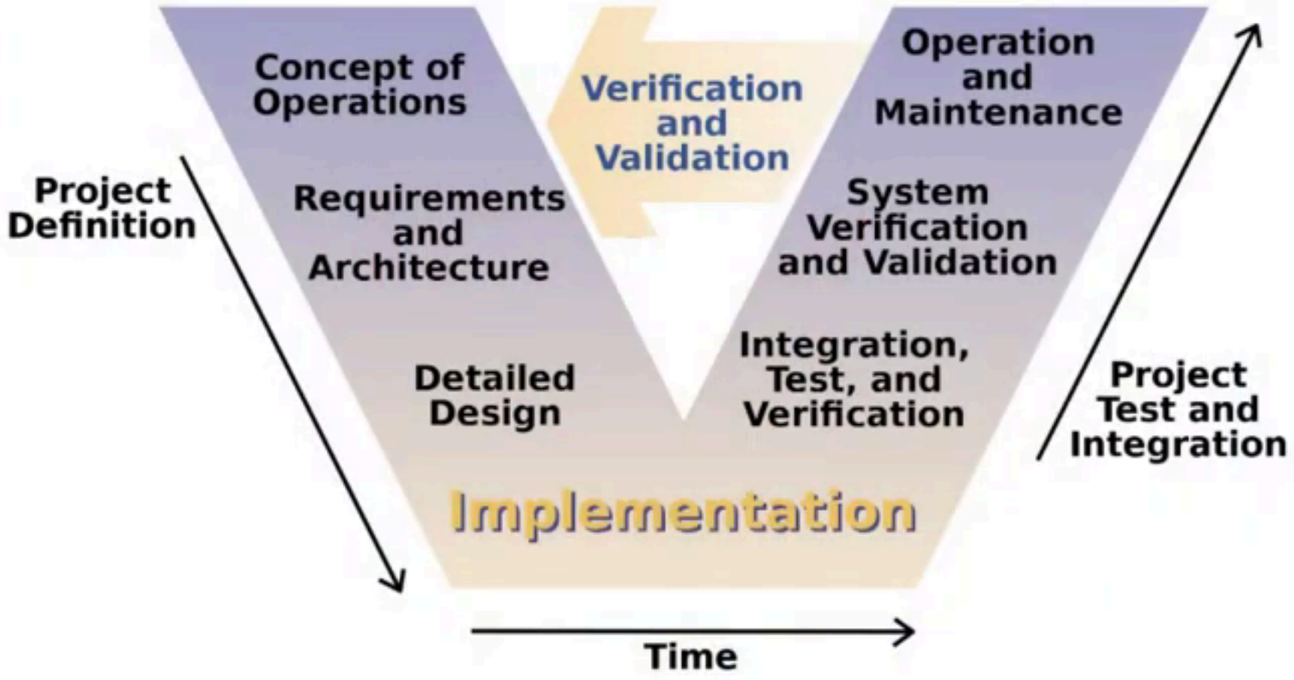
IIT BOMBAY

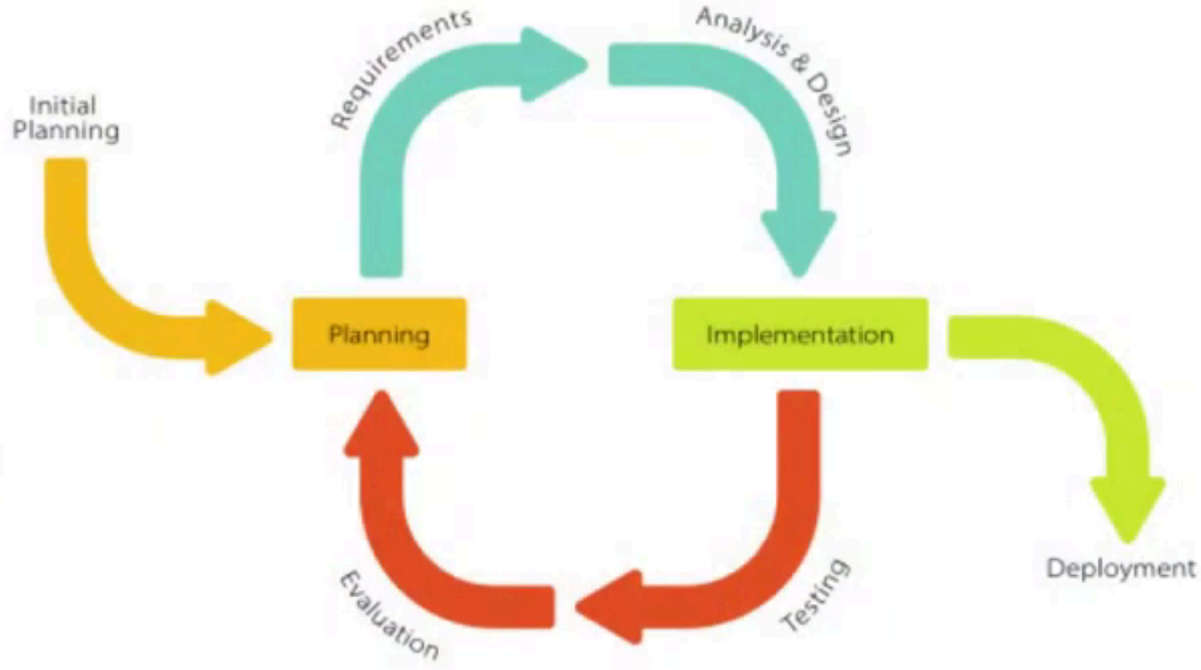
What are the software development models we learnt last week?

Please pause the video and written down your responses

Recap: Software Development Models

IIT BOMBAY





Apply Learnings to Design a Software

- Apply learnings from the different phases and models to a software development problem

Mood Based Music Player

IIT BOMBAY

- Automatically detects mood
- Plays music according to the mood
- Provides secure authentication
- Remember's user's choice of music
- Recommend music based on mood and user's choice

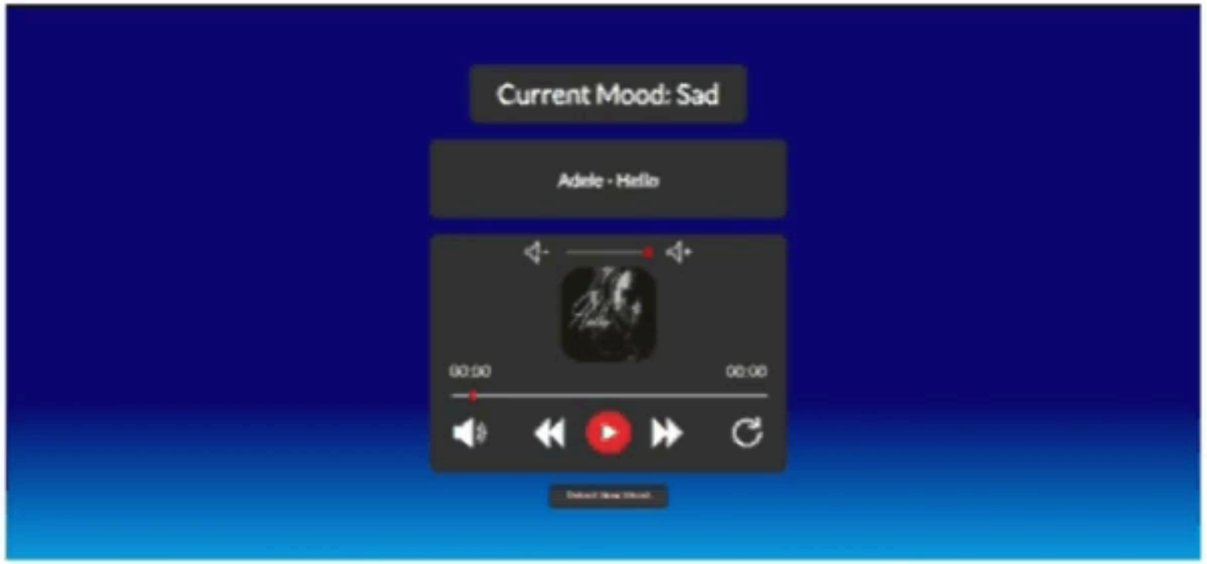


Image Credits - https://devpost.com/software/emotion-based-music-player

Reflection Spot

IIT BOMBAY

How do you think experienced software developers will go about designing such a mood based music player?

Please pause the video and written down your responses

FBS in mood based music player - Why?

IIT BOMBAY

- Purpose is deliberated from requirements
- Features of the solution emerge
- Functions - features of the solution

Mood Detection	F

Playlist Creation	F

User Authentication	F

FBS in mood based music player - What?

IIT BOMBAY

- Realize functions require components
- Physical as well as logical components
- Structure - Components of the solution

Emotion Recognition Camera	S

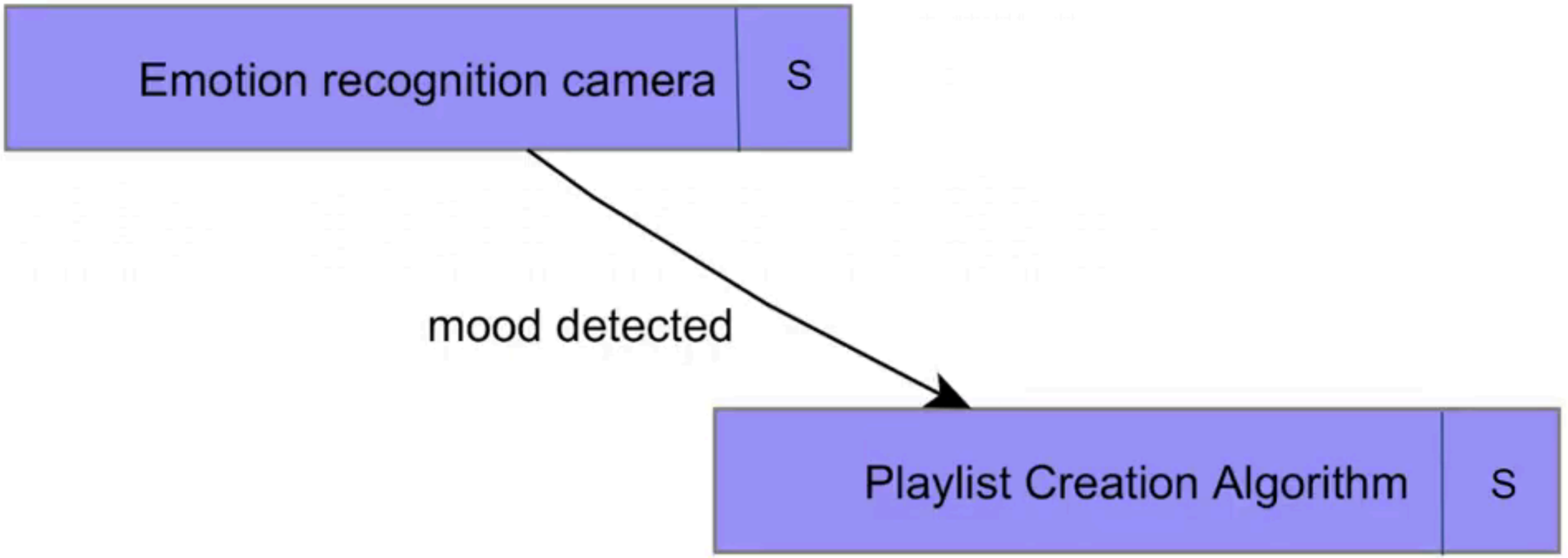
Playlist creation algorithm	S

Emotion tagging of songs	S

FBS in mood based music player - How?

IIT BOMBAY

- Structures interact with each other to achieve functions
- Behaviour - interaction between structure that achieves the functionality



Summary: Functions, Structures, Behaviours

- **Functionalities** that the software should contain
- These functionalities are realised by various **structures**
- Structures interact with each other → **Behaviours**

Reflection Spot

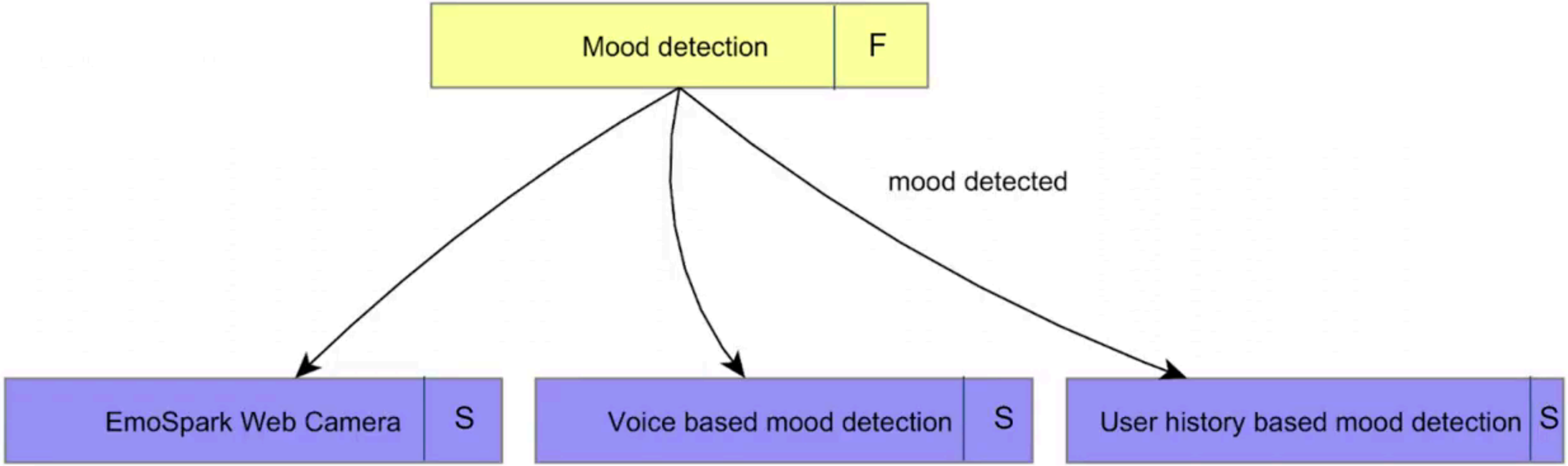
IIT BOMBAY

What will happen if we do not integrate function, behaviour and structures in the design?

Please pause the video and written down your responses

Incoherent and incomplete solutions

IIT BOMBAY



Integration of F-B-S is Essential

IIT BOMBAY

Incoherent solution if we do not integrate Function, Structure and Behaviour

Benefits of Integrating F, B and S

IIT BOMBAY

- Create an integrated view of the design solution
- Ensure that the solution fulfills all the requirements
- Communicate within the team with a unified view
- Avoid getting stuck while creating a design

How to integrate FBS?

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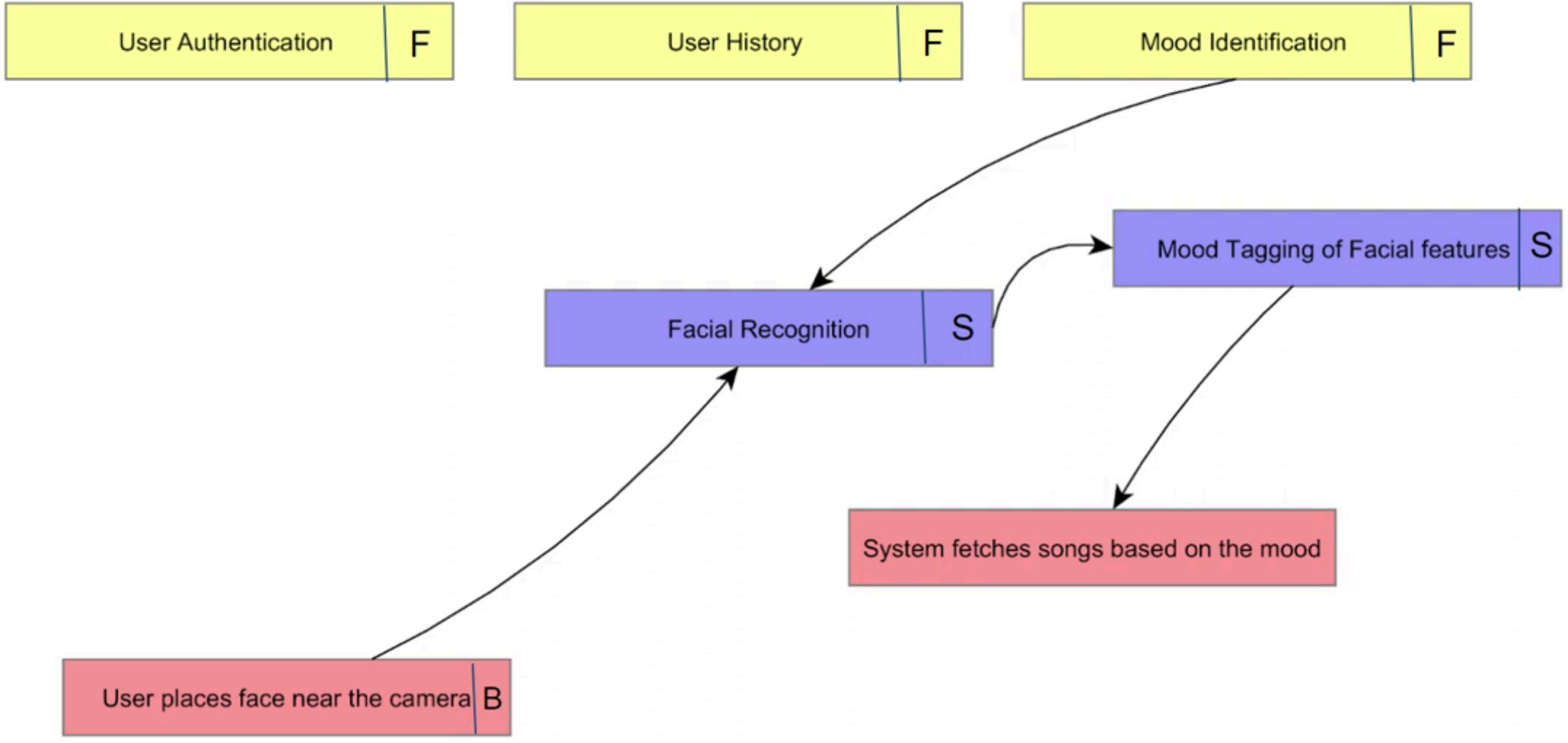
Function

Structure

Behaviour

How to create & map FBS?



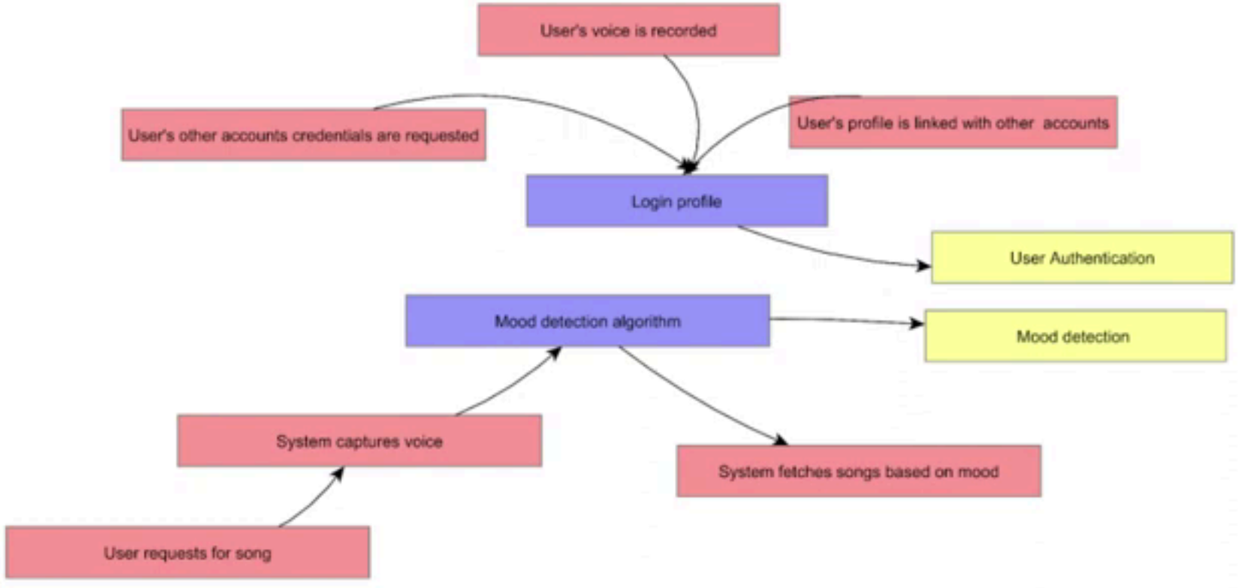
Reflection Spot

IIT BOMBAY

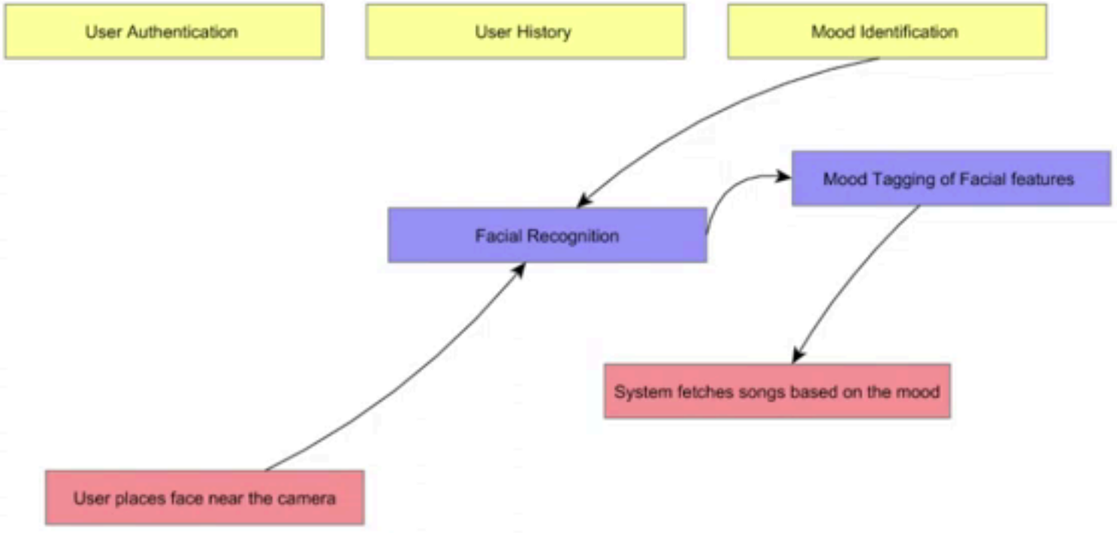
What are different combinations of FBS graph that can be created?

Please pause the video and written down your responses

Combinations of FBS graph



Behaviour-Structure-Function Graph

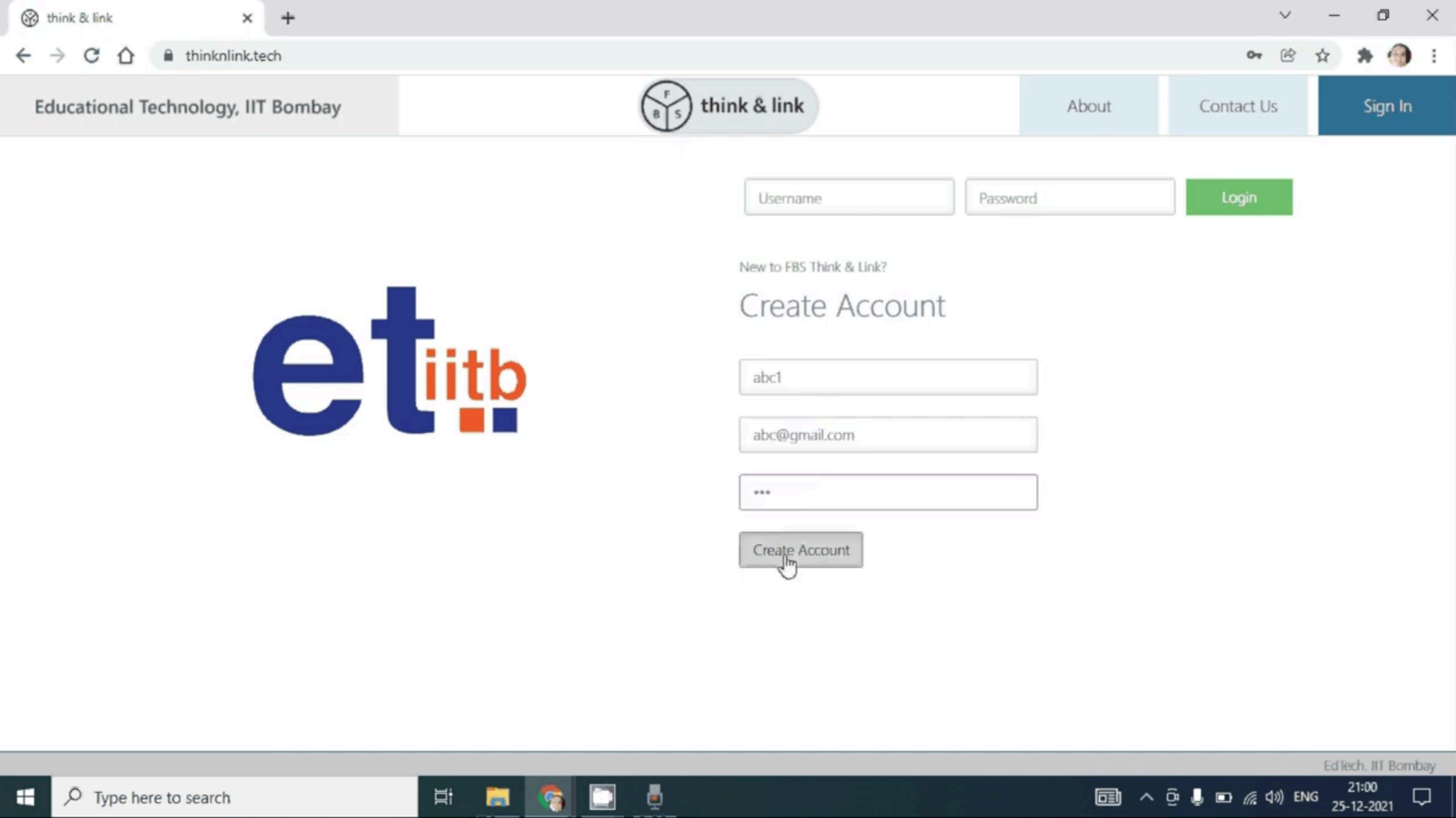


Function-Structure-Behaviour Graph

Summary

IIT BOMBAY

- Importance of integrating F-B-S while creating a design
- FBS graph
  - Nodes form the function, structure and behaviour
  - Connected via links
- Combinations of FBS graph possible while creating a software design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00:00:20

Hello. I am your Conceptual Design Assistant aka CASA.

In this phase you will be assisted to build Function-Behaviour-Structure understanding.

You will do this in the context of software conceptual design for the design problem- mood based music player system.

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Problem

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00:00:43

What do you think you will learn in this phase?

How will it be useful in software conceptual design?

Save Answers

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Problem

Introduction Ver 2-Video

think & link

Watch later Share

Task to do

A series of questions will be given in 'Worksheet' tab which you need to complete

These questions will help you understand the concept of FBS as well as the FBS graph

Watch on YouTube

https://thinknlink.tech/introduction_phases.php#note_tab

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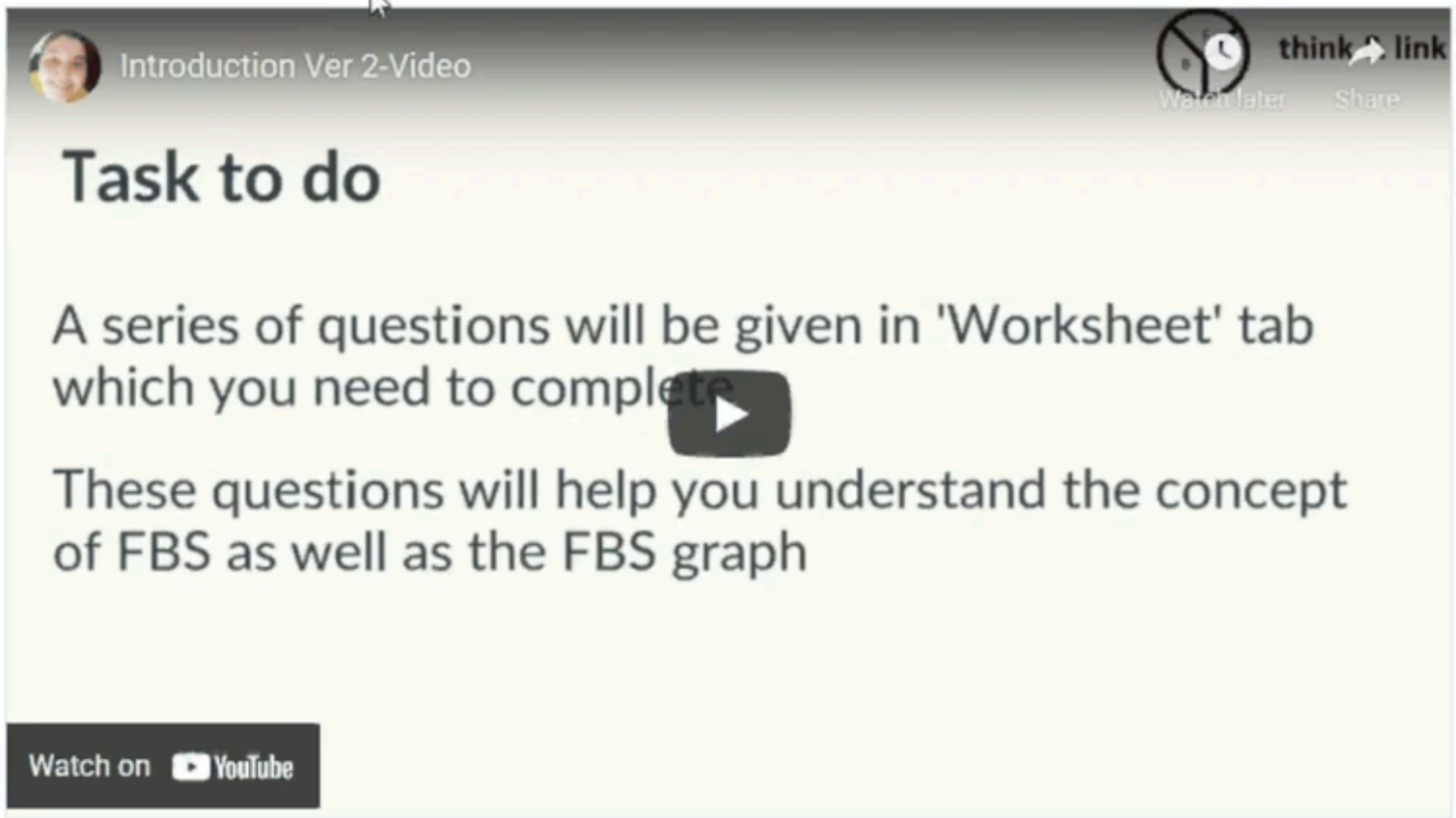
Answers saved, now you can click on other tabs to explore and fill the Worksheet.

Notes

Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Problem



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00:02:30

Here you will get to look at the FBS graph to answer questions in Worksheet.

View the 'FBS Graph - Video' to get to know what is a FBS graph.

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Graph

https://thinknlink.tech/introduction_phases.php#introduction_graph_task

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00:02:35

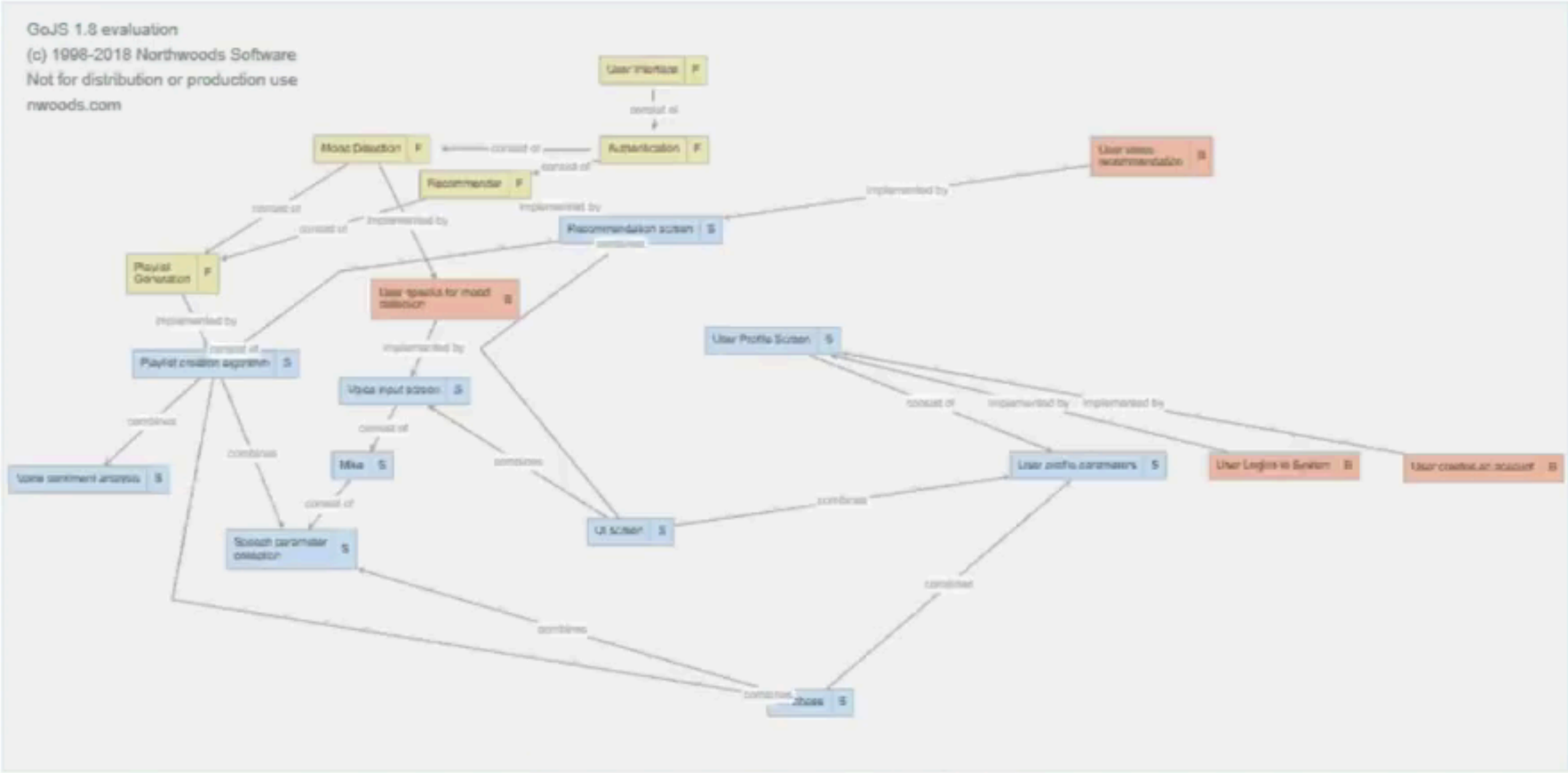
Use the information page and the FBS graph to identify the different types of the nodes.

more

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Graph



https://thinknlink.tech/introduction_phases.php#introduction_graph_task

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00:02:38

Use the information page and the FBS graph to identify the different types of the nodes.

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
5
6

Task1

Preamble - In this task you will get to look at the FBS graph and identify the different types of the nodes. Use the visual cues in the FBS graph and the pointers provided by CASA.

Q. Using the FBS graph for mood based music player and list 3 function node elements

Q. Using the FBS graph for mood based music player and list 3 structure node elements

Q. Using the FBS graph for mood based music player and list 3 behaviour node elements

Next | Previous

Type here to search
ENG 21:04 25-12-2021

00:02:41

Identify the FBS element in the statement, refer to the FBS graph and definitions in the information page to evaluate and answer.

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

Task2

Preamble - In the previous task you characterised the different types of nodes visually. In this task you will get to look at the FBS graph on screen and reason the purpose of different types of nodes. Use the visual cues in the FBS graph and the pointers provided by CASA.

1

Q. Evaluate the statement - "Automatic mood detection is a functionality provided by the proposed mood based music player"

○ True ◉ False

2

Give support for the choice (true/false)

3

Q. Evaluate the statement - "Voice recognition algorithm is a logical component in the proposed mood based music player"

○ True ◉ False

4

Give support for the choice (true/false)

5

Q. Evaluate the statement - "The proposed mood based music player implements the behaviour which allows the user to request for any song"

○ True ◉ False

6

Give support for the choice (true/false)

Next | Previous

Type here to search

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00:02:42

Use the previously identified FBS (Function, Behaviour, Structure) design elements, definitions and your understanding of FBS to answer the question.

Notes

Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
5
6

Task3

Preamble - Until now you have reasoned and characterised the different types of nodes in FBS graph. In this task you will explicitly state your understanding of the terms F, B and S. Use the previously identified FBS (Function, Behaviour, Structure) design elements, definitions and your understanding of FBS to answer the following questions

Q. Give description of your understanding of function

Q. Give description of your understanding of structure

Q. Give description of your understanding of behaviour

Next | Previous

Type here to search
ENG 21:04 25-12-2021

00:02:43

Use the information page and the FBS graph to list the different connectors based on visual information.

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
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6

Task4

Preamble - In this task you would list the different connector types of the nodes in the FBS graph. Look at the connectors/links in the provided FBS graph and identify the different types of connectors and the types of nodes that they connect. Use the visual cues in the FBS graph and the pointers provided by CASA.

Q. Note down all the identifying the connectors that connects the different FBS design element pairs (F-S, S-B, F-B, S-F, B-S, F-F, S-S, B-B) . For e.g., Combines : B-B

Next | Previous

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Look at the provided FBS graph and identify the pairs of (F-S, S-B, F-B, S-F, B-S, F-F, S-S, B-B) e.g. S-F: Connectivity Mechanism implements Login functionality.

more

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
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6

Task5

Preamble - In this task you would list the concrete examples of pairs like (F-S, S-B, F-B, S-F, B-S, F-F, S-S, B-B) and trios (FBS, SBF, FSB, BSF) from the provided FBS graph. Use the visual cues in the FBS graph and the pointers provided by CASA.

Q. Identify as many pairs of F-S, S-B, F-B, S-F, B-S, F-F, S-S, B-B from FBS graph and list them. For e.g., S-F: Connectivity Mechanism implements Login functionality

Q. Identify as many trios of FBS, SBF, FSB, BSF from FBS graph and list them. For e.g., FSB: Mood detection is implemented by Voice recognition algorithm which is utilised when user logins into the system

Next | Previous

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00:02:46

Look at previous concrete examples of trios and abstract their relation like,e.g. Function is implemented by Structure which gets utilized during user Behaviour

Notes

Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
5
6

Task6

Preamble - In the previous task you worked on specific examples of pairs and trios from the graph. In this task you are required to abstract the relationship of the trio F, S and B.

Q. Create sentences from the word bank below such that it explains the relationship between the trio F, S and B. For e.g., Function is implemented by Structure which gets utilised during user Behaviours

Next | Previous

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This video states the relationship between the FBS graph and UML diagrams.

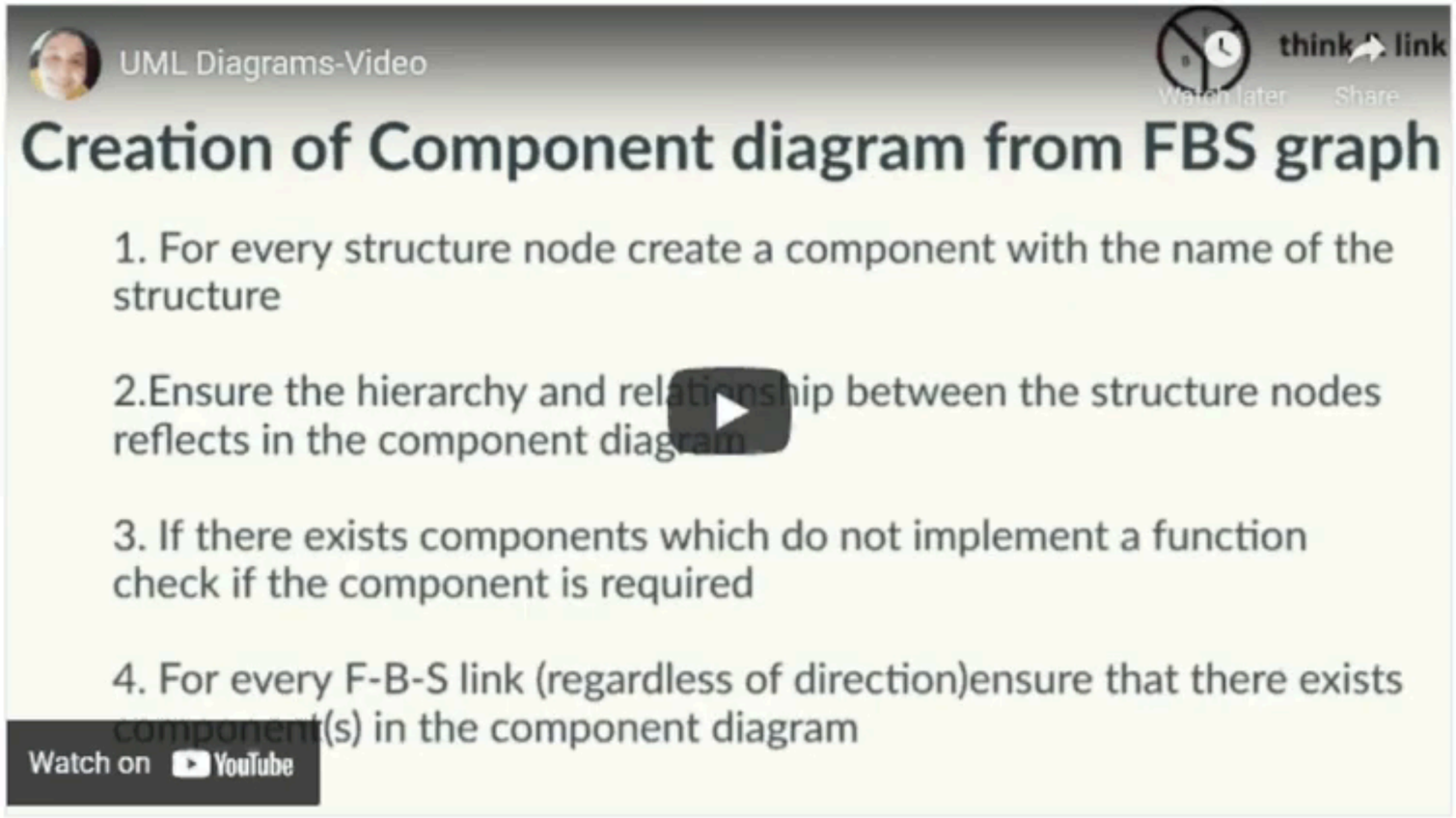
It provides the procedural steps to derive a UML diagram from the FBS graph.

Notes

Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro



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Use the information page and the FBS graph to identify the different types of the nodes.

Notes

Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Task

1
2
3
4
5
6

Task6

Preamble - In the previous task you worked on specific examples of pairs and trios from the graph. In this task you are required to abstract the relationship of the trio F, S and B.

Q. Create sentences from the word bank below such that it explains the relationship between the trio F, S and B. For e.g., Function is implemented by Structure which gets utilised during user Behaviours

Next | Previous

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00:01:30

Here you will get to look at the FBS graph to answer questions in Worksheet.

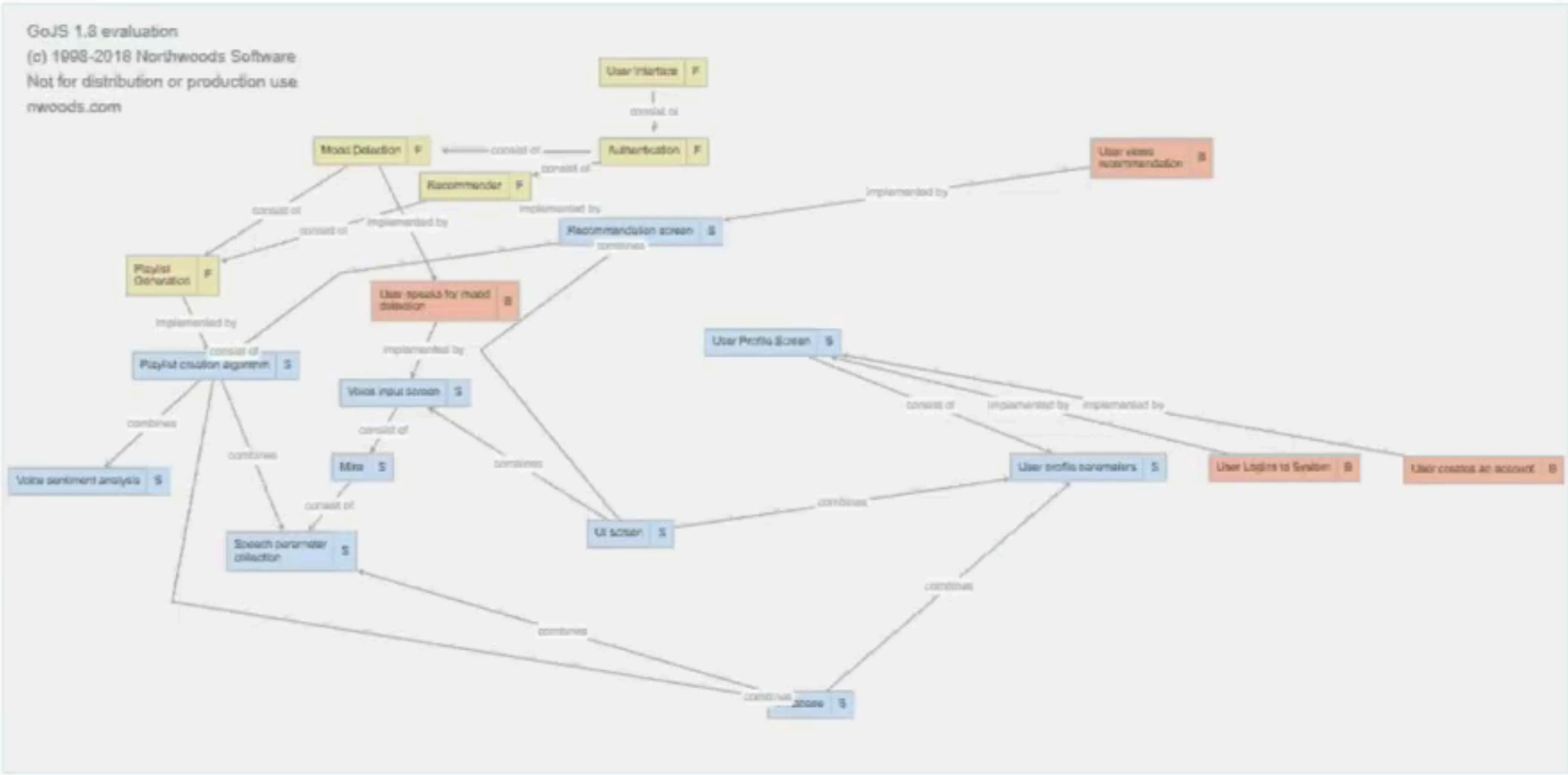
View the 'FBS Graph - Video' to get to know what is a FBS graph.

Notes

Your Mentor
CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Graph



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Type here to search

ENG 21:08 25-12-2021

think & link

thinknlink.tech/introduction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

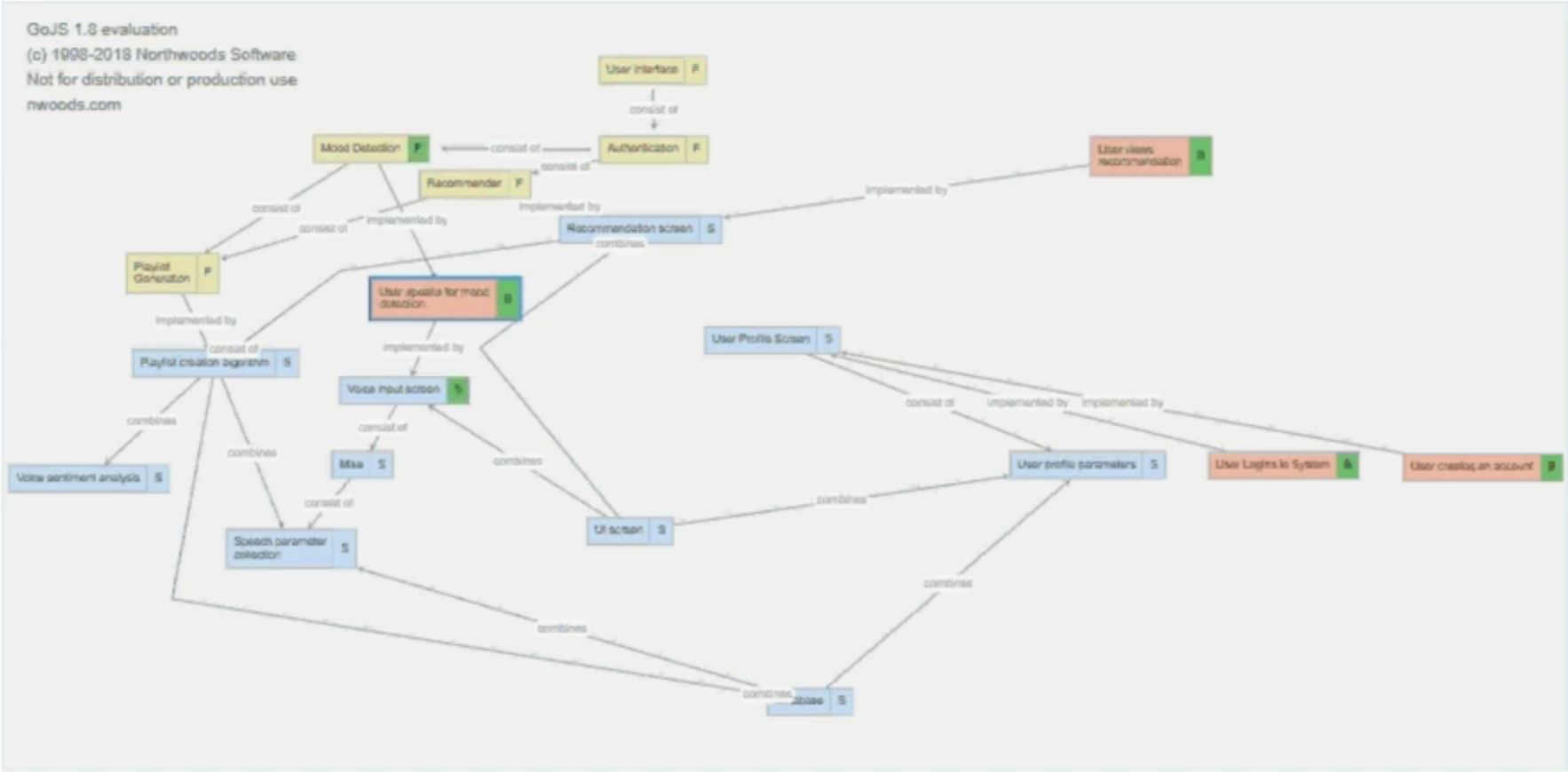
00:01:49

You have to first select link, to highlight similar links

Notes | Your Mentor CASA

Design Problem | FBS Graph | UML Diagrams | Worksheet

Intro | Graph



Toggle Highlights:

Similar Node

Similar Link

Adjacent Node

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ENG 21:08 25-12-2021

think & link

thinknlink.tech/introduction_phases.php

Introduction | Induction | Ideation | Iteration | Information | abc1

thinknlink.tech says

Are you really completed this phase?

OK | Cancel

00:02:05

View all the tabs 'Design Problem', 'FBS Graph', 'UML Diagrams', 'Worksheet' before this phase ends.

Notes | Your Mentor CASA

Design Problem

Task | End Phase

1
2
3
4
5
6

Task6

Preamble - In the previous task you worked on specific examples of pairs and trios from the graph. In this task you are required to abstract the relationship of the trio F, S and B.

Q. Create sentences from the word bank below such that it explains the relationship between the trio F, S and B. For e.g., Function is implemented by Structure which gets utilised during user Behaviours

Function is implemented by Structure which gets utilised during user Behaviours

2

Next | Previous

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ENG 21:09 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:03:19

Hello. I am your Conceptual Design Assistant aka CASA.

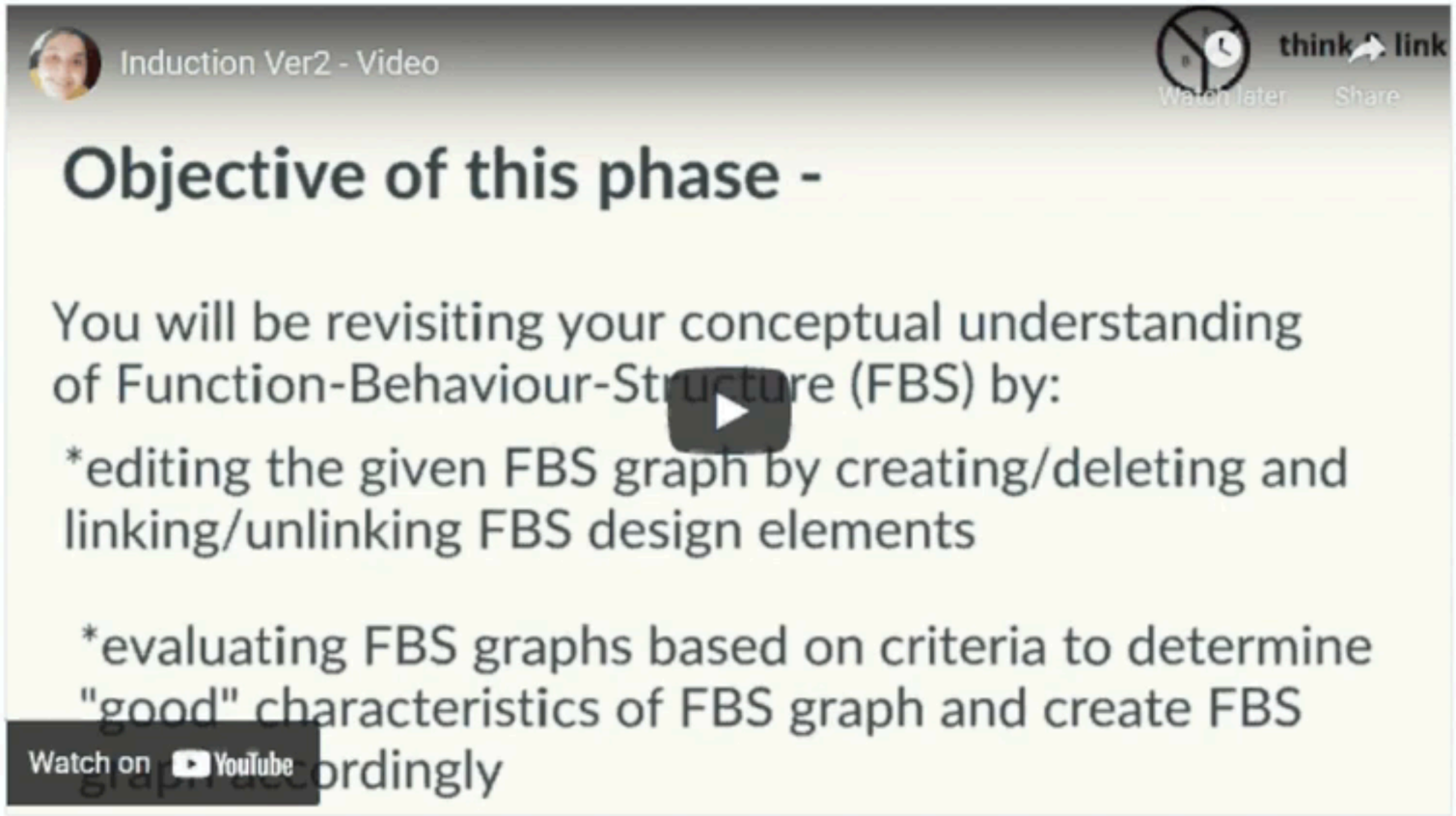
You have completed a previous phase 'Introduction'.

Based on your understanding from that phase answer the question in 'Task' and 'Save Answers'.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task | Problem



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ENG 21:10 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:03:22

What do you think you will learn in this phase?

How will it be useful in software conceptual design?

Save Answers

Notes

Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task | Problem

https://thinknlink.tech/induction_phases.php#note_tab

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ENG 21:10 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

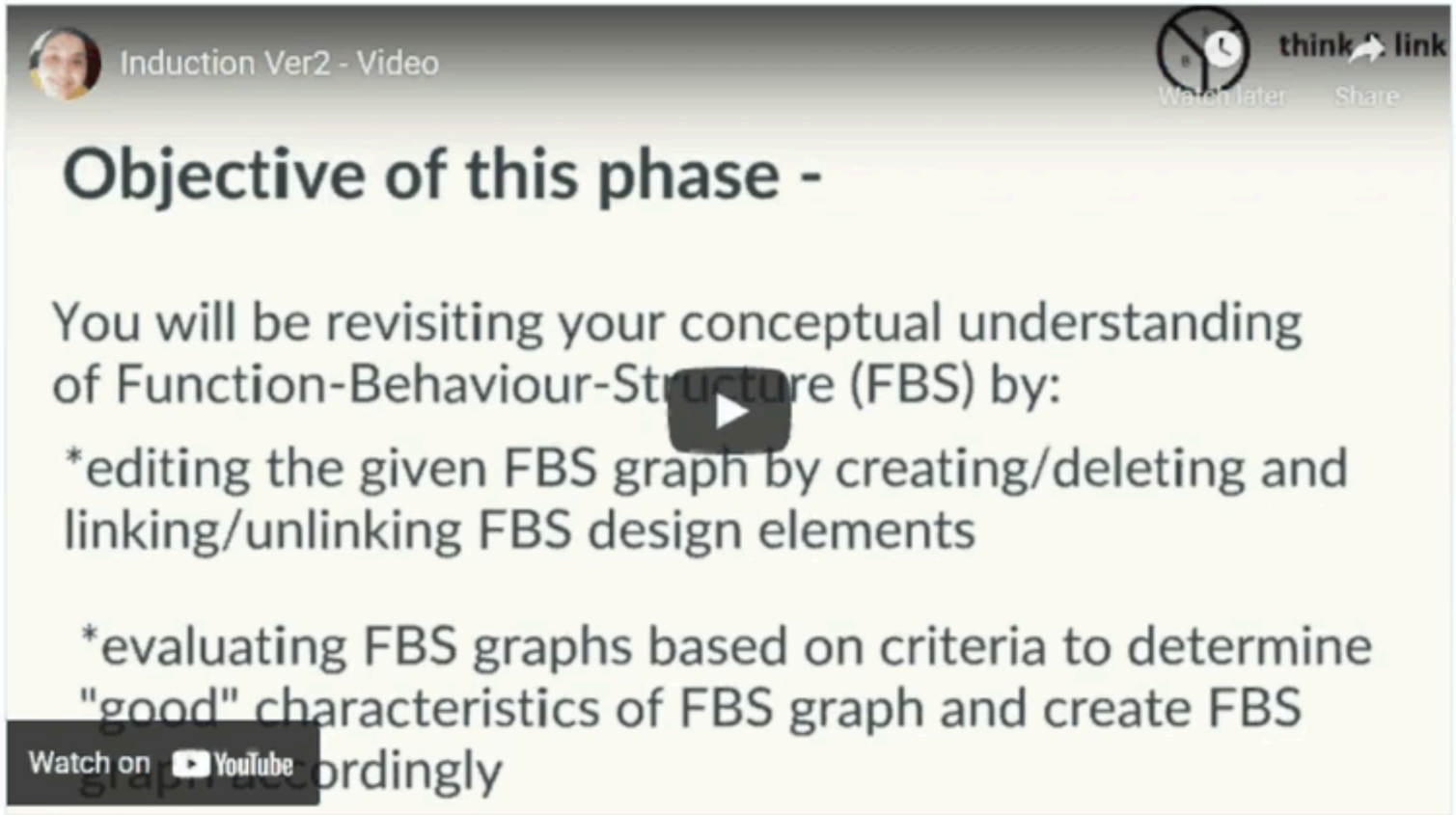
00:04:51

Answers saved, Now you can click on other tabs to proceed.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task | Problem



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ENG 21:12 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

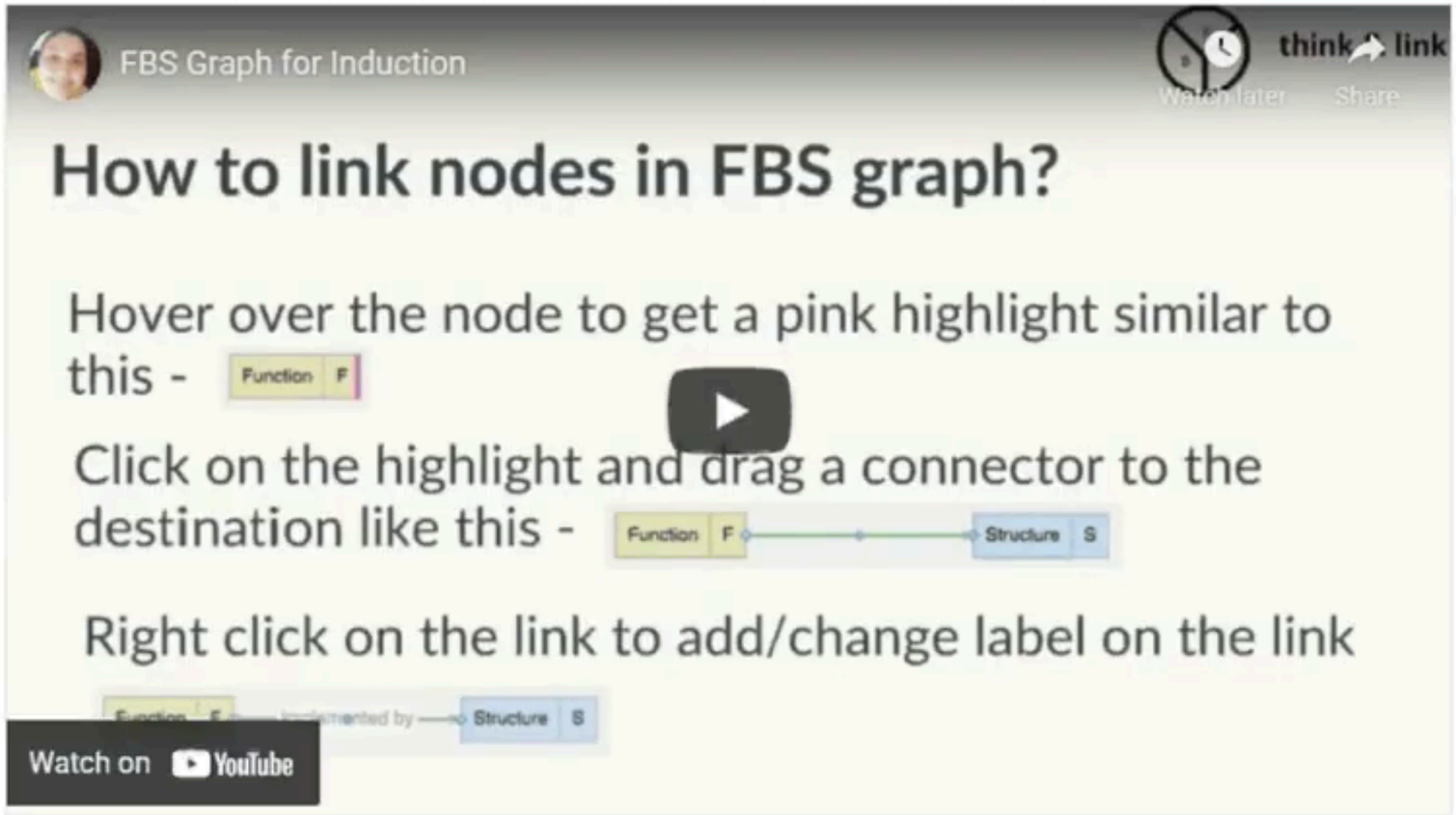
00:04:59

View the 'FBS Graph - Video' to get to know what is a FBS graph.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Graph



Processing request...

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Type here to search

ENG 21:12 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:06:01

You can edit the FBS graph.

You could populate the graph with the different nodes & links to create conceptual design.

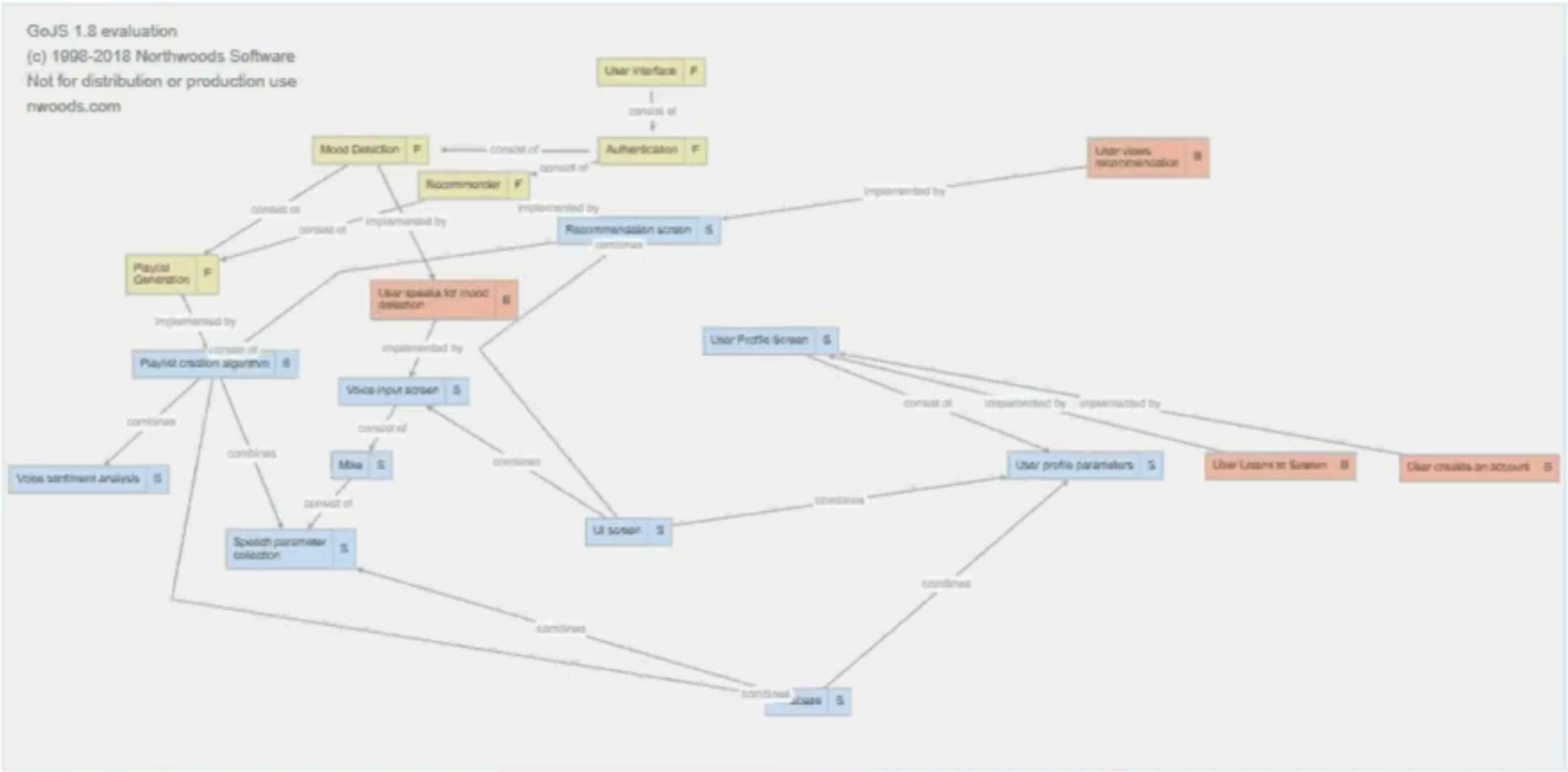
Use the information page and the FBS graph to identify the different types of the nodes.

You could delete the nodes in the graph to edit the graph and populate the

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Graph | Save Graph



https://thinknlink.tech/induction_phases.php#induction_graph

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Type here to search

ENG 21:13 25-12-2021

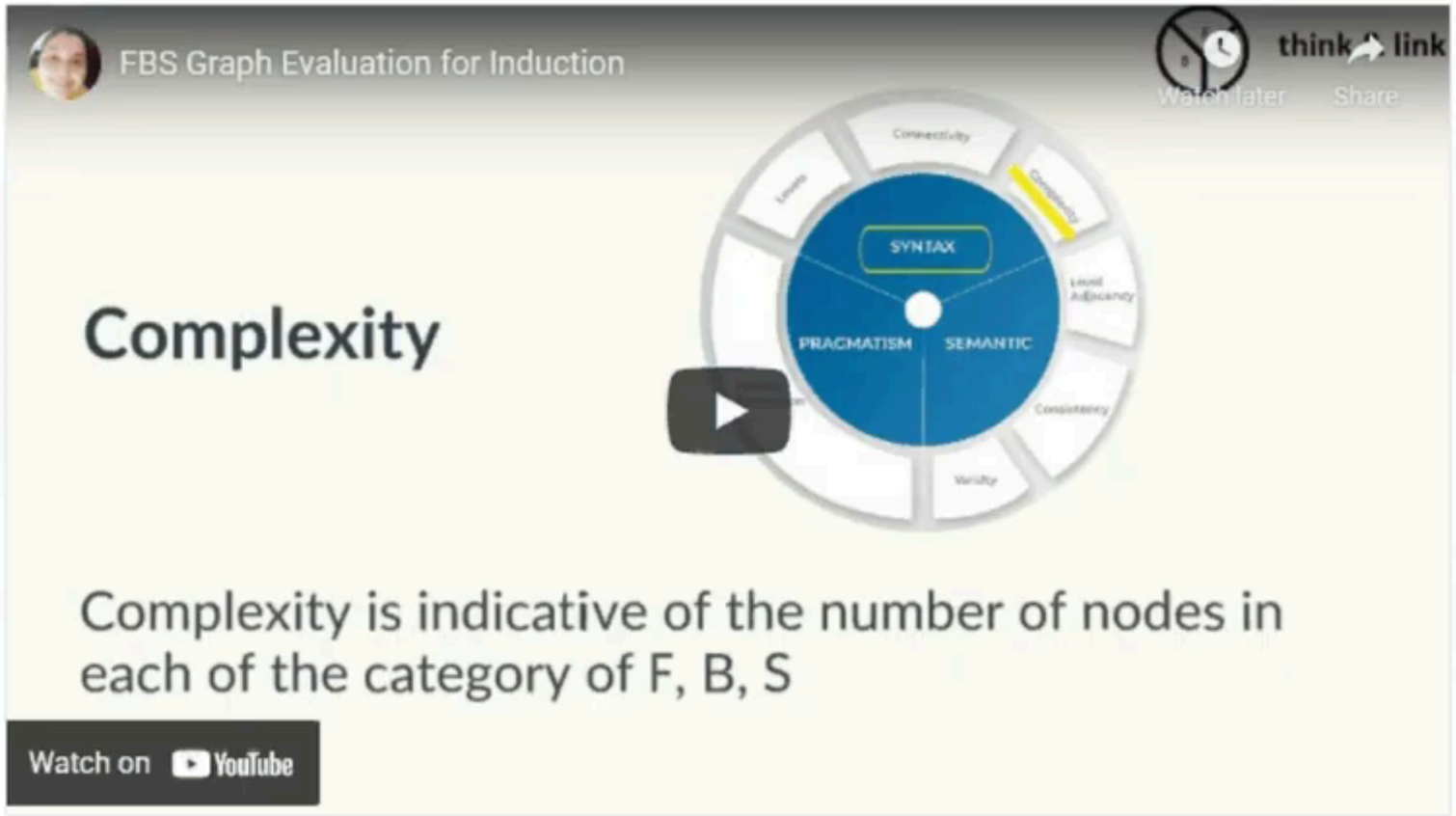
00:05:29

View the 'FBS Graph Evaluation-Video' to get to know what does evaluating a FBS graph entail.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task



Type here to search
ENG 21:12 25-12-2021

think & link

thinknlink.tech/induction_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:06:05

The wheel is clickable. Click on each part of the the wheel.

The text box below the wheel, gives the explanation to the part of the wheel clicked on.

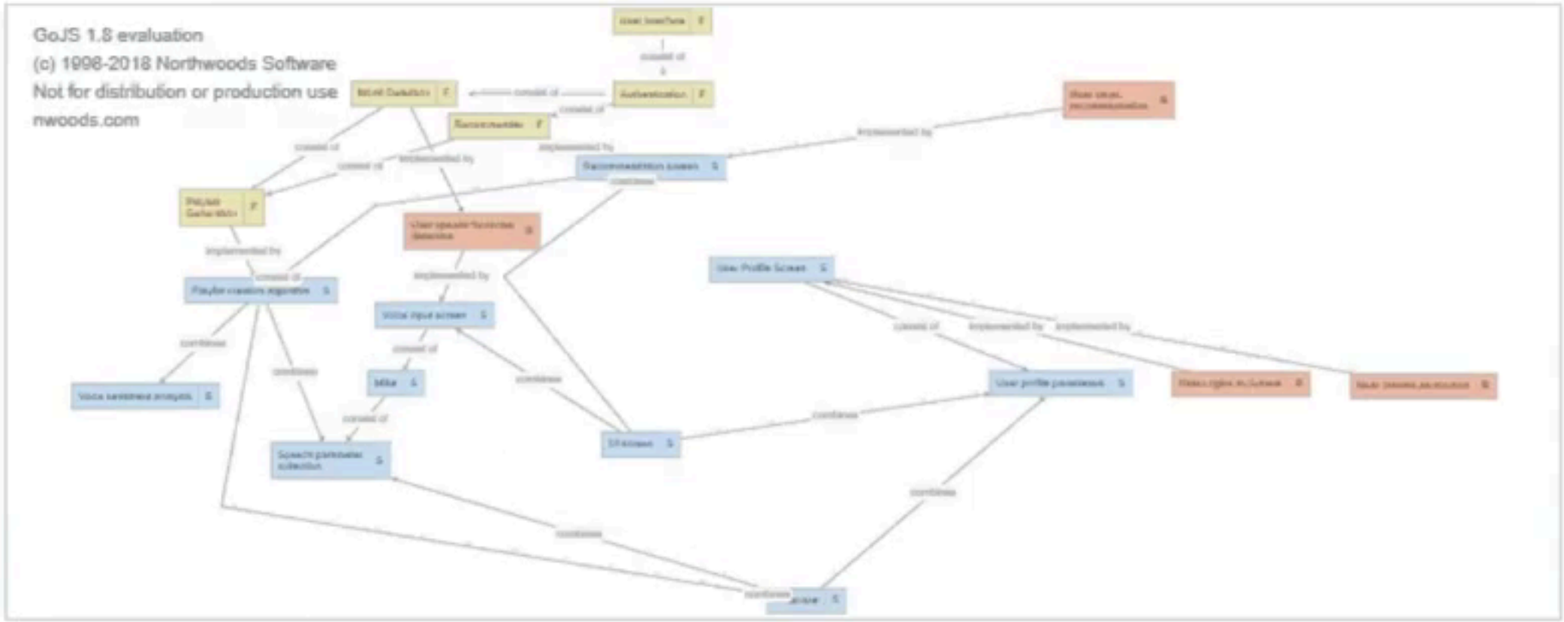
It provides example of the category Meets Expectation / Needs Improvement / Inadequate / Missing

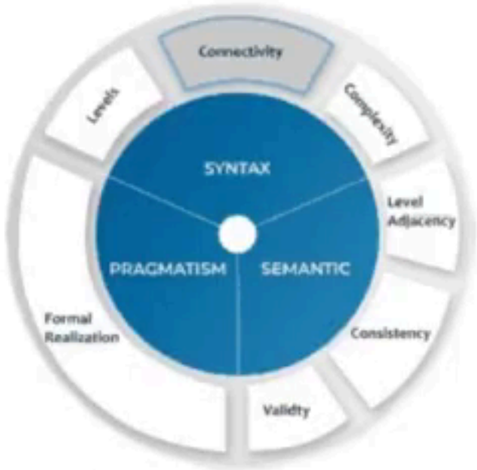
more

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task





- (●) Meets Expectation
- ( ) Needs Improvement
- ( ) Inadequate
- ( ) Missing

Support your choice: Level - Meets Expectation

Support your choice with reason

What are the changes you will make in the FBS graph Based on the Evaluation?

Connectivity is indicative of linking purpose, concept, component and mechanism in the conceptual design, Meets Expectation in 'Connectivity' indicates that - All the

Waiting for thinknlink.tech...

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ENG 21:13 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:09:27

Hello. I am your Conceptual Design Assistant aka CASA.

You have completed a previous phase 'Induction'.

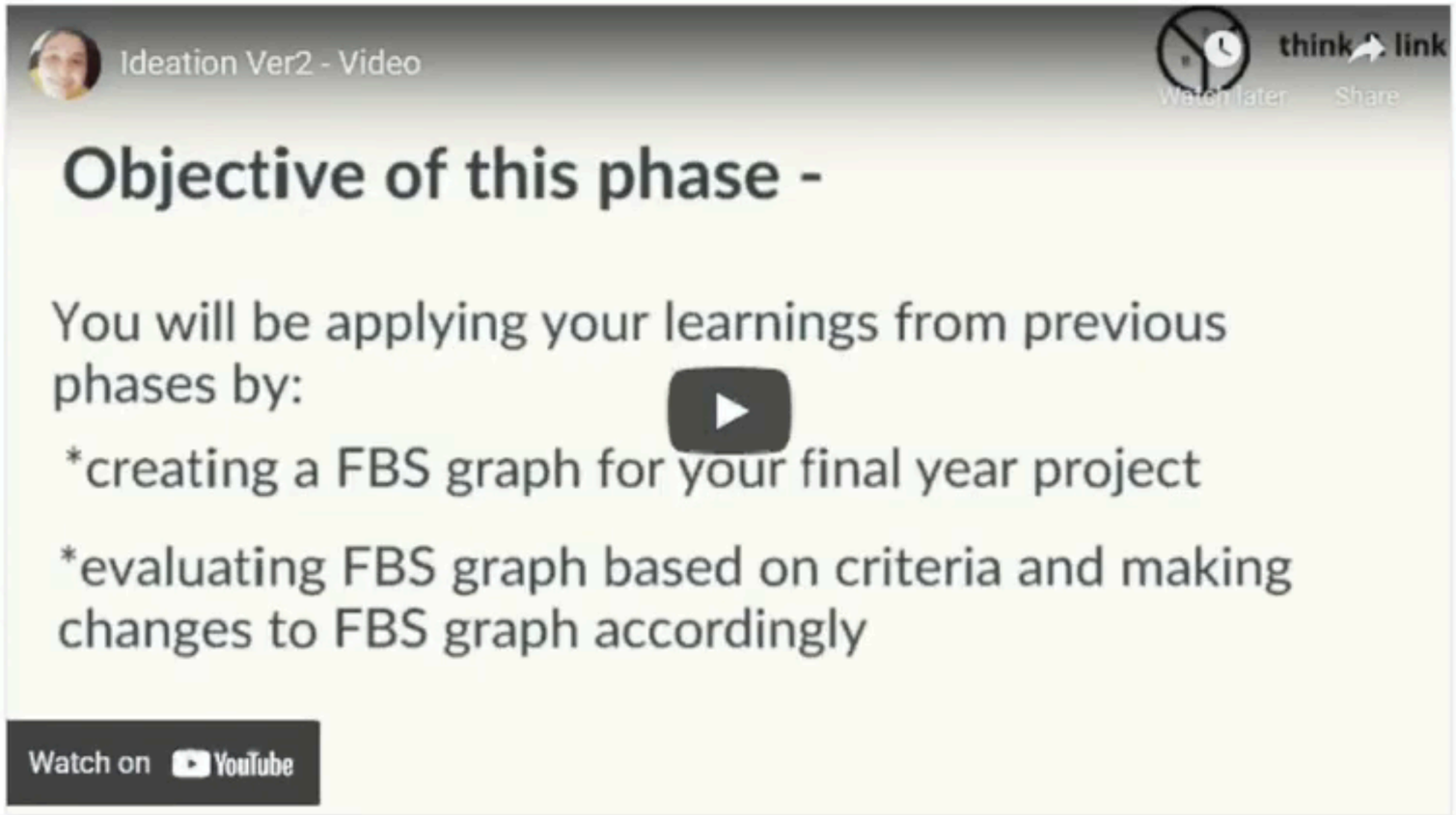
View the task video in the 'Intro' tab and fill the answers to the questions in 'Notes'.

Once you submit the answers other tabs will be enabled.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Problem



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Type here to search

ENG 21:16 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:09:36

What do you think you will learn in this phase?

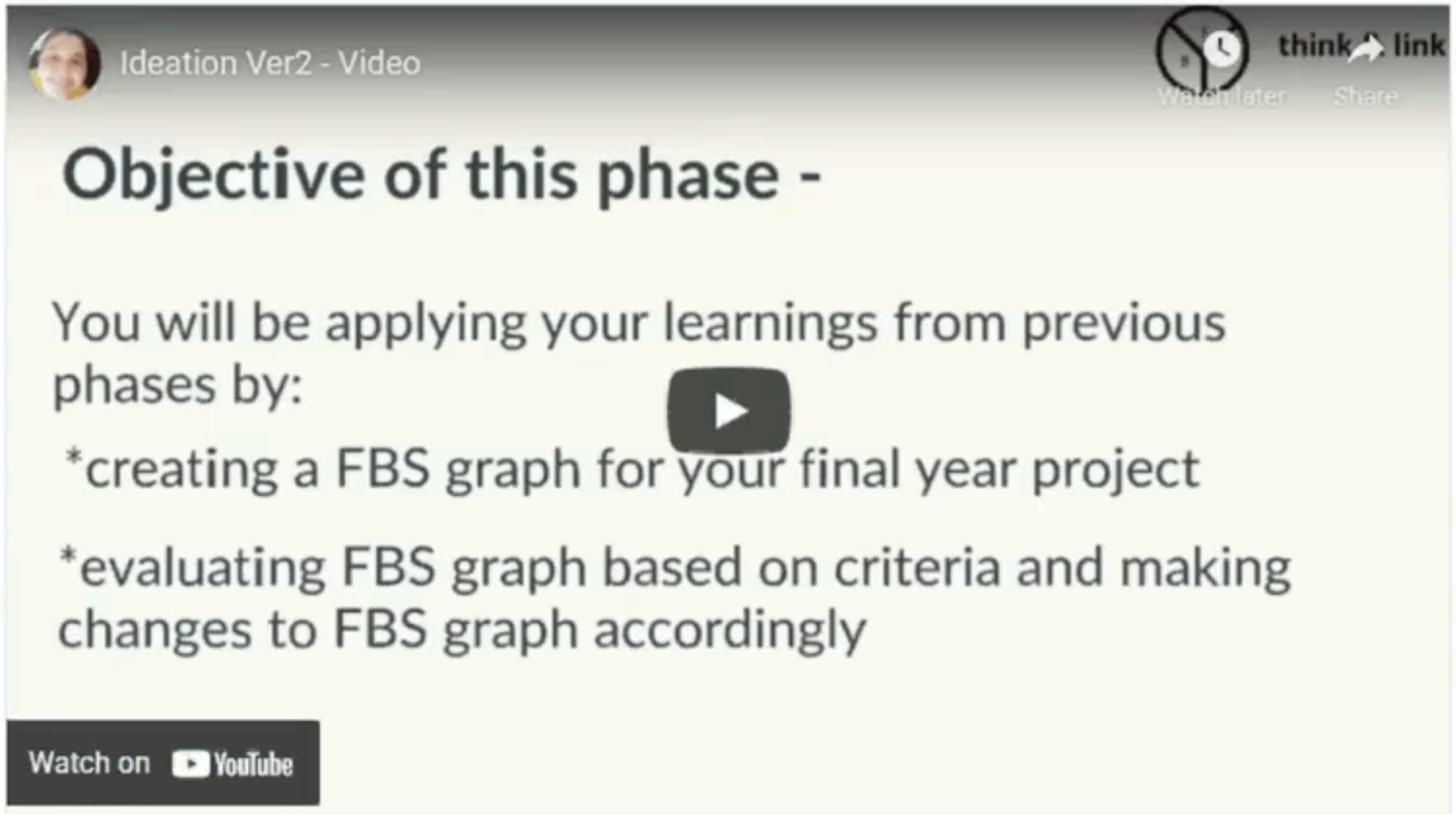
How will it be useful in software conceptual design?

Save Answers

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Problem



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ENG 21:16 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:10:50

Answers Saved

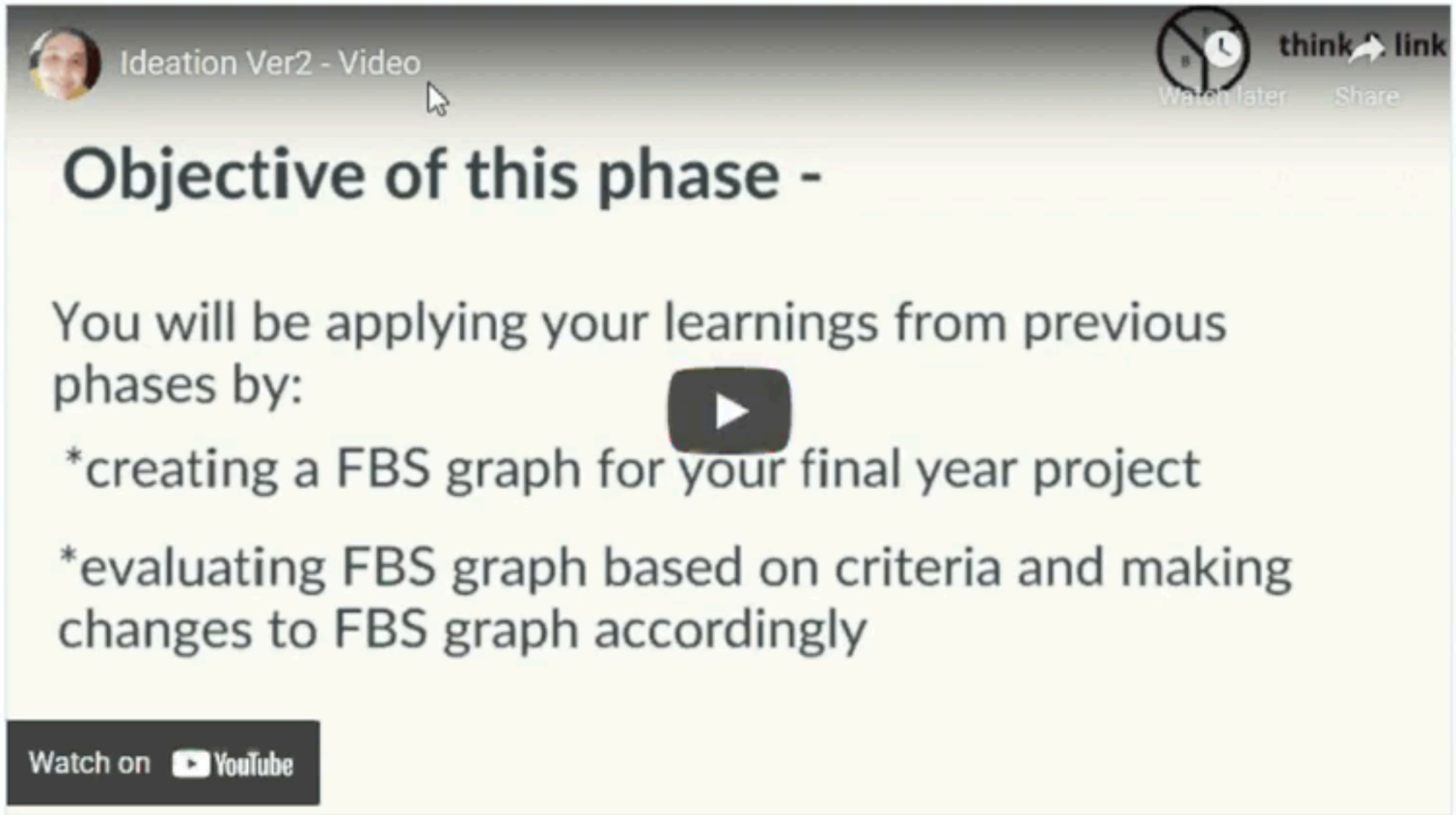
You are to create a software conceptual design for your final year project problem.

Click on 'Problem' Tab and fill in the details of the problem in the problem text box and save it.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Problem



Type here to search

ENG 21:18 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:10:55

Hello. I am your Conceptual Design Assistant aka CASA.

You are to create a software conceptual design for your final year project problem.

Fill in the details of the problem in the problem text box and save it.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Problem | Save Problem

Enter your design context problem here?

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Type here to search

ENG 21:18 25-12-2021

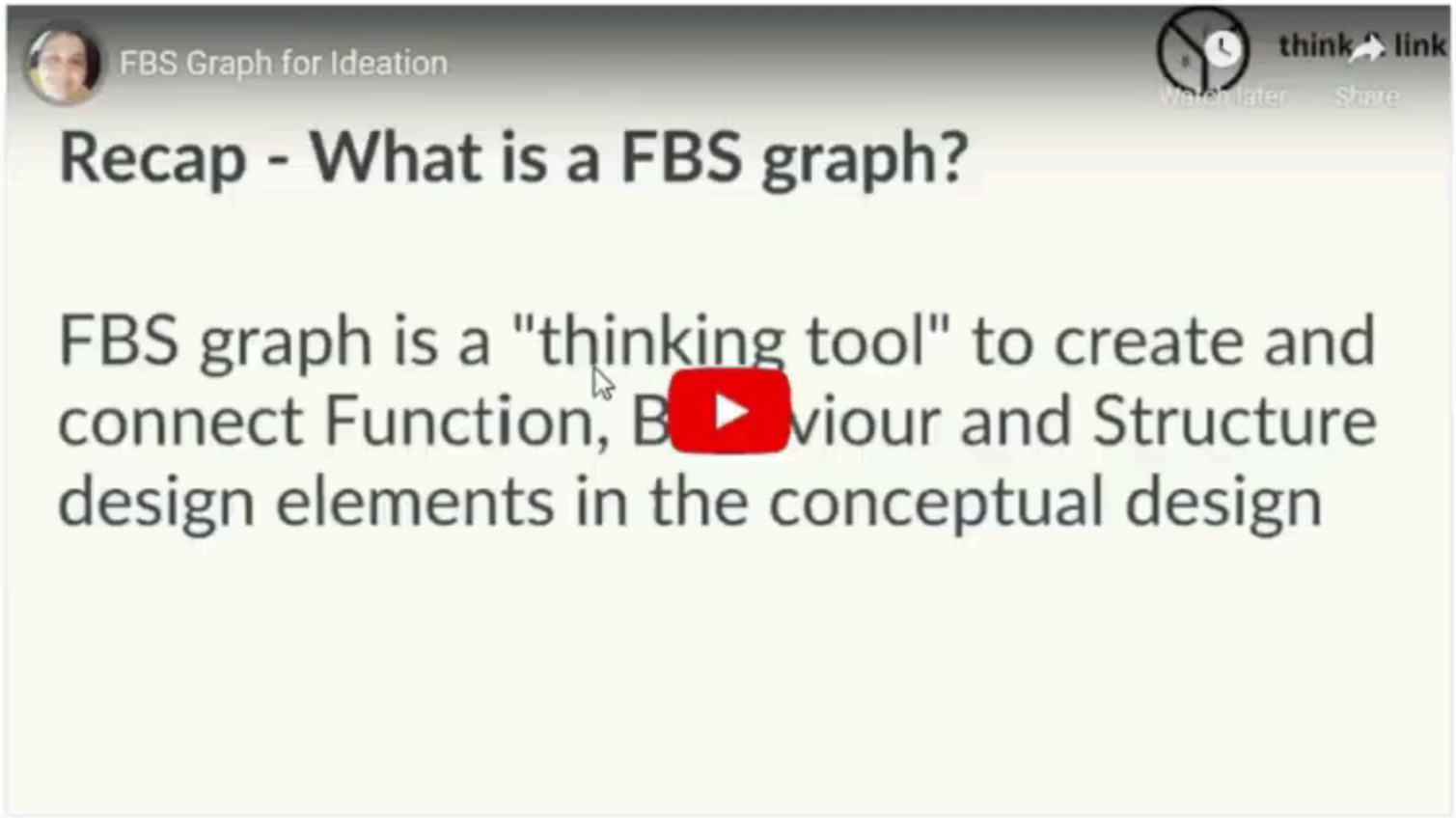
View the 'FBS Graph - Video' to get to know what is a FBS graph.

Also you could refer to the FBS graph in the 'Induction' phase.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Graph



Type here to search

ENG 21:18 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:11:08

You can edit the FBS graph.

You could populate the graph with the different nodes & links to create conceptual design.

Use the information page and the FBS graph to identify the different types of the nodes.

You could refer to the FBS graph that you created in the 'Induction' phase.

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Graph | Save Graph

GoJS 1.8 evaluation
(c) 1998-2018 Northwoods Software
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nwoods.com

Add Node:

Add Function

Add Structure

Add Behaviour

Toggle Highlights:

Similar Node

Similar Link

Adjacent Node

https://thinknlink.tech/ideation_phases.php#ideation_graph_task

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Type here to search

ENG 21:18 25-12-2021

00:11:13

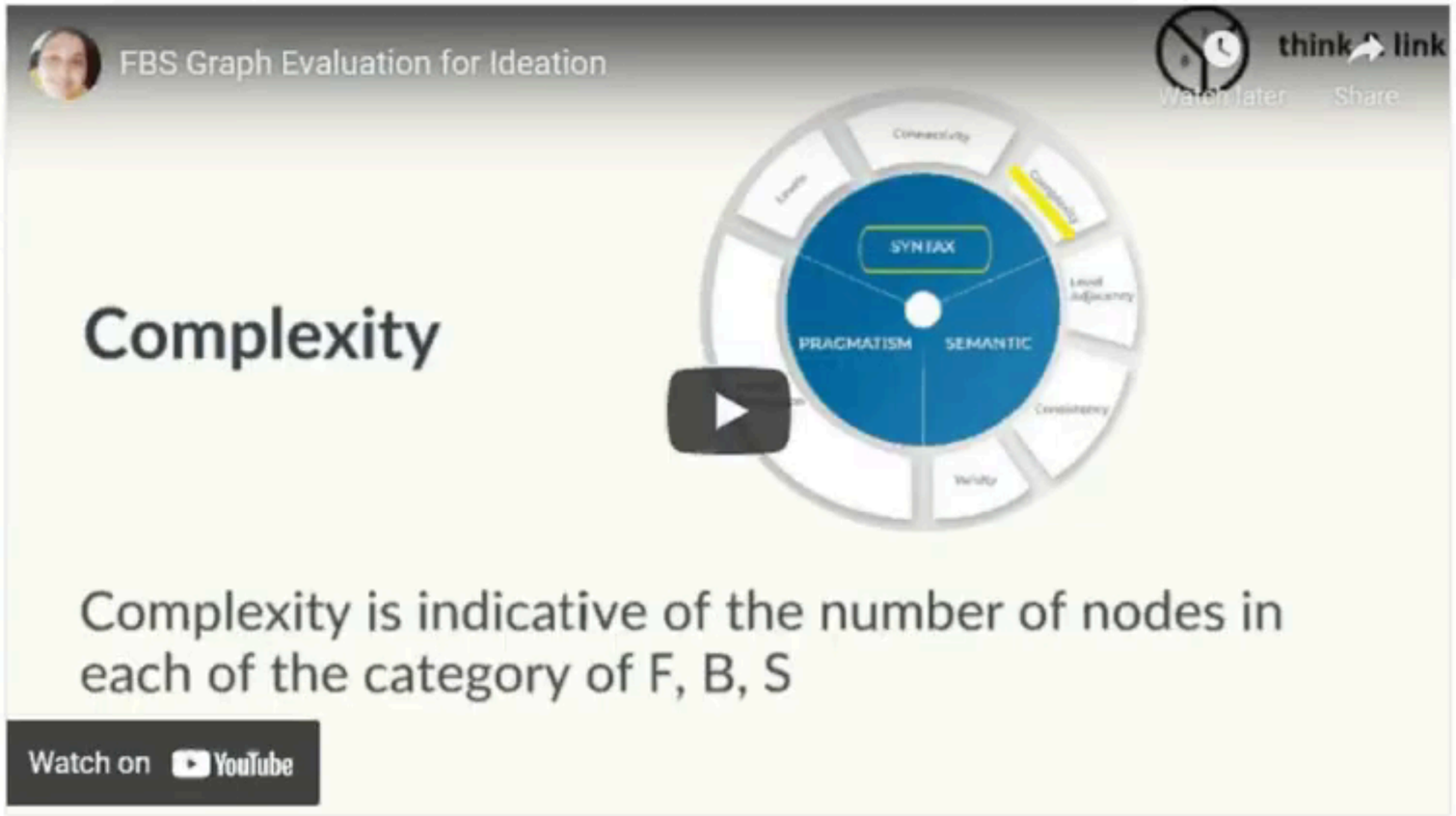
View the 'FBS Graph Evaluation-Video' to get to know what does evaluating a FBS graph entail.

Notes

Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task



Type here to search

ENG 21:18 25-12-2021

think & link

thinknlink.tech/ideation_phases.php

Introduction | Induction | Ideation | think & link | Iteration | Information | abc1

00:11:21

The wheel is clickable. Click on each part of the the wheel.

The text box below the wheel, gives the explanation to the part of the wheel clicked on.

It provides example of the category Meets Expectation / Needs Improvement / Inadequate / Missing

more

Notes | Your Mentor CASA

Design Problem | FBS Graph | Evaluate FBS Graph | UML Diagrams

Intro | Task

GoJS 1.8 evaluation
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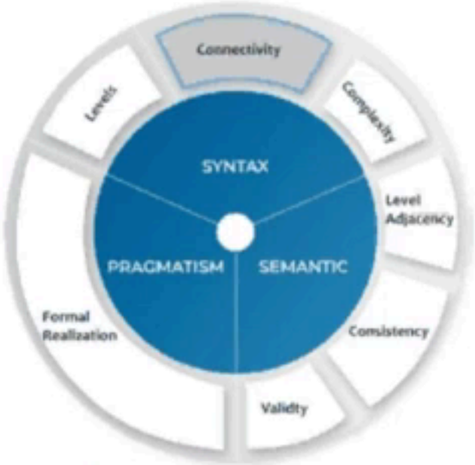

- (●) Meets Expectation
- ( ) Needs Improvement
- ( ) Inadequate
- ( ) Missing

Support your choice: Level - Meets Expectation

Support your choice with reason

What are the changes you will make in the FBS graph Based on the Evaluation?

Connectivity is indicative of linking purpose, concept, component and mechanism in the conceptual design, Meets Expectation in 'Connectivity' indicates that - All the

https://thinknlink.tech/ideation_phases.php#ideation_eval_task

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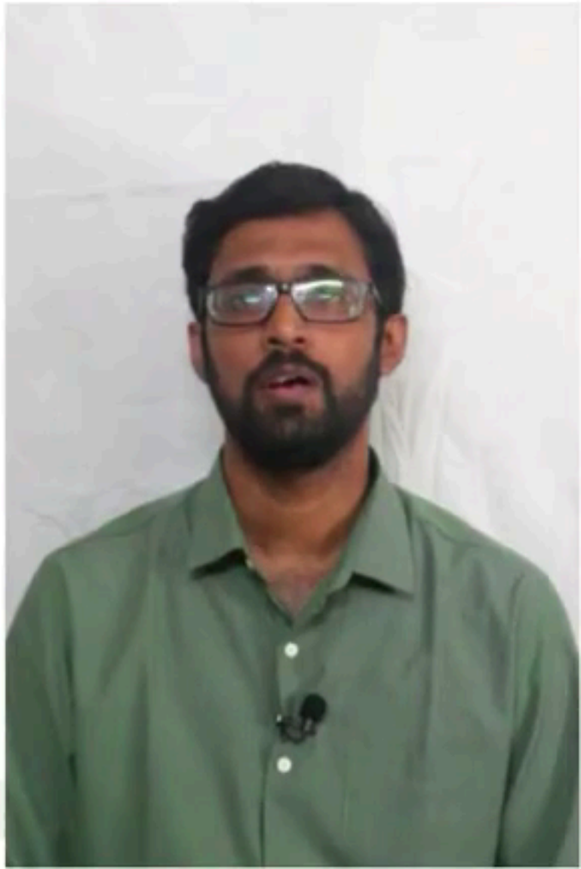
ENG 21:18 25-12-2021

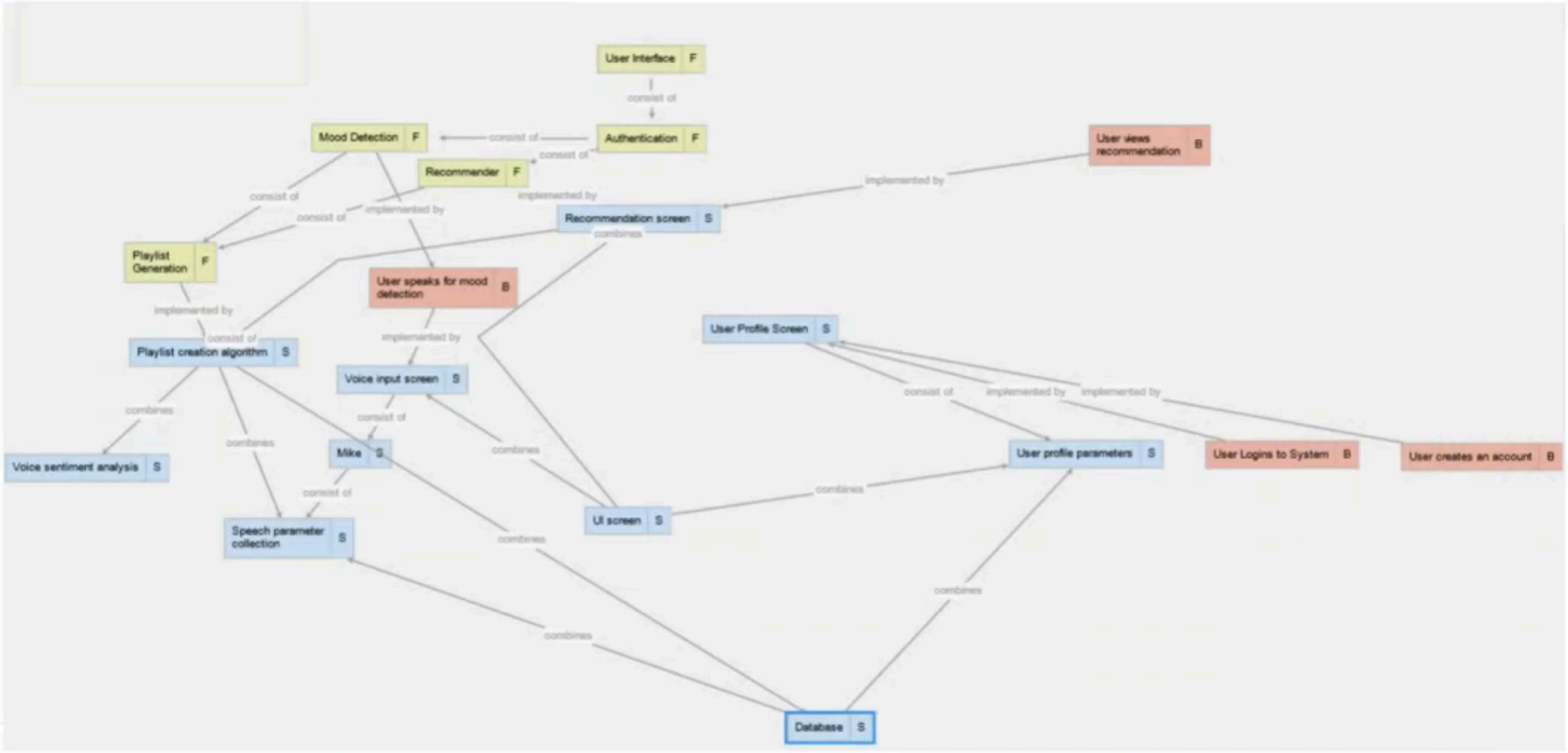
Modelling

IIT BOMBAY

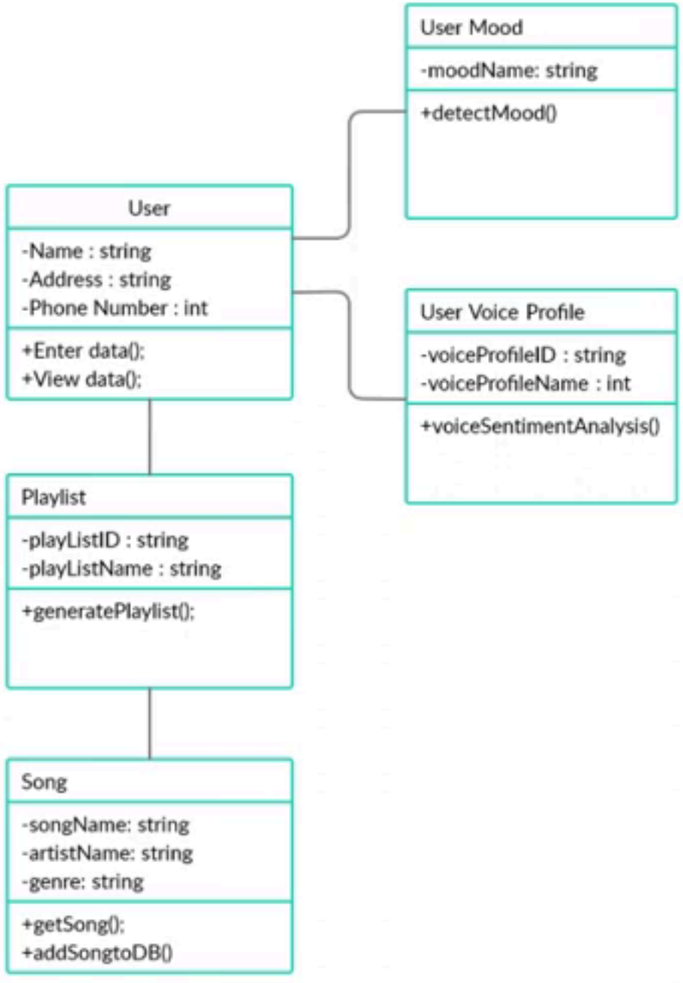
- Modelling - Creating an external, explicit representation of the system to be built



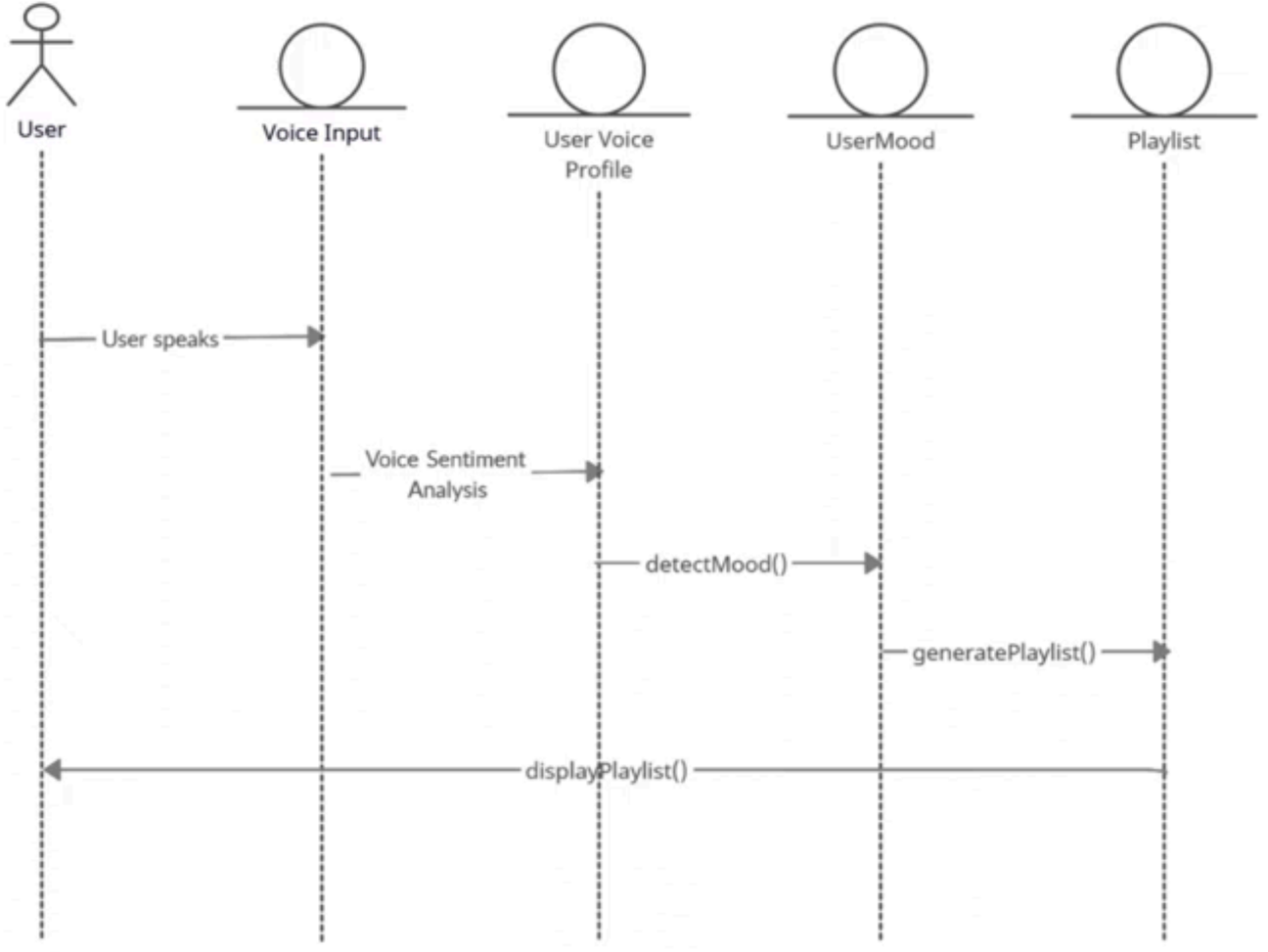
Modelling using FBS



Modelling using UML



Class Diagram



Sequence Diagram for Mood Detection

Reflection Spot

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We looked at software modelling techniques like the F-B-S representation and UML modelling.

What purpose do each of these representations serve?

Please pause the video and write down your responses

Modelling at Different Levels of Detail

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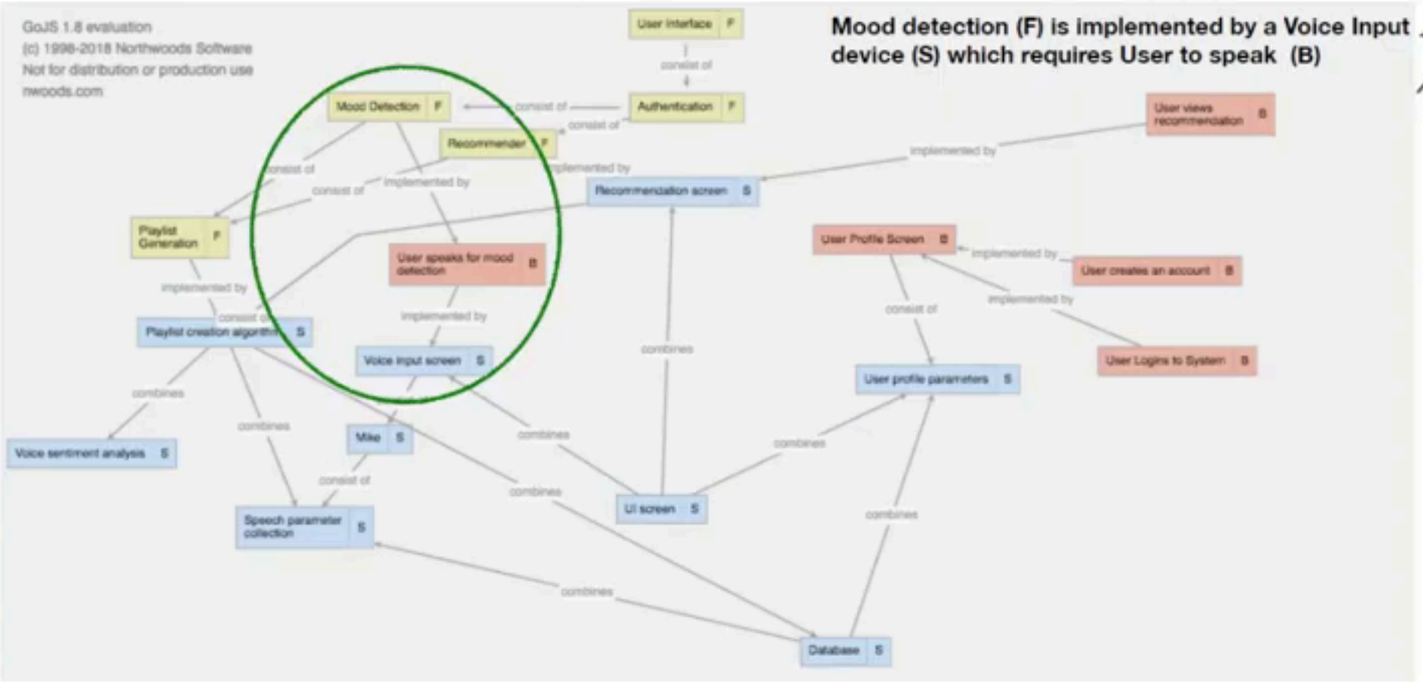
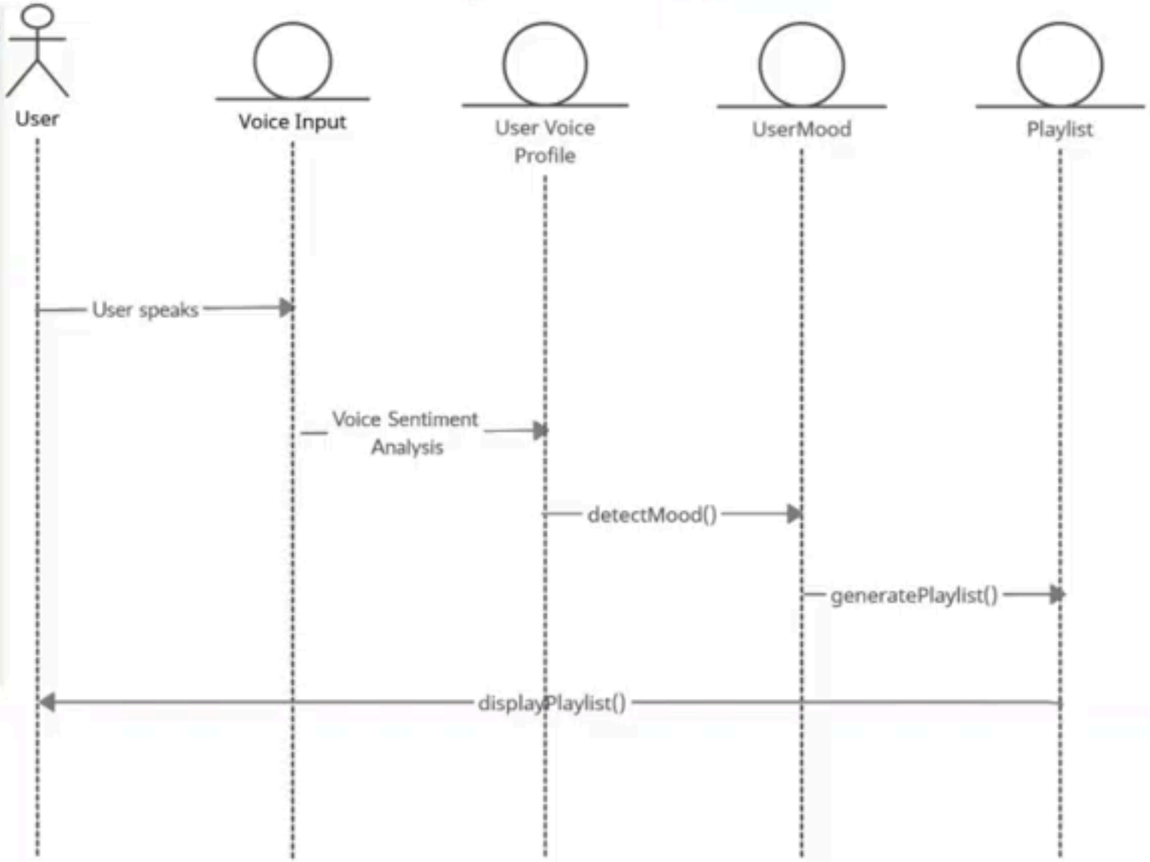
Sample FBS graph for the mood based music player design problem



Sequence Diagram



Modelling

IIT BOMBAY

- Modelling - Creating an external, explicit representation of the system to be built

- Unified Modelling Language (UML) - Help represent the software design via
  - Multiple views
  - Greater level of detail

Important Views of a Software System

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- Structural view
- Dynamic behavioural view

Structural View

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- Structure/components of the software system, and relationships
- Describes logical parts of the system - classes, data and functions

Structural View: Class Diagram

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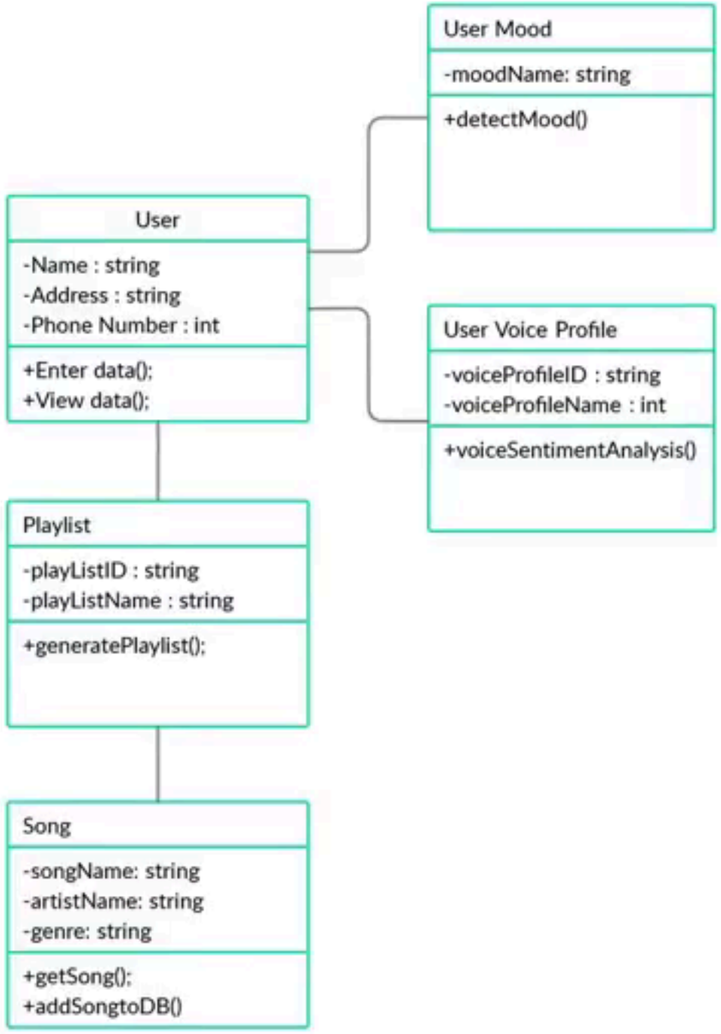
- Describes the structure of the system
- Describes system's classes, attributes, operations, relationships among objects.

Class Diagram for Mood based Music Player



Dynamic Behavioural View

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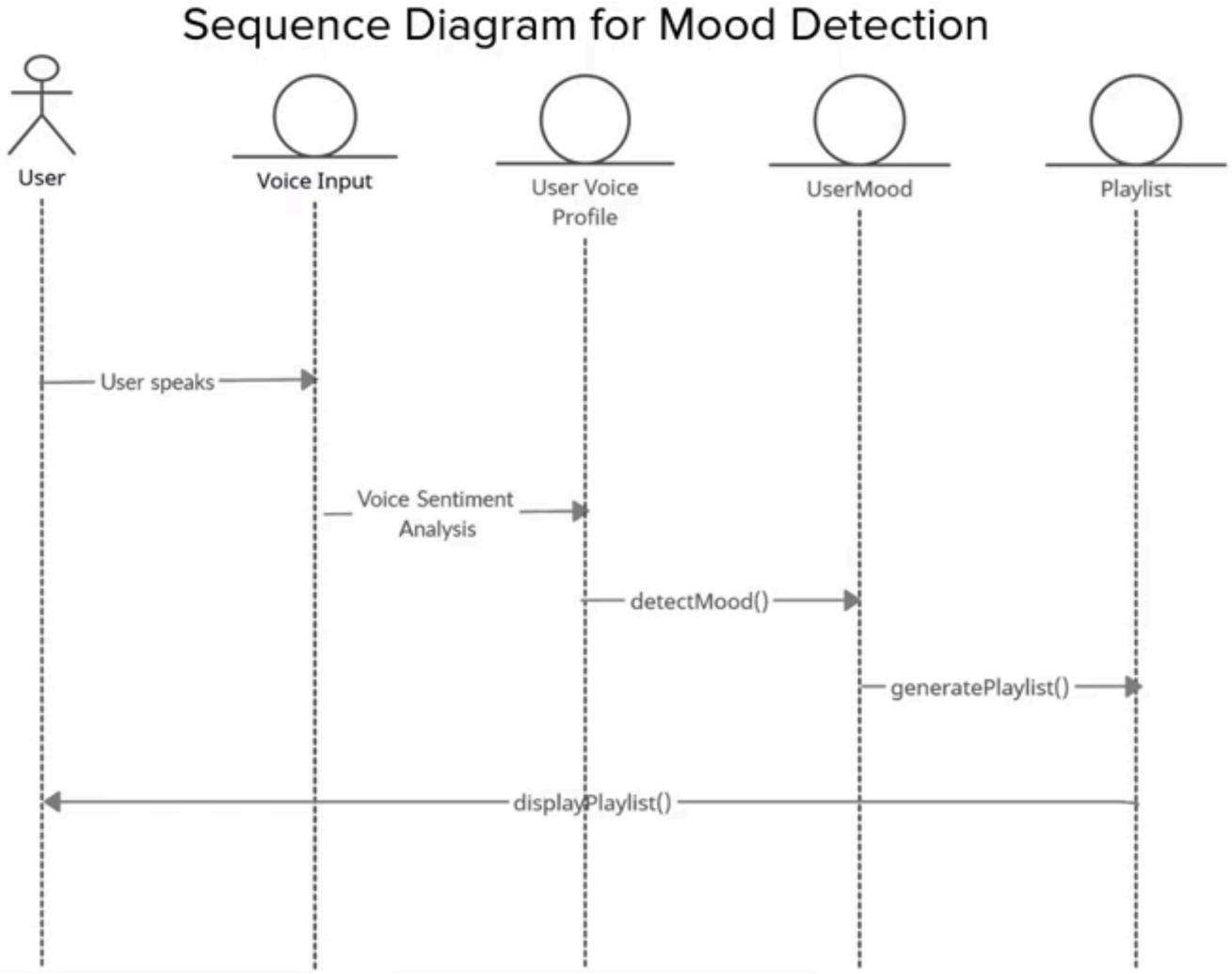
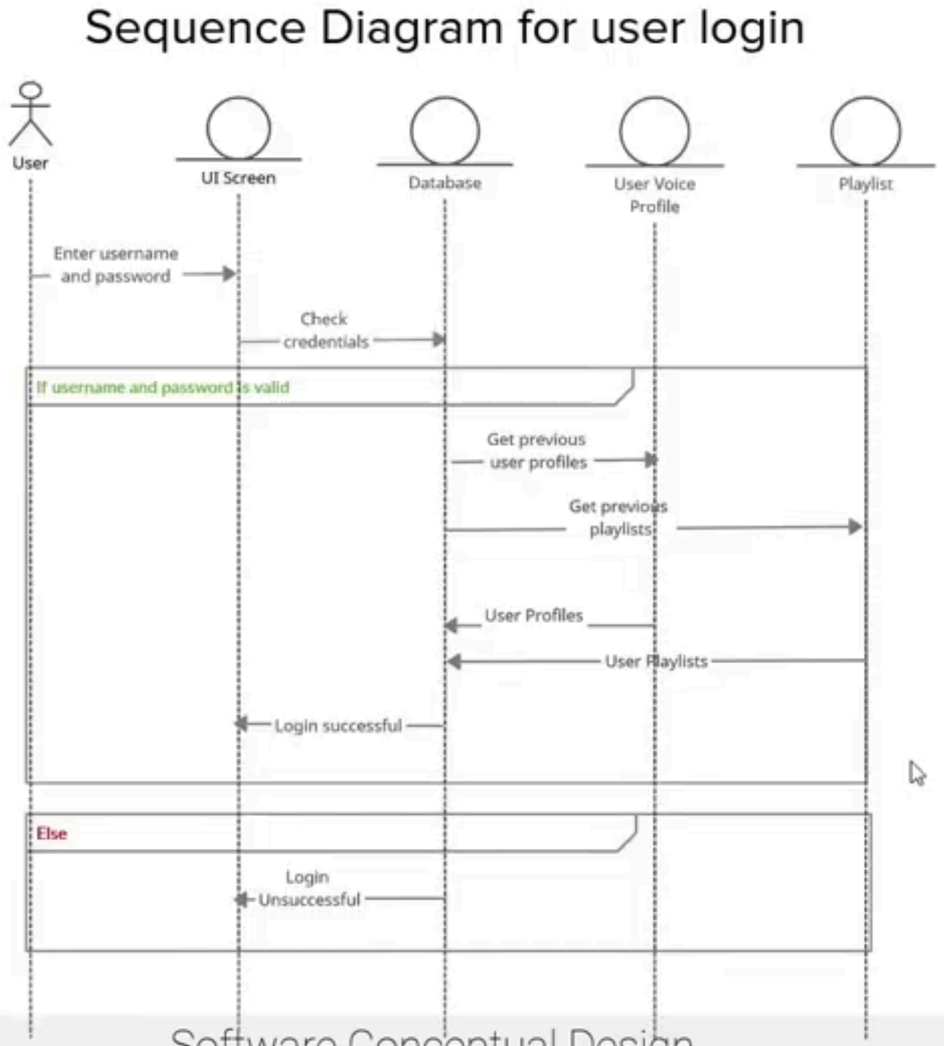
- Describes behaviour of the system over time
- Further classified into:
  - State Machine View - models different states of an object of a class
  - Activity View - models flow of control among computational activities
  - Interaction View - sequence of message exchanges among parts of a system

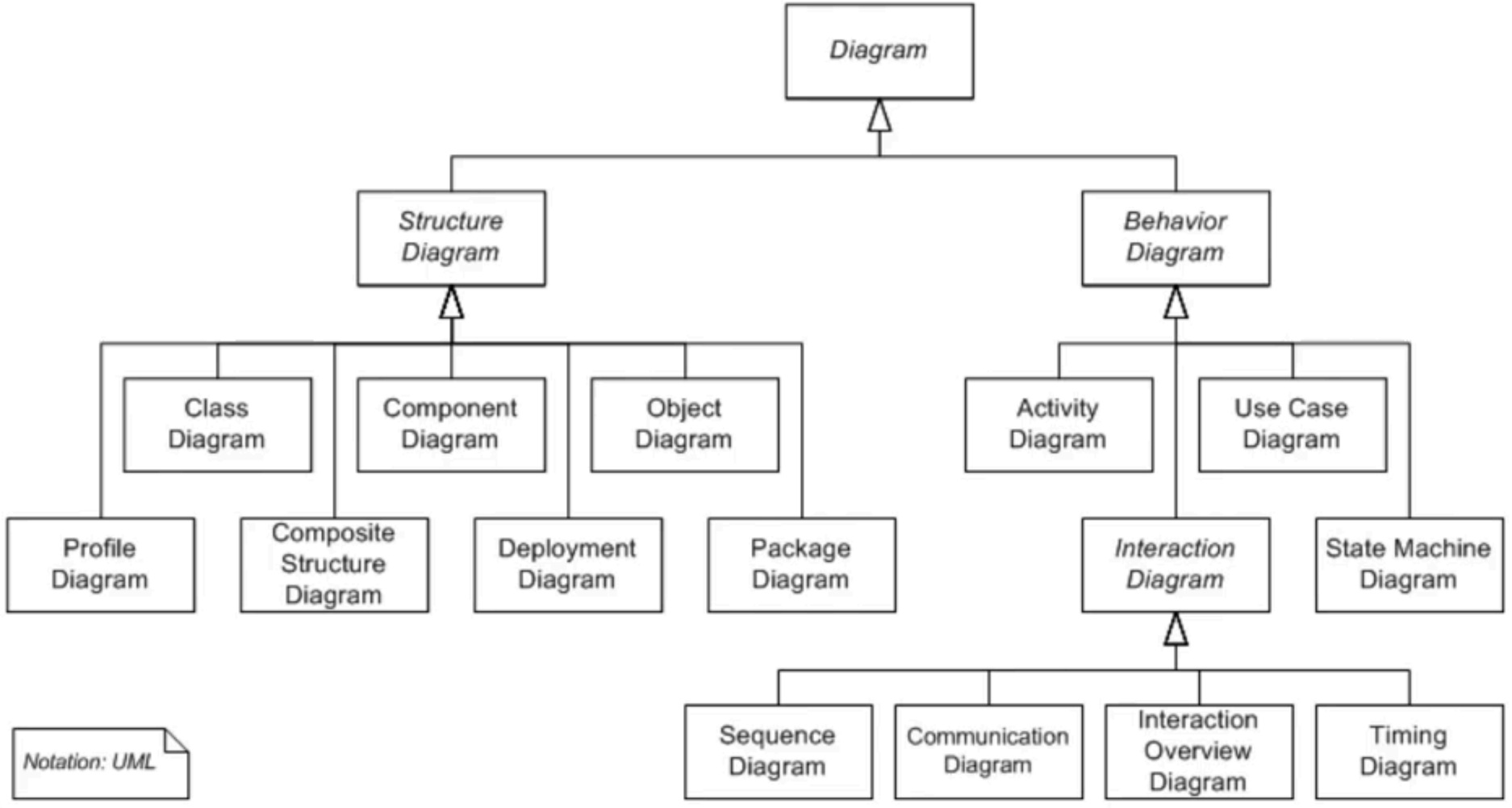
Behavioural View: Sequence Diagram



Behavioural View: Sequence Diagram



UML Diagrams



Paulo Merson, Public domain, via Wikimedia Commons

Design Comprehension in the Workplace

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Software developers usually

- Work on existing large, complex systems
- Write additional features based on new requirements

Require them to explicitly comprehend software designs

Adding New Features

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Mood based music player

Amazon

Inventory module

Payment Gateway module

Video streaming module

Module X

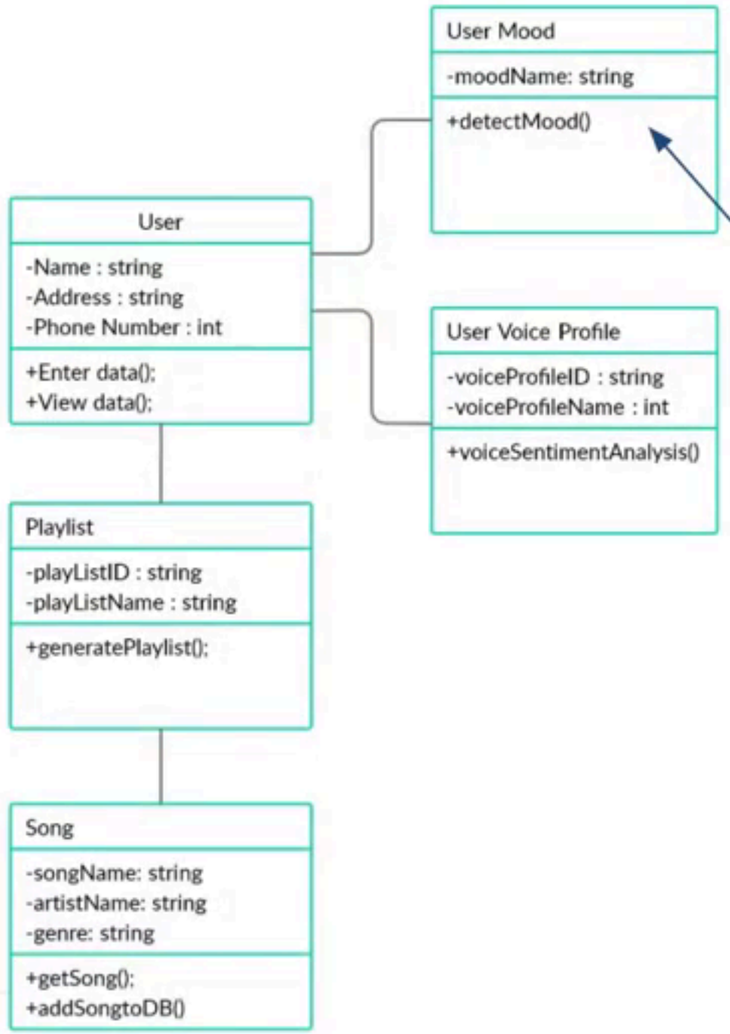
Reflection Spot

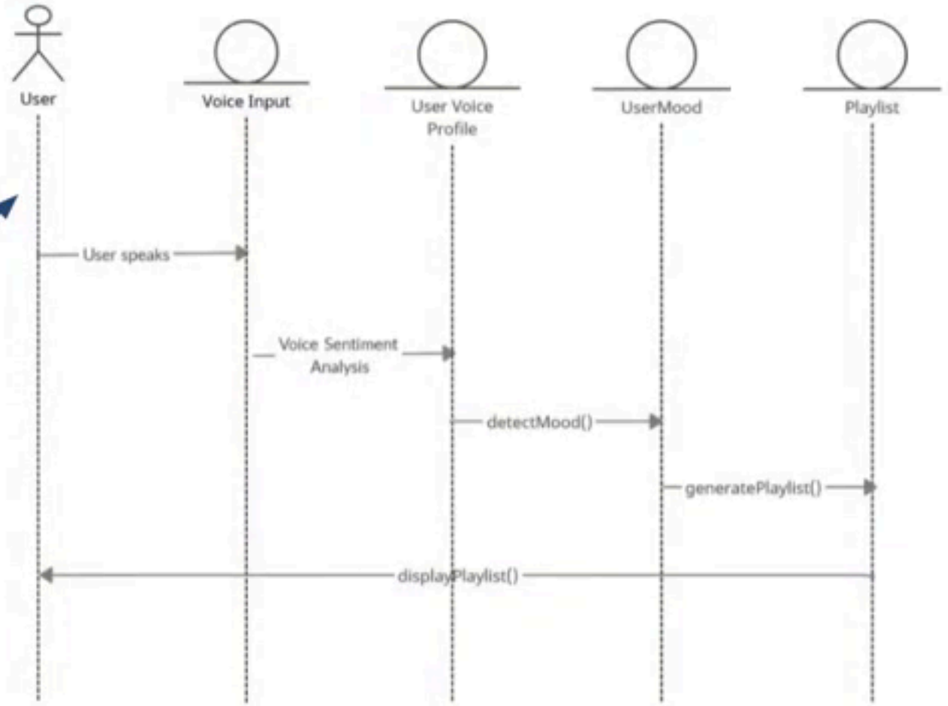
IIT BOMBAY

What difficulties do you think one faces while trying to analyze different UML diagrams in a design?

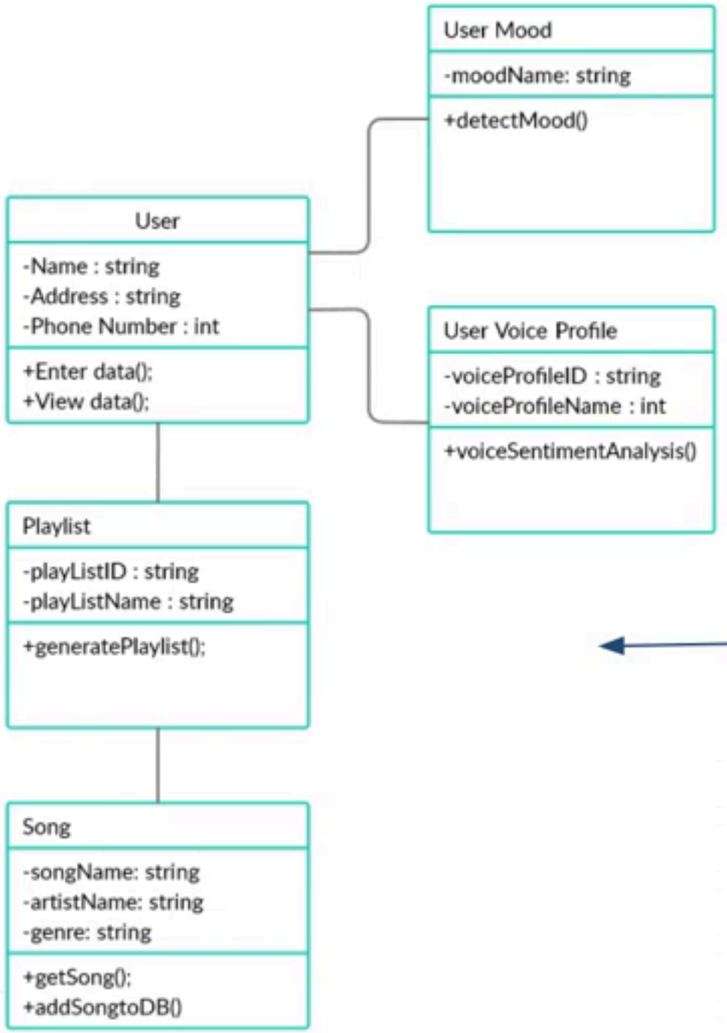
Please pause the video and write down your responses

Difficulties in Understanding Notations





Difficulties in Mapping Requirements to Design



Requirements of Mood Based Music Player

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NPTEL

Difficulties in Making Connections between Diagrams

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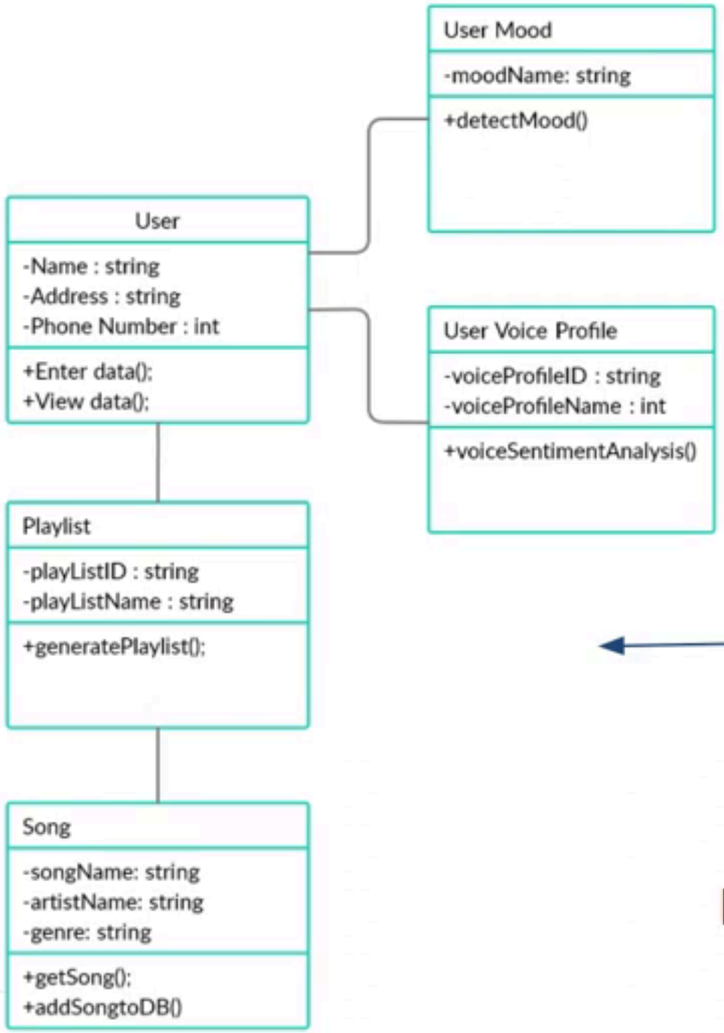
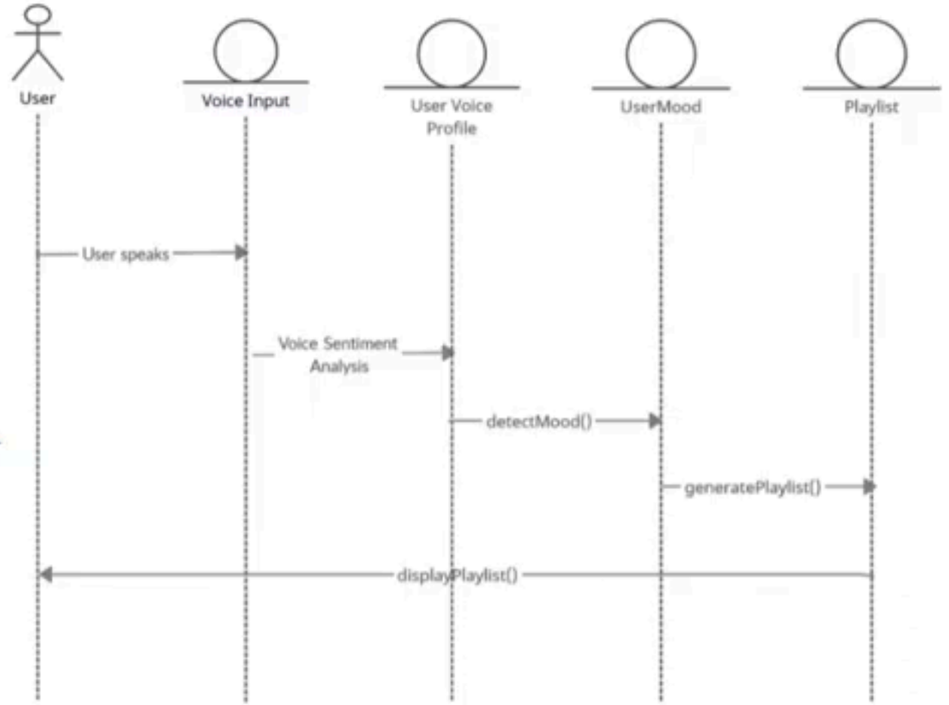
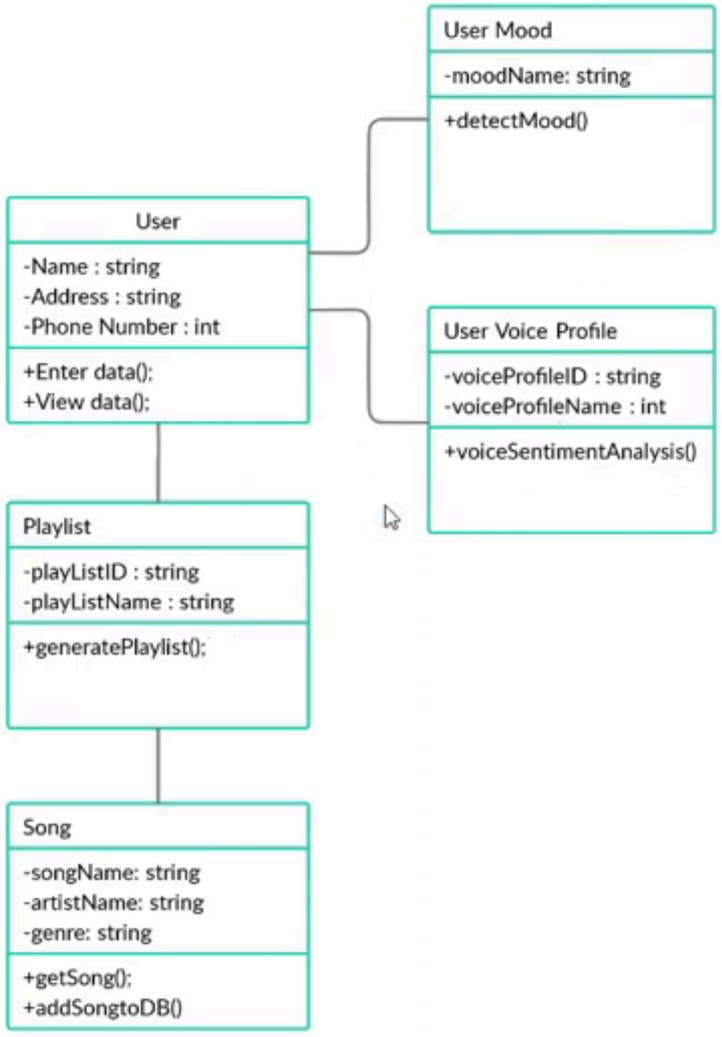


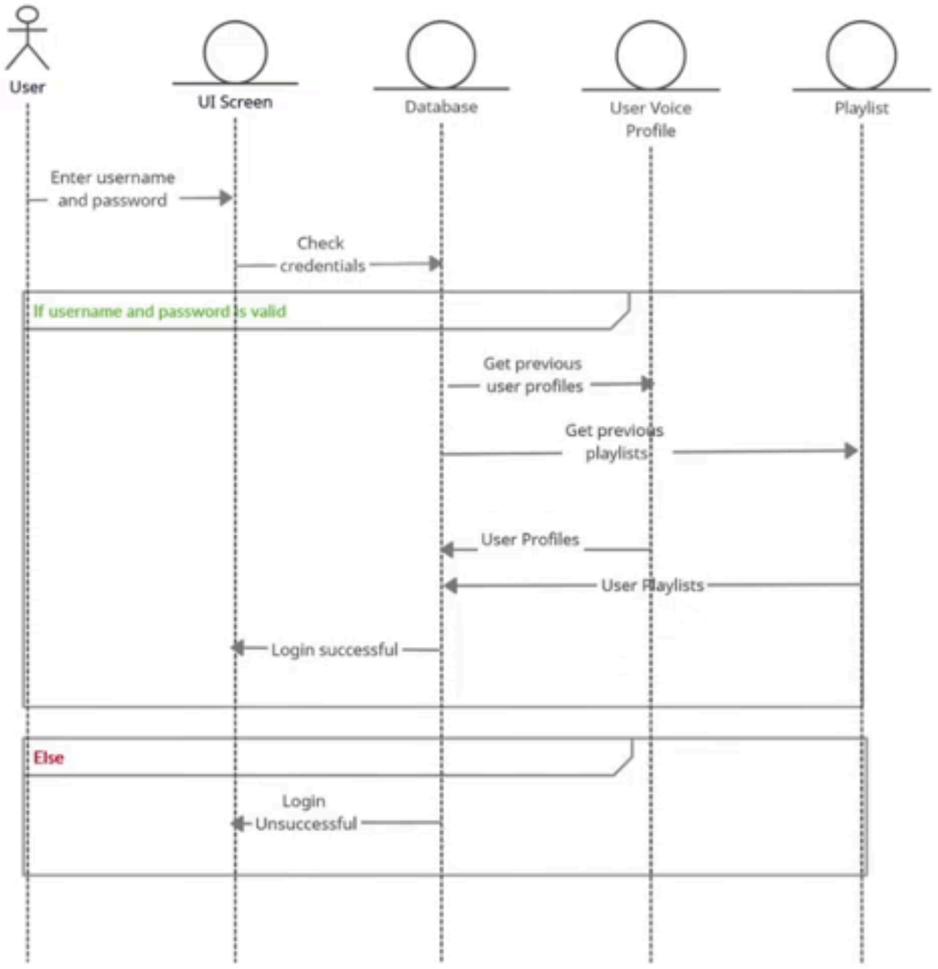


Relationship between different diagrams

1. Domain knowledge - e.g. knowledge about music players
2. Design diagram knowledge
3. Main goals
4. Dynamic behaviours in the design



Sequence Diagram for user login



Design Comprehension Process

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To effectively perform software design comprehension, one requires knowledge of:

- The domain
- The design diagrams
- Main goals present in the diagrams
- Dynamic behaviours in the design

Unified Modeling Language Diagrams

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- UML diagrams - describe different views of the system

- VeriSIM Learning Environment -
  - **Veri**fying designs by **Sim**ulating Scenarios
  - develop an integrated understanding of class and sequence diagrams

Software Design Context in VeriSIM

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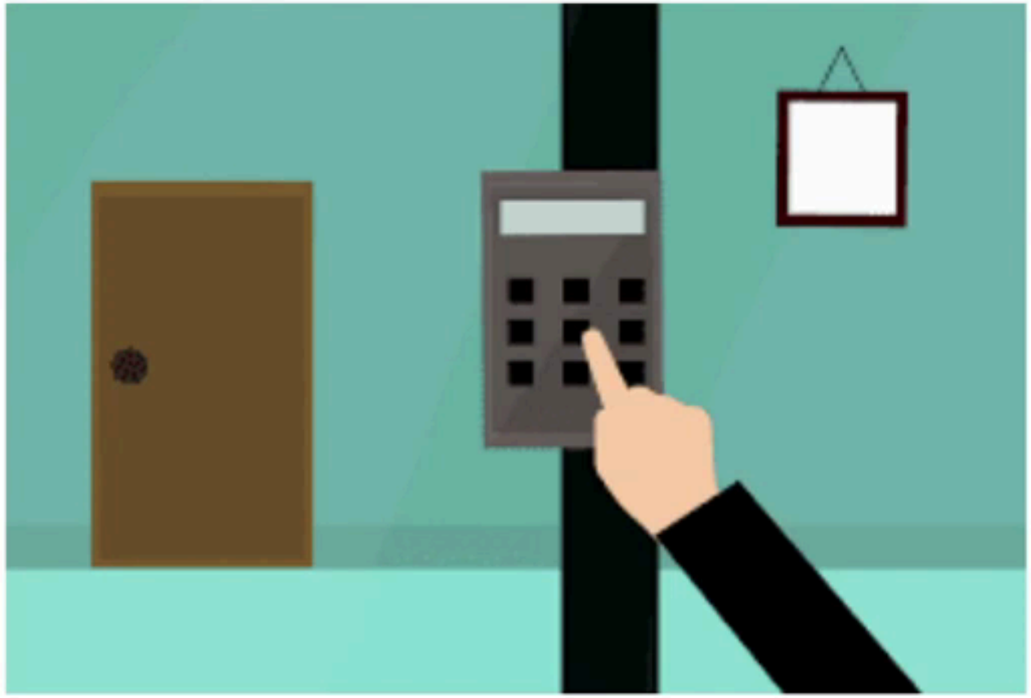
Automated Door Locking System:



Requirements:

1. If the passcode hasn't been set yet, the user can **register** and enter a required passcode.
2. When the user chooses the **lock option**, and enters the correct passcode, the door should lock. If the passcode is incorrect, the door remains unlocked.
3. When the user chooses the **unlock option**, and enters the correct passcode, the door should unlock. If the passcode is incorrect, the door remains locked.

UML Diagrams

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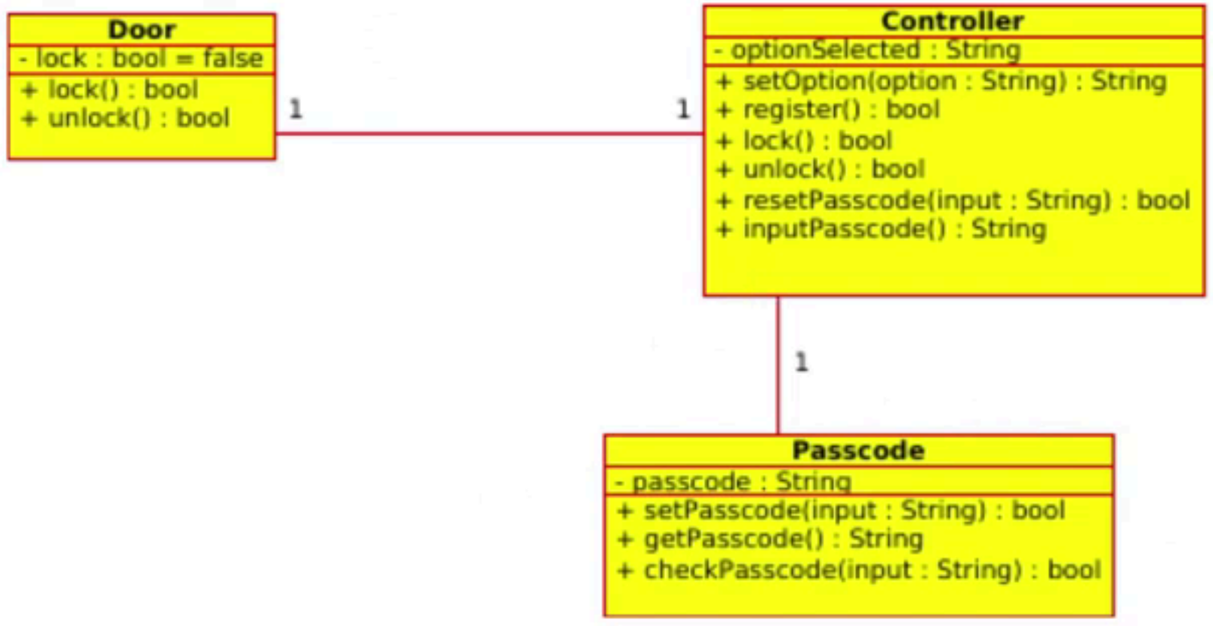
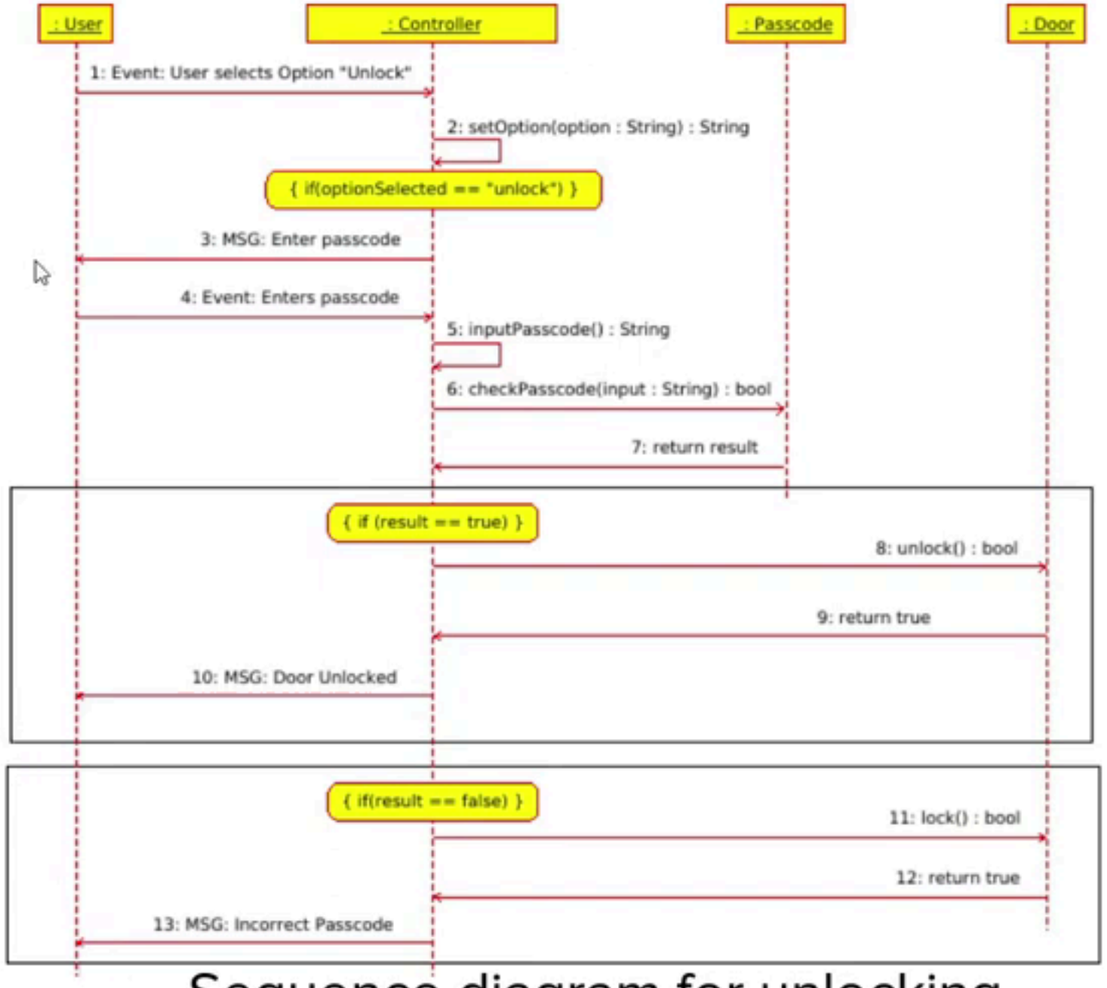
Requirement: When the user chooses the **unlock option**, and enters the correct passcode, the door should unlock. If the passcode is incorrect, the door remains locked.



Class diagram



Sequence diagram for unlocking

Design Tracing Strategy in VeriSIM

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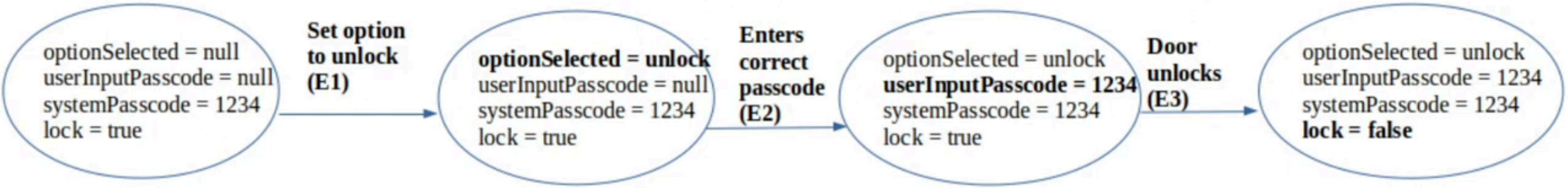
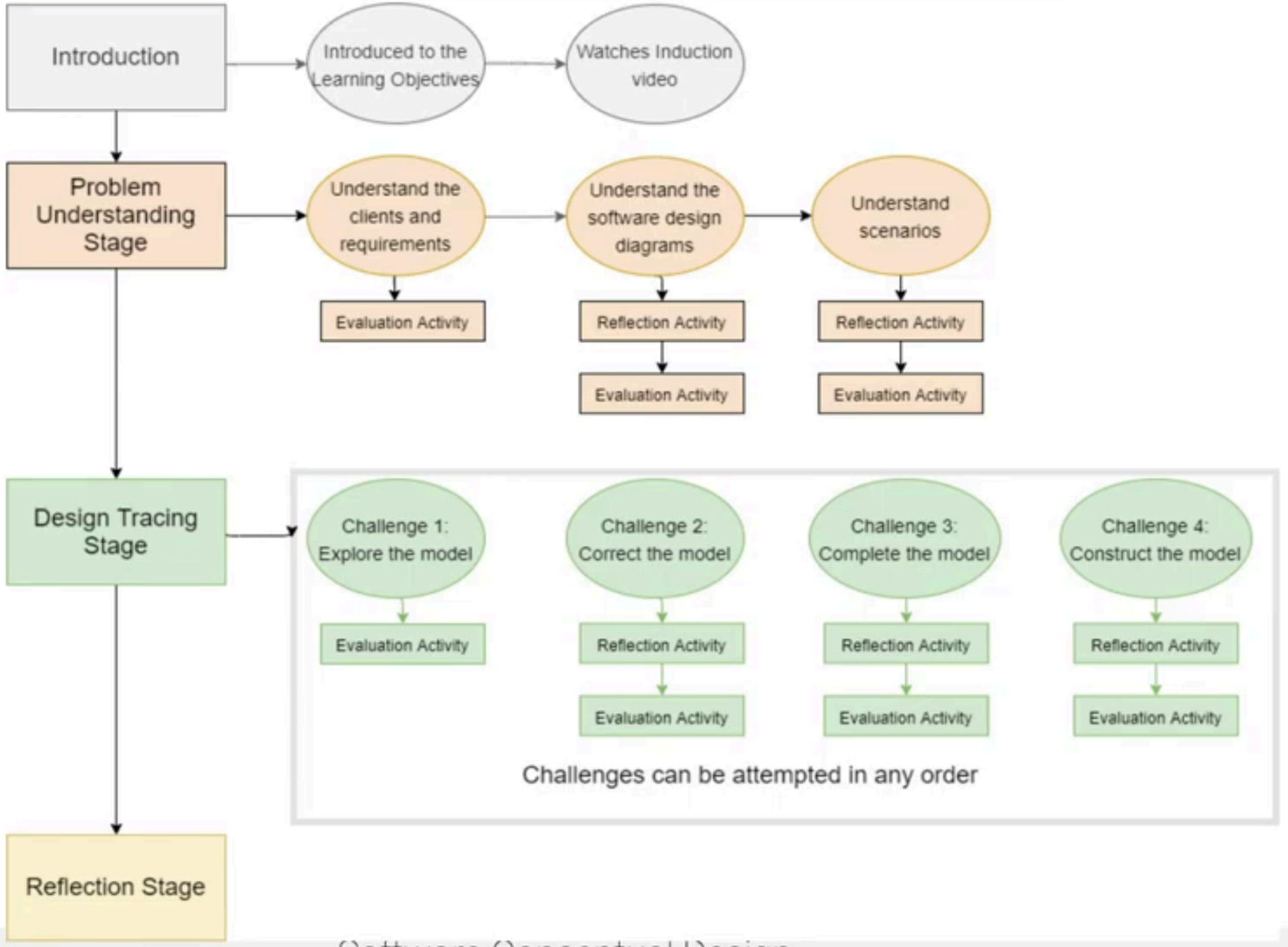
Scenario:

When the door is initially locked and the user selects the unlock option and enters the correct passcode, the door unlocks"



Construct a state diagram which models the scenario

Activities in VeriSIM



Reflection Spot

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You are paying money for a tiffin service, which provides you with three meals - breakfast, lunch and dinner everyday.

What are different scenarios for which you will **not** be satisfied with the tiffin service?

Please pause the video and written down your responses

Correctness

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Correctness is the extent to which a program behaves according to its specification

Functional Requirements

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- **Correctness** - one of the functional requirements of the system
- Functional requirements describe **what the system should do.**
- We need to ensure that all the functional requirements are satisfied by the design

Reliability

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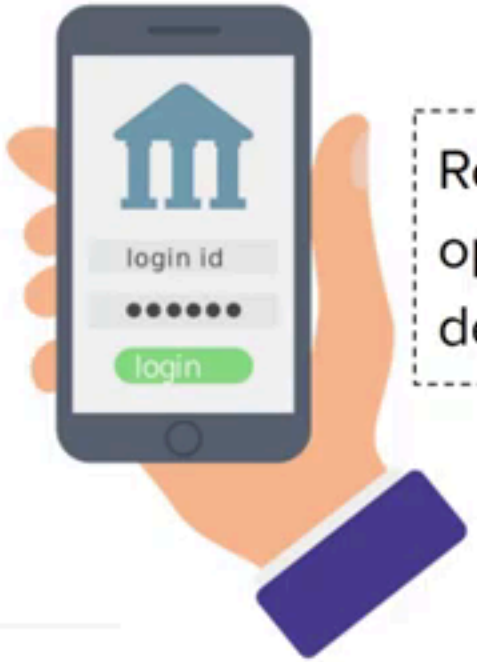
Reliability is the extent to which a program behaves the same way over time in the same operating environment



Reliably handle important operations like withdraw and deposit

Robustness

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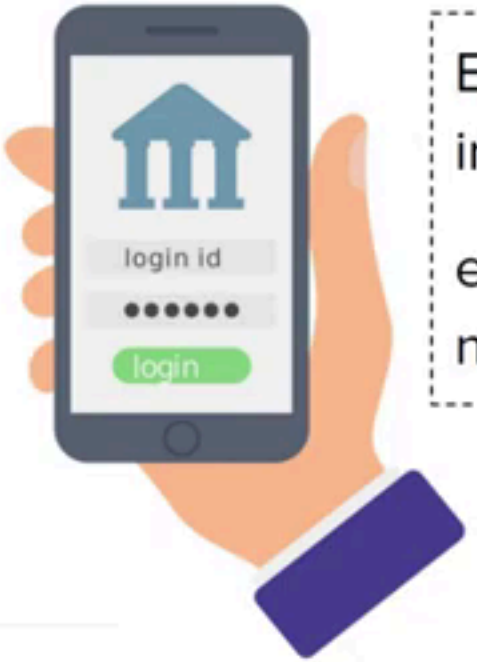
Robustness is the extent to which a program can recover from errors or unexpected input.



Ensure handling of unexpected input

e.g. - negative numbers, large numbers, high traffic etc.

Functional and Non-Functional Requirements

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Functional requirements - describe what the system should do.

Non-functional requirements - specify how the system should behave

Modularization

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- Divide a software system into multiple independent modules
- Capable of carrying out the desired functionalities

Reflection Spot

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Why did we learn about the various phases in software design and the software development models?

Please pause the video and write down your responses

Function-Structure-Behaviour

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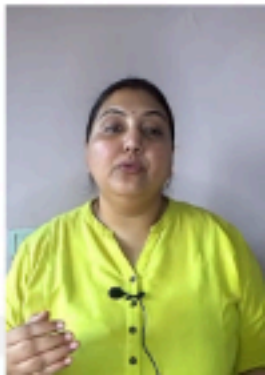
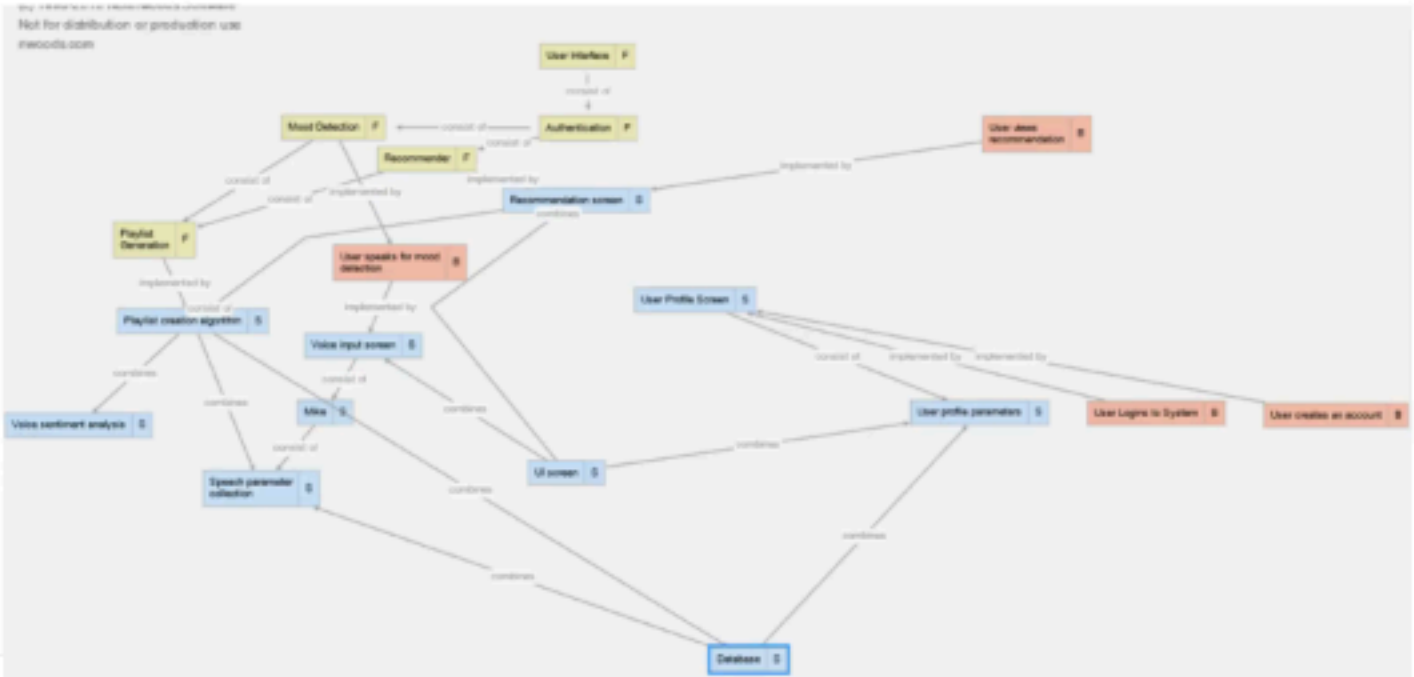
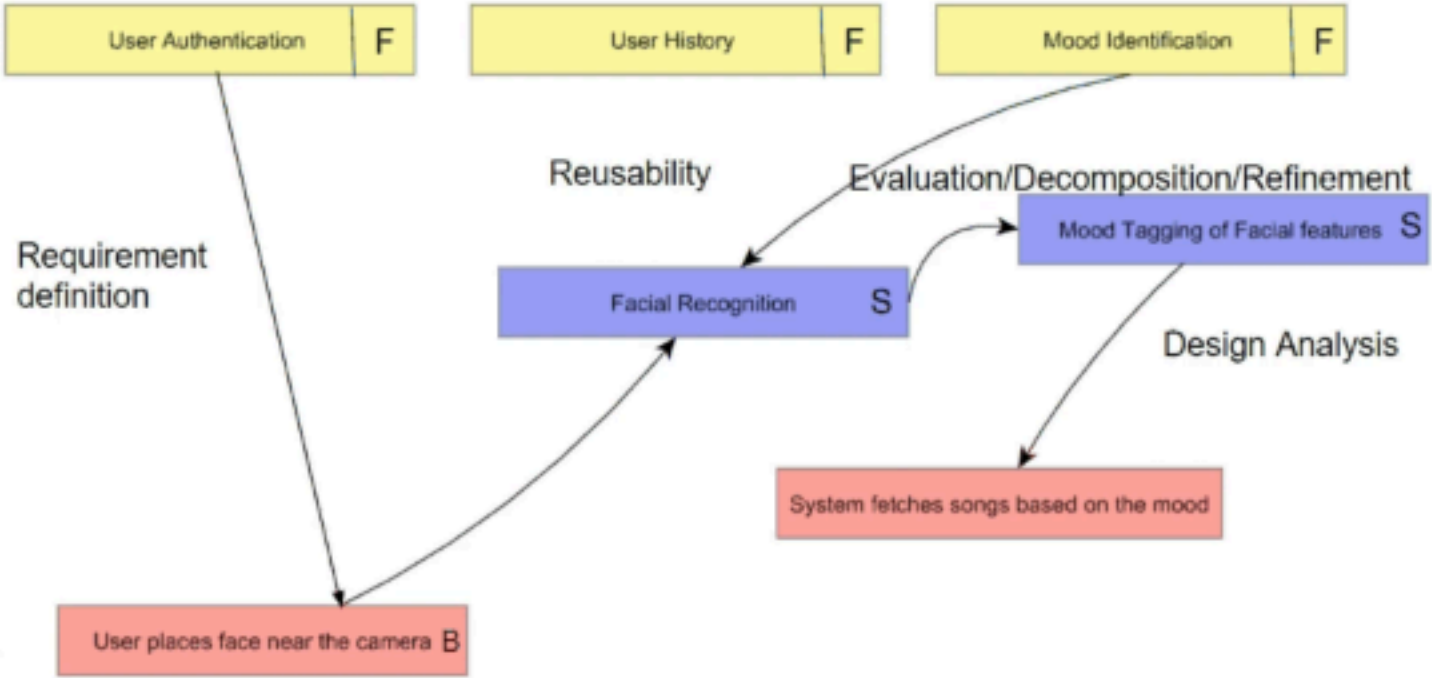
- **Function** - Functionalities and features in the design
- **Structure** - physical and logical components of the solution
- **Behaviour** - structures interact to achieve the functionality



Modelling using FBS



Mapping of FBS design elements



Mapping of FBS design elements

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Mapping FBS elements	Software Engineering Process (Rational Unified Process)
Function & Behaviours	Requirement definition
Structure & Behaviours	Design analysis, testing, implementation and requirements change
Behaviours & Behaviours	Evaluation, decomposition and refinement
Structure & Structure	Evaluation, decomposition and refinement
Function & Structure	Reusability and change in needs

Abstraction in Design

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Definition:

Hiding details in the design at each stage of the design process

High Level Abstraction

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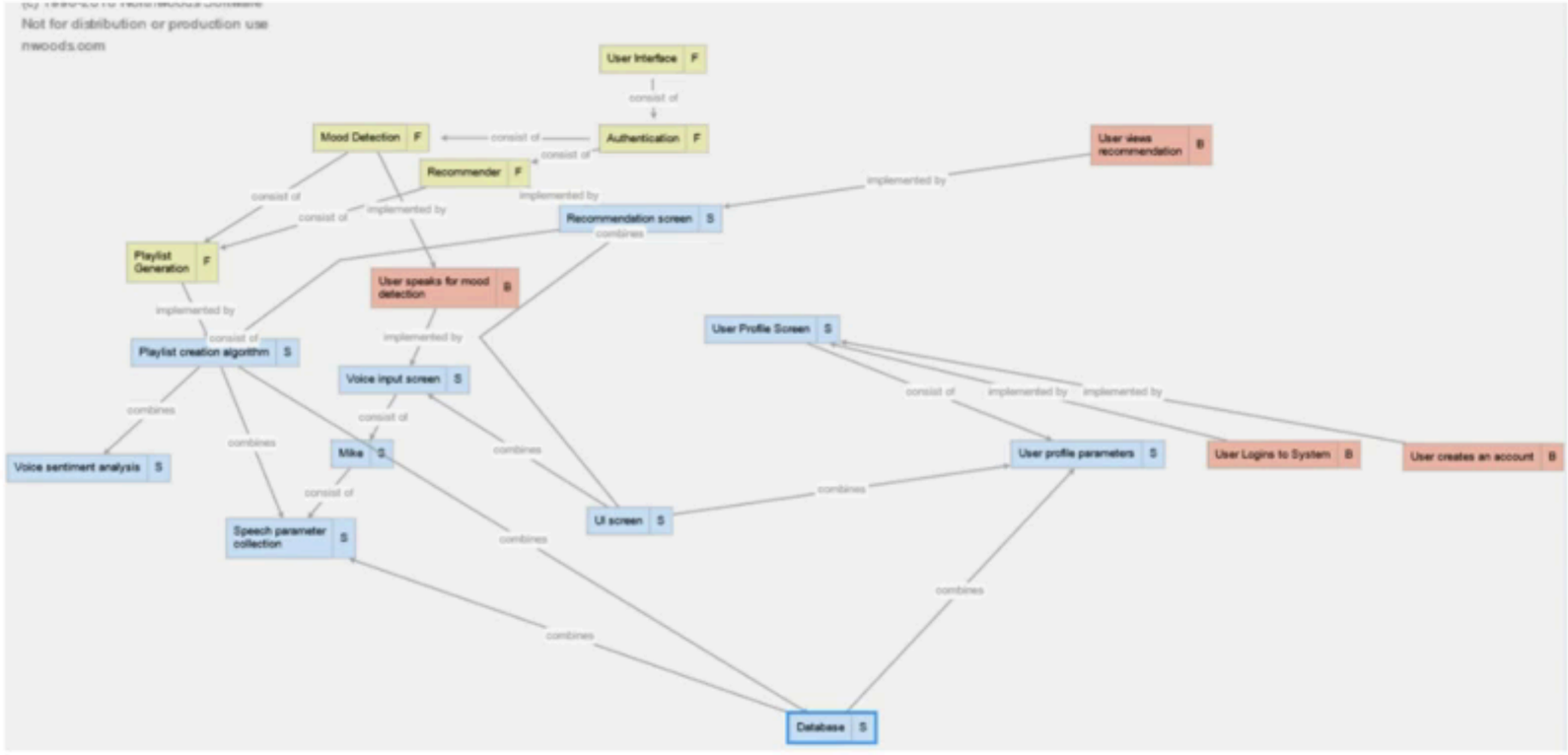
E.g. - FBS representation



Medium Level Abstraction

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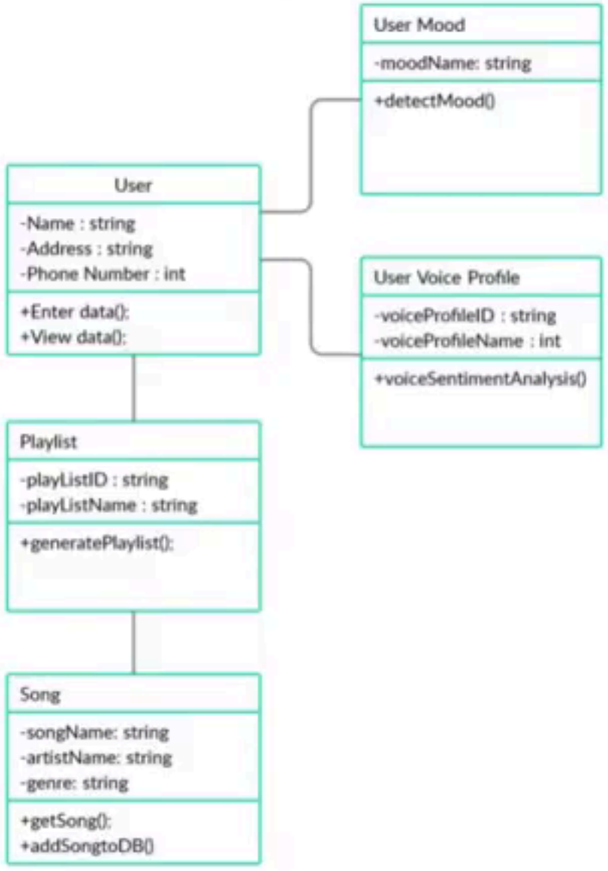
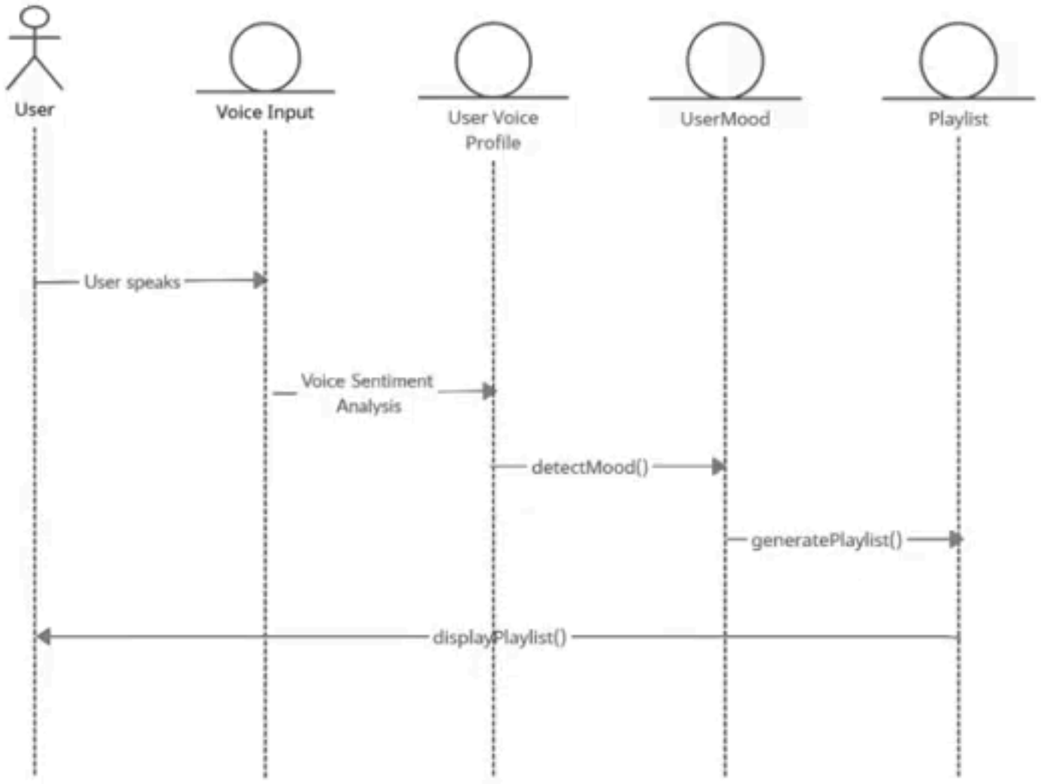
E.g. - UML diagrams





Modelling at Different Abstraction Levels

- Functions - Use cases
- Structures - Class diagrams detailed from the structures
- Behaviours described by sequence diagrams

Implementation

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- Close correspondence with the implementation (code)

- E.g. - Every class in a model → map onto one or more classes in the implementation code

UML to Code

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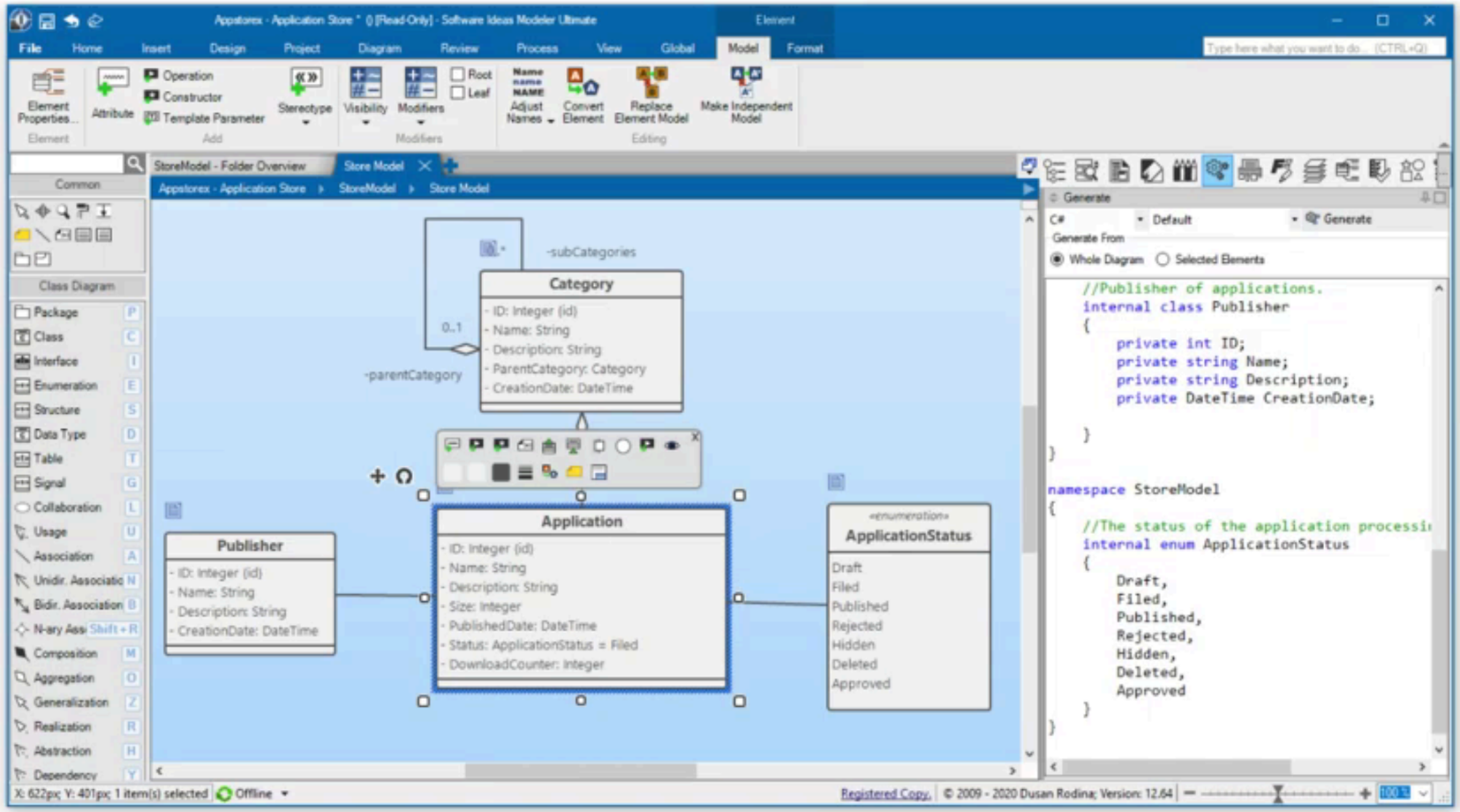

https://www.softwareideas.net/code-generation-diagram-to-code

Reflection Spot

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What are different purposes for which these modelling techniques are used?

Please pause the video and write down your responses

Purposes of Modelling

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- Vehicle for communication and idea generation
  E.g. - FBS Graph

- Analysis and comprehension
  E.g. - UML models

- Blueprint for Implementation
  E.g. - Detailed UML models